



D4.1: 'First wave' of DEDALUS Digital Twins and data-driven services for residential DR

D4.1: 'First wave' of DEDALUS Digital Twins and data-driven services for residential DR
WP4



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List of Acronyms

ADA	Adaboost
API	Application Programming Interface
BAG	Bagging
DB	Data Base
DDWT	Derivative Dynamic Time Warping
DHN	District Heating Network
DLT	Distributed Ledgers Technology
DM	Demand Response
DR	Demand Response
DT	Digital Twin
DT	Decision Tree
DTW	Dynamic Time Warping
ED	Euclidean distance
FR	Functional Requirement
HAC	Hierarchical Agglomerative clustering
HLUC	High Level Use Case
ISP	Integrated Scheduling Programming
KNN	K-Nearest Neighborhood
MQTT	Message Queuing Telemetry Transport)
NB	Naïve Bayes
RC	Resistance-Capacitance
REMAP	Real-time Energy Metering and Actuation Platform
RF	Random Forest
sPMV	simplified Predicted Mean Vote
TE	Technological Enabler



TRL	Technology Readiness Level
TSO	Transmission System Operator
UC	Use Case
WP	Work Package



Executive summary

This deliverable reports about the first wave of DEDALUS Digital Twins and data-driven services for residential DR, corresponding to the 1st technology development cycle completion, MS4 due at M13. D4.1 has a twofold objective: to describe the software artifacts released, as DEDALUS Technology Enablers (TE) and services, while also providing the technical specifications of TEs and more in general of services that are outcome of WP4. Indeed, this deliverable is also conceived as technical specification, addressing as intended audience, mainly services developers. Therefore, for each TE the following items are provided:

- Overview of the service.
- Architectural diagram compliant with the DEDALUS architecture formally provided in D2.3.
- Data input required for the execution of the service.
- Service's output.
- Main benefit and innovation.
- Result of the development activity as achievement of MS4 1st technology development cycle completion.
- Outlook for the future expected release as plan for the foreseen developments.

The development activities under the scope of Task 4.1 *Digital Twins for optimal assets vs consumers flexibility planning* involves two TEs: **TE-8 - Digital Twins for consumer-aware optimal flexibility planning** and **TE-5 blockchain and smart contracts integrated DT**. The first one is a tool designed to optimise the management of various assets and mainly to support consumers in the optimal management and assessment of flexibility resources; it is applied across different scopes in pilots 2 (DK), 3 (GR), and 4 (IT). In the first two pilots, MS4 consists in a data analysis process as prerequisite for developing the Simulation Engine component; regarding the pilot 4, a first version of Middleware API is available. Going into detail, the Danish pilot focuses on the demonstration of smart heating management and flexibility services of apartments buildings connected to the public heating network; the data analysis performed in this pilot, works as prerequisite for having a clear understanding on the behavioural of flexibility assets and how they can be actively involved in the provision of flexibility services. Starting from pilot historical data available, a preliminary assessment of DHW consumption has been performed and, to evaluate the degree of correlation between different temperatures, in order to evaluate the possibility to train Machine Learning or Deep Learning forecasting models. The Jupiter Notebook with all the aforementioned analysis is available in the project's GitHub repository. Regarding the Greece pilot, the development activity mainly consists in a deep analysis of pilot data to gain insight on the flexibility potential of the Greek pilot participants once the DR services are deployed.

The development activity in the Italian pilot mainly consists in the first version of Middleware API allowing the interaction with the service responsible for providing recommendations and alerts to the building manager concerning energy consumption while considering occupants' comfort. The middleware API is developed using FastAPI and it is available in the GitHub repository set up for the project.

The development of TE-5, exploited in the Romanian pilot, is a digital twin model for the students' apartments building that incorporates data from heterogeneous sources with the purpose of supporting decision making in demand response programs and simulating the execution of apartment level DR actions. The first version of TE-5 consists in the initial implementation of



Analytics component, of Apartment models and the DR Optimisation Decision Making algorithm as Python function, available on the TUC private GitLab repository.

Focusing on Task 4.2 *Optimal trade-off among comfort, energy and building performance*, the MS4 involves the development of Comfort-based flexibility model that is linked to **TE-9 Comfort-based flexibility models** and aims to ensure comfort in indoor environments by harmonizing Indoor Environmental Quality (IEQ) factors such as thermal comfort with energy consumption and costs. It facilitates Demand Response (DR) services while emphasizing the preservation of health and well-being during their implementation. The development activity achieved as MS4 is the analysis of historical data from the Danish pilot and the development and application of ML models with preliminary results. This development is done in Python and the associated codes are available in the DEDALUS GitHub repository.

The software released in the context of Task 4.3 *Optimal DR management strategies (implicit and explicit) adaptable to different segments of residential energy consumers* is linked to **TE-10 Nudging application** and **TE-11 Multi-value DR service** and corresponds to 7 services (named from A to G) aiming to develop optimal DR strategies in the Greek and Austrian pilot sites. **Service A** *Design of optimal DR strategies for the gas pilot* aims to calculate/simulate the energy consumption performance of each household under all possible configurations in order to identify the households that present potential of energy savings, without of course violating the comfort levels of the end users. The first version of the simulator is available on the DOM-X cloud system and will be mainly exploited in the Greek pilot. **Service B** *Building performance and end-user profiling for the gas pilot* has the main goal to unveil concealed intricacies within heating load profiles and therefore to have a more distinctive comprehension of houses' energy consumption dynamics and heating system operations. The current development is related to the preparation of proper data set and the application of clustering algorithms. The software code is available on the DEDALUS GitHub. **Service C** *Energy disaggregation for the electricity pilot* relies to Non-Intrusive Load Monitoring (NILM) techniques to disaggregate energy consumption data at the appliance level; the first version of this service is available, and it is the input of DR services and the Flexibility Modelling Tool in the Greek pilot.

Service D *Development of tariff system for implicit and explicit electricity DR services* aims at generating a tariff system based on multiple conditions and their combination. The first implementation of this service consists in the development of an Application Programming Interface (API) which provides relevant information (data from ISP) for further developing the DR and Digital Twins services. **Service E**, the *OurPower Nudging Application*, enhances user engagement and optimises energy consumption within communities by providing personalized advice and enabling user responses to energy-saving opportunities. It is applied in the Austrian pilot and mock-ups of the user interface are provided as first version of the application committed on the OurPower private Github repository. **Service F** *Multi-value Demand Response for virtual clusters of building* provides Implicit DR recommendations or explicit DR actions given remote control capabilities. The development of this service consists in the definition of data structure of the DR recommendation or action, moreover the optimisation problem is provided. Finally, **service G** *Integration of Implicit Demand Response service - Recommendation in the DT is*, aims at integrating the outcome of service F within the context of the Digital Twin TE-8 in Task 4.1. The integration of DR Programs in the Digital Twin will follow the implementation of the respective TE; for this development cycle, a first version of middleware is available.

The service developed in the context of Task 4.4 *Local flexibility management for energy communities and self-consumption/cost optimisation* focuses on optimising and managing local energy flexibility within communities and districts, targeting economic, social, and environmental



benefits. Leveraging **TE-12 P2P digital platform** and **TE-13 District level DR optimisation**, the service enhances demand response (DR) management through a sophisticated decentralized management framework and enables trading flexibilities on the flexibility marketplace. The development activities in the reported timeframe of the project have focused primarily on conceptualizing the service and planning the integration of its various components, setting a solid conceptual foundation for future development and implementation phases. Even if the same approach is shared for the service definition, TE-12 will focus on the Austrian demo while TE-13 will focus on the Spanish demo.

Task 4.5 *Optimal Aggregation of flexible energy resources for heat vs electrical flexibility valorisation* involves the development of services linked to the **TE-14 Aggregation of flexible energy resources for district heating management** extending the NEOGRID Platform and applied in the Danish pilot. The implementation activity consists in the technical specification of services with regards to the input and output data.



1. Introduction

1.1. Purpose and audience of the document

The main purpose of this document is to provide information about the first wave of the *DEDALUS Digital Twins and data-driven services for residential DR*, as main outcome of development activity of WP4 *Value added energy and non-energy data-driven services for DR in energy-efficient residential buildings*. Indeed, D4.1 corresponds to the first technical release MS4 (May 2024). Moreover, this report provides relevant information with regards of the functional and technical specification of each service and details about the architectural aspects and the designed algorithms.

The information presented in this document is useful for all DEDALUS partners, but the audiences mainly composed by technical partners responsible for the development activities and therefore involved in the extension of DEDALUS Technology Enabler (TE) to be delivered in the involved pilot.

1.2. Relation to other activities

This deliverable exploits the functional specification formally provided in the WP2. Indeed, use cases and functional requirements provided in D2.1 and further detailed and updated in D2.3 have been used as starting point to specify the DTs and data-driven services from technical point of view and therefore to guide the development of the first software release. The outcome of this deliverable will be mainly exploited by the WP5 *Demonstration of multi-value and energy carrier-agnostic residential DR, and social acceptance* responsible for the overall planning, management, and evaluation of the DEDALUS pilots.

1.3. Structure of the document

This document is structured as follows:

- Chapter 2 provides information about the development activities under the scope of Task 4.1 *Digital Twins for optimal assets vs consumers flexibility planning*. It is mainly related to TE-8 *Digital Twins for consumer-aware optimal flexibility planning and to the Apartment building Digital Twin for blockchain integration* that is delivered as part of TE-5 *blockchain and smart contracts integrated DT*
- Chapter 3 provides provide information about software expected to the developed as outcome of Task 4.2. This development is linked to the TE-9 *Comfort-based flexibility model*
- Chapter 4 gives a detailed description of software delivered as outcome of Task4.3 corresponding to several services of optimal DR strategies.
- Chapter 5 is about the local flexibility management for energy communities and self-consumption/cost optimisation service as outcome of Task 4.4.



- Chapter 6 provides information about the software expected to be developed as outcome of Task 4.5 and corresponding to the TE-14 *Aggregation of flexible energy resources for district heating management*.



2. Digital Twins for optimal assets vs consumers flexibility planning

This section provides information about the development activities under the scope of Task 4.1 *Digital Twins for optimal assets vs consumers flexibility planning*. The Technological Enablers (TE) that are involved are:

- TE-8: Digital Twins for consumer-aware optimal flexibility planning;
- TE-5 blockchain and smart contracts integrated DT

Focusing on the second one, the development is related to the Apartment building Digital Twin for blockchain integration. Both TEs are extended in the context of the project according to the functional requirements introduced in D2.1 *DEDALUS gap analysis and use case capitalisation* and further detailed in D2.3 *High-level requirements for the DEDALUS social, technological and business framework*.

2.1. Digital Twins for consumer-aware optimal flexibility planning

2.1.1. Overview of the service

As powerful tool designed to optimise the management of various assets, Digital Twins (DTs) are commonly known for facilitating the oversight of buildings, district heating networks, dispatching grids, and similar infrastructures. In literature [6], the term of DT is used to identify the virtual representation of real system (e.g., building, district heating) and it is characterized by a two-way interaction (manual or automatic) between the physical and digital entity. The DT is fed by historical and real-time data and any changes in the real system is mirrored in the DT and vice versa. Moreover, the DT can offer a simulation environment to support the decision-making process enhanced by machine learning and optimisation techniques. The possibility of controlling the real system through the DT depends on the level of maturity of the DT itself spanning from 0 to 5 according to the level of integration between the physical system and its virtual representation. According to the report presented by the IET [7] and shown in Figure 1, the level 3 defines the minimum requirements for having a DT since it encompasses the real-time interaction with sensors, connected devices and IoT technologies.



Maturity element (logarithmic scale of complexity and connectedness)	Defining principle	Outline usage
0	– Reality capture (e.g. point cloud, drones, photogrammetry, or drawings/sketches)	– Brownfield (existing) as-built survey
1	– 2D map/system or 3D model (e.g. object-based, with no metadata or BIM)	– Design/asset optimisation and coordination
2	– Connect model to persistent (static) data, metadata and BIM Stage 2 (e.g. documents, drawings, asset management systems)	– 4D/5D simulation – Design/asset management – BIM Stage 2
3	– Enrich with real-time data (e.g. from IoT, sensors)	– Operational efficiency
4	– Two-way data integration and interaction	– Remote and immersive operations – Control the physical from the digital
5	– Autonomous operations and maintenance	– Complete self-governance with total oversight and transparency

Figure 1: Digital Twin maturity level according to IET

In the context of the project and with regards to TE-8, Digital Twins are addressed to residential consumers in order to improve their awareness and active participation in Demand Response services. The primary objective is to empower District Heating Network Managers, Building Managers, Facility Managers, and similar roles, to fully harness the potential of Digital Twins by supporting the management of diverse infrastructure types, such as heat districts or power distribution networks. This support extends to highlighting flexibility opportunities and assisting operators in their day-to-day management tasks. Within the context of the DEDALUS project, Digital Twins will be applied across different scopes in pilots 2, 3, and 4 as Front Runners, as well as in pilot 6 and 7 as Multiplier. The development process is guided by HLUC4 *Digital Twins prediction of user energy profile and flexibility for consumer-aware optimal flexibility planning*, defined in D2.1 and specific use cases defined in D2.3, including UC6, UC7, UC8, UC9, UC10, and UC11. Table 1 reports the complete list of Functional Requirements defined on top of the previously mentioned Use Cases.

Table 1: Service's overview

Functionalities (brief textual description)	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated pilot site	Development Status
Digital twin with an interface that will visualize the different cluster of consumers and the	TE8-FR1	UC10	TE-1, 4, 8	Pilot 3 – GR	Not Started



flexibility assessment for each cluster.					
Digital twin will deliver analytics (environmental/energy) to the energy manager.	TE8-FR2	UC9	TE-8	Pilot 4 - IT	Not Started
Digital Twin will provide recommendations/alerts (not direct control) to the energy manager based on energy consumption, while considering their comfort preferences.	TE8-FR3	UC10	TE-8	Pilot 4 - IT Pilot 3 - GR	Not Started
Digital Twins will enhance the district visualization by providing a 3D visualization basing on IFC Files, enabling direct connection with monitoring real time data of different assets.	TE8-FR4	UC7	TE-8	Pilot 2 - DK	Not Started
Digital Twin will provide a simulation environment in order to optimise the management of district heating networks and enable the possibility to develop and integrate Digital Models of assets.	TE8-FR5	UC11, UC8	TE-8	Pilot 2 - DK Pilot 3 - GR	Not Started

2.1.2. Architectural diagram

Figure 2 show the high-level architecture of the service underlining its components and how they are mapped into the DEDALUS Architectural Diagram reported in D2.3. Each of the components will be tailored on the pilot's needs following the previously mentioned FRs but the development will be compliant with DEDALUS Architecture following the interoperability guidelines provided in the context of Task 3.2. Indeed, data will be provided by each respective pilot and will be exploited by the Digital Twin through the FIREWARE Orion Context Broker.



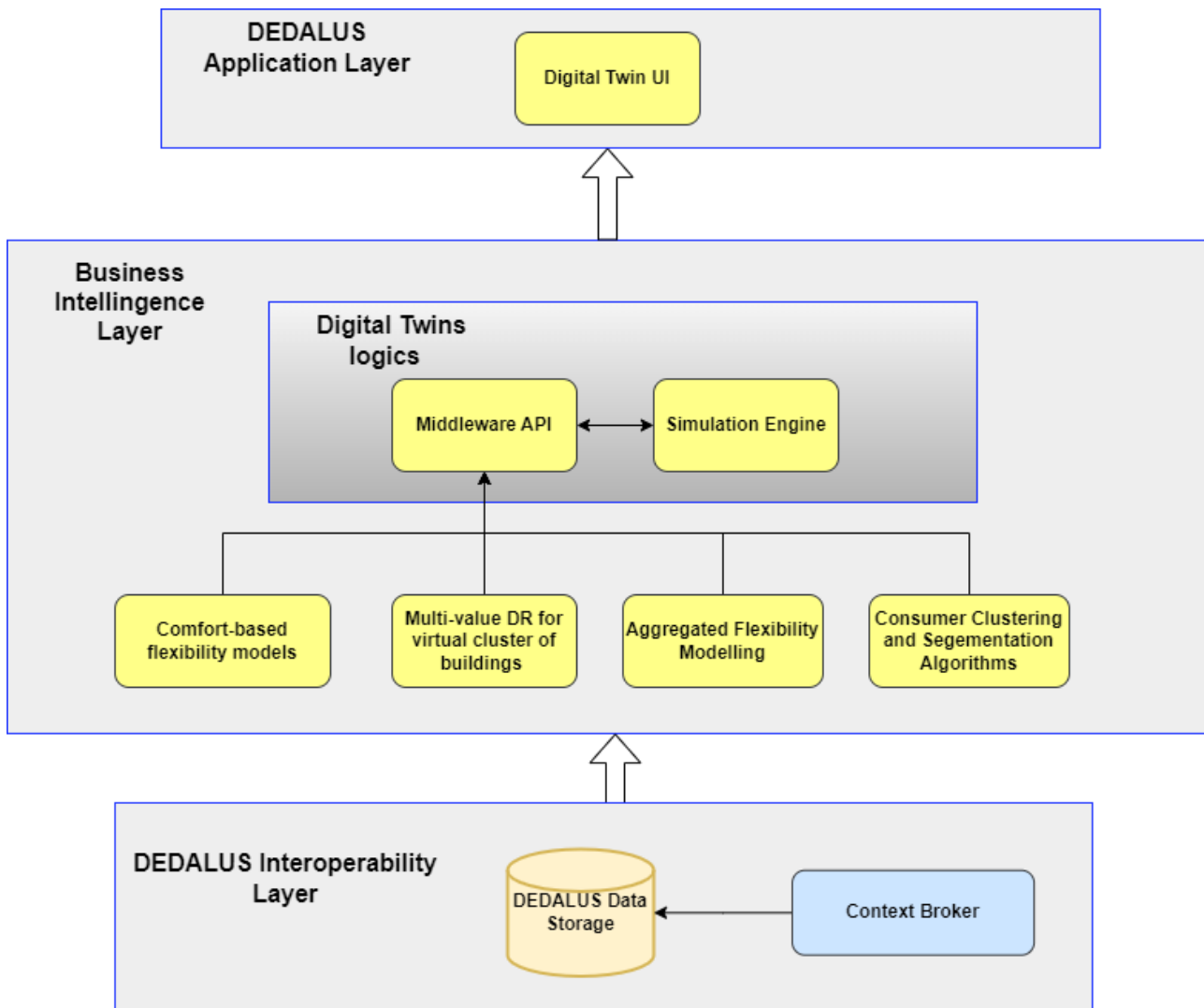


Figure 2: Digital Twins for consumer-aware optimal flexibility planning- architectural diagram

The Digital Twin is composed by the following components:

- User Interface (UI): it serves as the primary access point for the Digital Twin (DT).
- Middleware API: it is responsible for receiving all the communications coming into the DT, formalizing them into proper data structures to allow the communication and data sharing between different components. For instance, the results of the simulation engine will be provided to the middleware layer and subsequently to the UI, only once the Middleware has formalized the data. The co-existence of this layer with the DEDALUS Interoperability Layer allows a stronger and deeper integration between data and services and between service and service.
- Simulation Engine: it is in charge of performing different simulation scenarios. It can be utilized to evaluate 'what-if' scenarios by altering certain parameters to assess their impact on the infrastructure. Additionally, it can simulate Digital Models of assets, thereby enabling the Facility Manager to gauge the potential flexibility offered by the addition of specific assets.

2.1.3. Data for service execution

Data requirements for each service can be defined based on pilots and related use cases. The data to be employed for the development of the DT are determined based on the needs of the pilots and are appropriately utilized across different components. Table 2, Table 3 and Table 4 provide the data requirements for each pilot in which the DT will be applied, that are: Pilot 2 Denmark, Pilot 3 Greece and Pilot 4 Italy. For each identified data set, the related data format and protocol is defined; if the dataset is already available details are provided in the *Notes* column. Moreover, for each data set is specified the involved component.

Table 2: Data Requirements for Pilot 2 - DK

Dataset Name	Protocol	Format	Used By	Available	Notes
Energy Meter DHW	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September
DHW Circuit	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September
Energy Meter Main	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September
Heat Circuit	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September
Indoor Climate	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September
Weather Data	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September
Radiator thermostat	API	CSV, JSON	UI, Simulation Engine	yes	2023, August-September



Table 3: Data Requirements for Pilot 3 - GR

Dataset Name	Protocol	Format	Used By	Available	Notes
Smart Meter consumption data: energy, power, voltage, power factor	API	JSON, CSV	Context Broker	Yes	Access from HERON Energy Control (smart metering platform)
Smart plug data: energy, power, operation (on/off)	API	JSON, CSV	Context Broker	Yes	Access from HERON Energy Control (smart metering platform)
GR Day-ahead market scheduling and prices	API	CSV, JSON	Simulation Engine	YES	EnEx data filtered through a HERON API
RES, HYDRO and Load from GR ISP1 results for calculation of the Green %	API	CSV, JSON	Simulation Engine	YES	EnEx data filtered through a HERON API
HERON 500 KW PV	API	CSV, JSON	Simulation Engine	YES	HERON data for owned PV
brl_mod_level: Boiler Modulation Level	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
blr_t: Water Temperature Inside the Boiler	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
Heat: Heat Indicator	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
Flame: Flame indicator	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for



Dataset Name	Protocol	Format	Used By	Available	Notes
					household N° 53, 9 and 2
Water: Domestic Hot Water	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
t_out: Outdoor Temperature	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
t_ret: Inlet Temperature	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
t_r: Indoor Temperature	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
t_r_set: Indoor Setpoint	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
otc_cur: Heating Balance	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
t_set: Boiler Setpoint	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
otc_max: Mx Adaptive Setpoint	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of



Dataset Name	Protocol	Format	Used By	Available	Notes
					those data for household N° 53, 9 and 2
Bypass: Bypass Indicator	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2
Nodata: Nodata Indicator	API	JSON, CSV	Simulation Engine	Yes	domX provided a sample of those data for household N° 53, 9 and 2

Table 4: Data Requirements for Pilot 4 - IT

Dataset Name	Protocol	Format	Used By	Available	Notes
Building Consumptions	MQTT, API	CSV	UI	No	Sensor Installation in progress
IAQ Data	MQTT, API	CSV	UI	No	Sensor Installation in progress
Comfort Data	API	CSV	UI	NO	Sensor Installation in progress

2.1.4. Service output

The outcome of TE-8 is multiple, achieved through the implementation of various components tailored to different objectives. The primary access point is a user-friendly interface, presented as a web page, designed to assist the involved actors like the Building and the Facility Manager in addressing flexibility proposals and Demand Response (DR) programs.

The Digital Twin Logic models will produce specific outcomes for each pilot. For Pilot 2 - Denmark, the model will provide a set of optimised temperatures, enabling the Building/Facility Manager to assess the impact of temperature changes within the building. In Pilot 3 - Greece, a digital model of a storage system will be developed based on specifications provided by the pilot. This model will be utilized to devise potential usage strategies aimed at reducing energy usage peaks



and minimizing the carbon footprint, delivering a set of recommendations on how to optimally utilize the storage system.

2.1.5. Main benefit and innovation

The main innovation brought about by the Digital Twin is the ability to facilitate the management of different types of infrastructure in an immediate and intuitive manner. The DT ensures that the Building Manager, rather than the Facility Manager always can have a clear and detailed view of the infrastructure he has to manage, as flexible asset, via the UI, and can assess the planning of the resources at his disposal, facilitating DR and flexibility programs. A further strength of this technology, in the context of DEDALUS, is the integration of comfort models, which allow DR programs to be refined by also considering the well-being of the facilities' occupants. In addition, the use and integration of simulation, optimisation and Digital Models allow the near real-time evaluation of hypothetical conditions (what-if scenarios), future energy balances and the presence of assets that do not actually exist (Storage Systems' Digital Model), without any detrimental effect on the real infrastructure, adding value to energy usage planning and optimisation.

2.1.6. Development activity

The result of the development activity as achievement of MS4 *1st technology development cycle completion* is mainly related to Pilot 2 and Pilot 4. Regarding the Pilot 3, the work has been mostly focused on the technical specifications of the service. More details are provided in the following sections.

2.1.6.1. Pilot 2 – Denmark

1. Pilot Configuration and Data Description

For Pilot 2, the social housing company, Fællesbo in Denmark, provided the apartments from which NEOGRID is collecting and sharing data. This pilot site is composed by a sub-section of the housing department, consisting of four fully renovated buildings with a total of 692 flats. A dedicated area of 2000 m² within this district is included in the pilot.

Each building is connected to the public district heating network. Within the building a district heating unit is mixing down the temperatures of the district heating water circulated around the building to the apartments to the temperature required. A physical BMS controller is connected to the district heating unit controlling the forward temperature based on the outdoor temperature. A Neogrid gateway is connected to the BMS controller enabling datalogging and external control of the district heating unit possible. From each apartment, data from IoT sensors measuring the indoor climate and Smart radiator thermostats controlling each radiator is logged. Furthermore, data from apartment ventilation systems and apartment energy meters are logged and provided to the DEDALUS project. The data used for this analysis – and which will train the models being



developed — is sourced both from the BMS and from the Heating Controller device, responsible for storing data related to the DHN unit.

While NEOGRID cannot control the DHN, they can have an impact on the temperatures of the buildings. Figure 3 reports the architectural model of the simulation engine, also highlighting the input data.

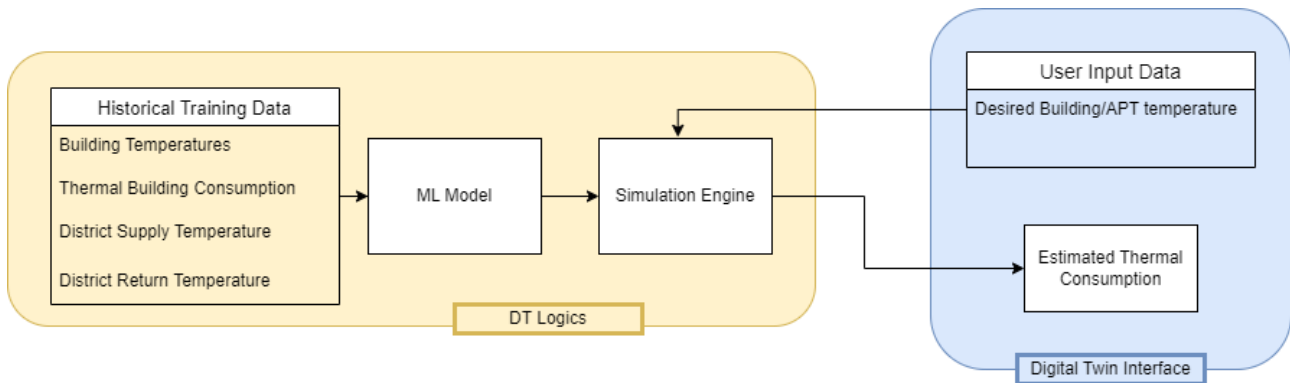


Figure 3: Simulation Engine – architectural model

The Simulation Model will be trained on a subset of the BMS data, storing the building related temperatures, and a set of data coming from District Heating Unit, storing the data coming from the public DHN. The Building Manager will be required to insert the desired building heating setpoint temperature, and the model will estimate the consumption given the input. This will allow the Building Manager to evaluate the impact, in terms of consumptions, of temperature changes in the building.

2. Simulation Engine Development

The development of Pilot 2 Digital Twin is related to the Simulation Engine component. In order to gain a comprehensive understanding of potential implementations, an exhaustive data analysis process has been undertaken, and this analysis has been conducted in order to calculate the thermal consumption related to buildings, to use it to train a prediction model, enabling the estimation of the variation of the consumptions, related to the variation of some operational parameters (e.g. setpoint temperatures). Particularly, a batch of historical data provided by NEOGRID, and stored in the project's repository, displaying some historical metrics related to a single location. The consumption calculation is performed for a single building, in these early stages, and then, once a more detailed version of the model will be developed, extended to the entire pilot.

As it was defined in the UC9 *Digital Twins as simulation environment for the District Heating Network* of D2.3, the main goal is to provide a simulation environment allowing the evaluation of the efficient operations of the DHN, in order to provide the Building/Facility Manager the possibility to assess the impact of temperature changes (e.g. supply/return temperatures) in the building, while taking into account also the comfort of the inhabitants. This will be achieved using a combination of Machine Learning models, to estimate the behaviour of the building given some

parameters, and optimisation models, aimed at achieving the optimal energy efficient status of the pilot.

To comprehensively grasp the pilot's behaviour, an extensive examination of temperatures profile of the building has been conducted, focusing – in a first step on DHW supply and return temperatures, resampled in 1-hour timesteps and setpoint temperatures. Given the intricacies of the dataset, a resampling technique was employed to aggregate data into two-hour slots, also facilitating the visualization. Figure 4 illustrates the outcomes of this detailed analysis. All the data available in the timeframe provided by the pilot (August – September 2023) has been considered.

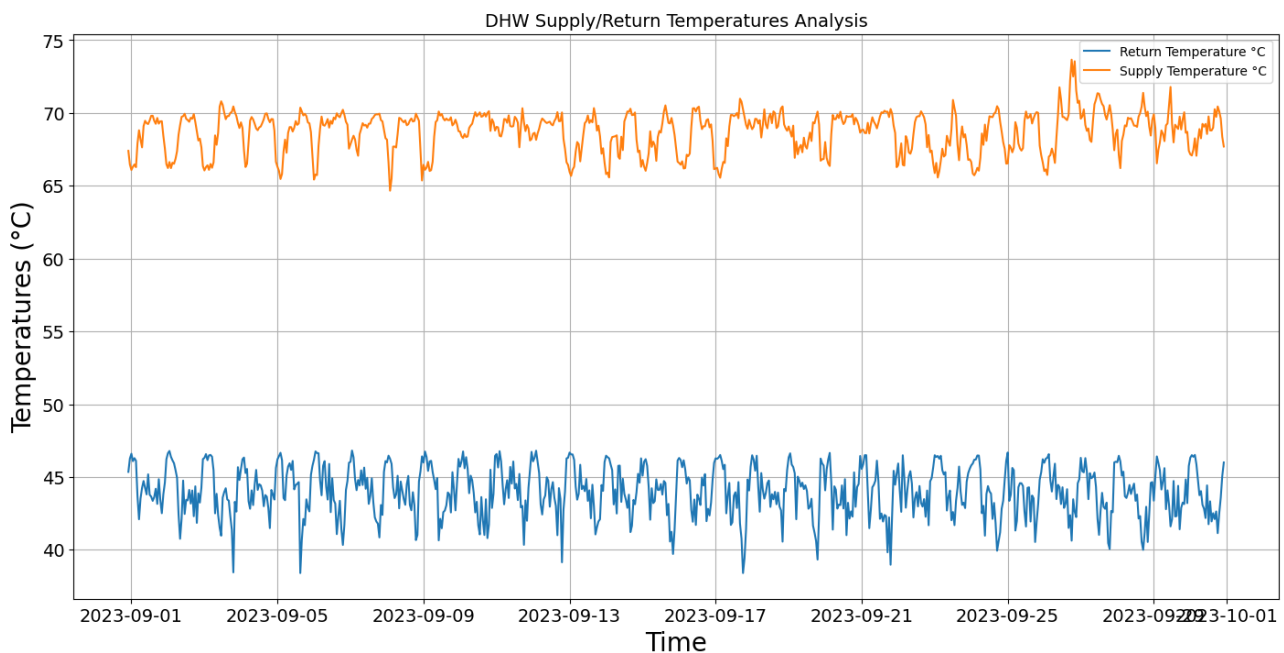


Figure 4: Comparison between Supply Temperature sent into the building and Return temperatures sent back to the grid.

Following in the analysis, a complete and aggregated dataset has been created in order to be compliant with what will be used for the simulation. To optimise the consumptions, the future trend of this temperatures should be estimated basing on this historical data. From the initial situation, Heat Circuit variables (the once has been considered in this second analysis), that depicts the temperatures of the water flowing across the buildings were also extracted, considering 5 minutes time slot, resampled using mean values in 1-hour steps. Heat Circuit Values, particularly the setpoint temperatures, are the values that are directly responsible for the space heating power that is then delivered to the apartments. Those are the main values that will be investigated to optimise the thermal buildings consumption while keeping the optimal distributed heating power across all the apartments. Figure 5 reports the results of this analysis. As it can be clearly seen, the trend of the temperatures resampled in this manner, shows a high degree of correlation, that it's exploitable for the development of a simulation model.



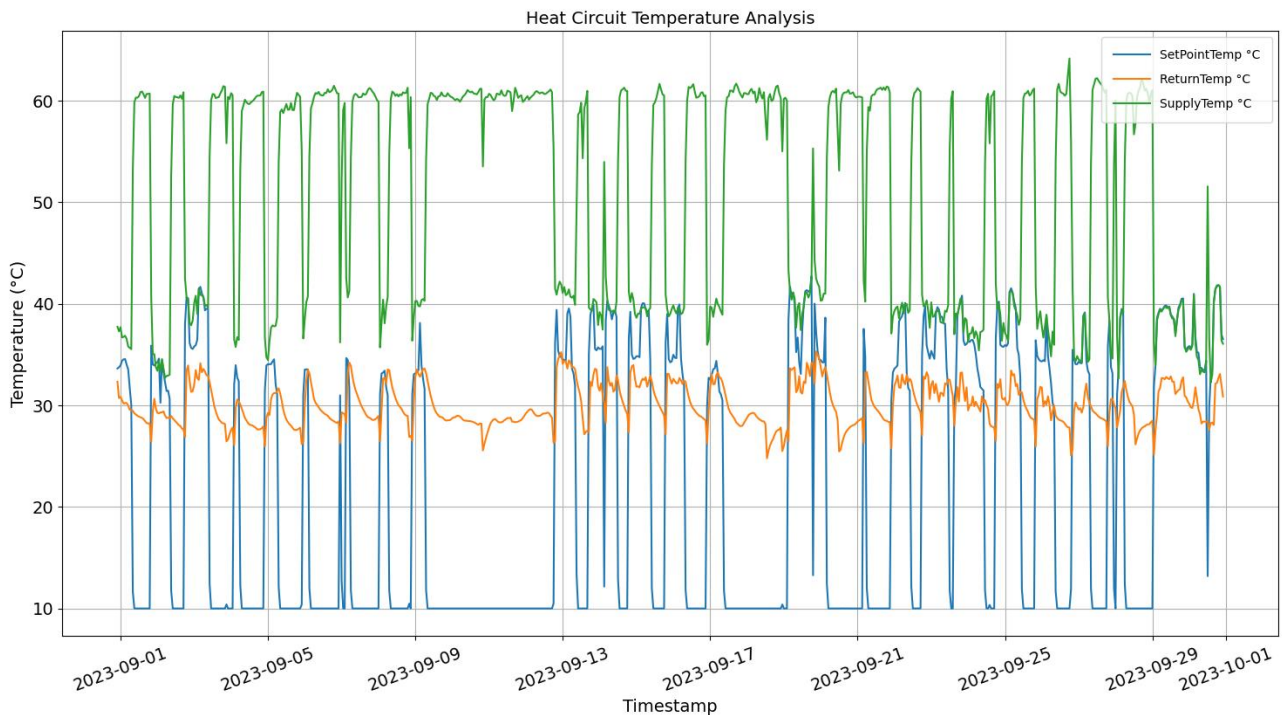


Figure 5: Heat Circuit Temperature Analysis: the blue line depicts the resampled trend of the SetPoint Temperature, the green line depicts the trend of the Supply Temperature while the orange one shows the trend of the Return temperature

Moreover, to add another fundamental variable to the calculation, using the Energy Accumulated Value, it is possible to estimate the DHW consumption. Figure 6 reports the trend calculated in this way while Figure 7 shows the consumption plot associated with a preliminary seasonality analysis. For this scope, the moving average was calculated using, for each value, 15 samples coming from the series to try to capture all the seasonal trends. As it can be seen, the presence of those seasonal trend is highlighted by the green line, and this can be exploited by some statistical model, like SARIMA and ARIMA, to estimate the future consumptions. The same analysis will be conducted on the Heat Circuit data once the thermal energy consumption data will be available.

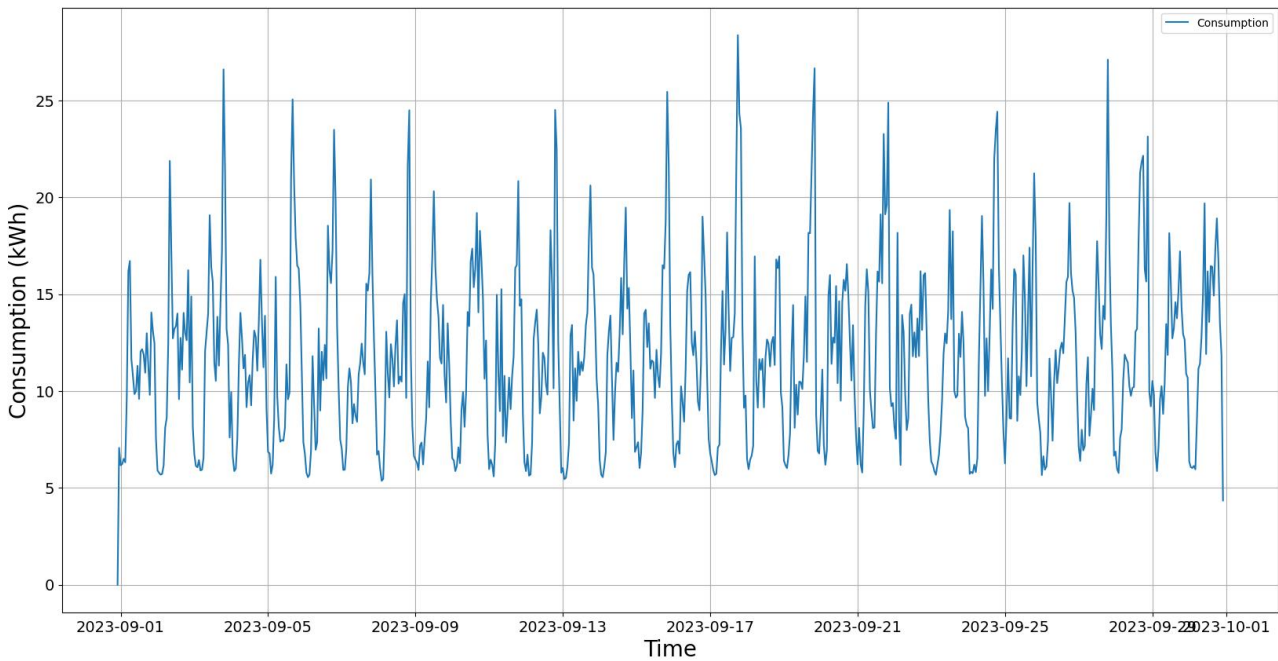


Figure 6: Danish Demonstrator DHW Consumption Plot

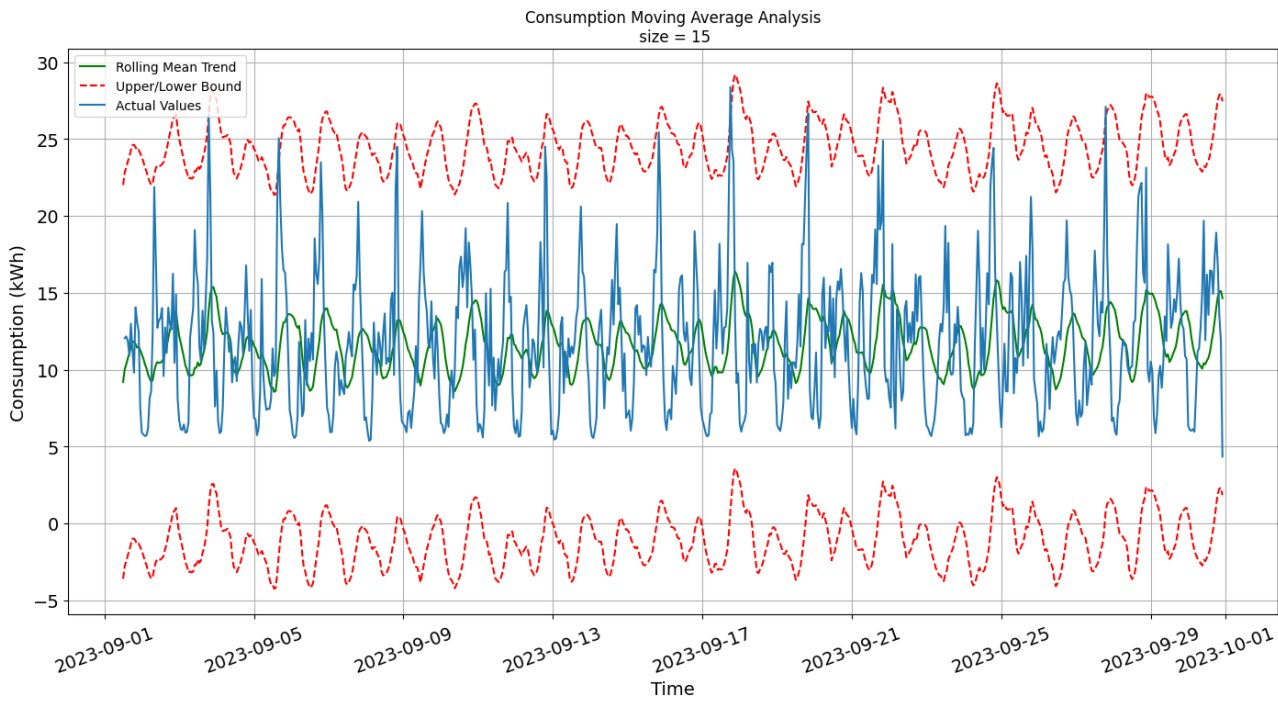


Figure 7: Seasonality Analysis of DHW Consumption



The insights gleaned from this analysis serve as a blueprint for crafting a simulator leveraging a Deep Learning (DL) algorithm. This simulator will take into account the different temperatures evaluated during the analysis, each exhibiting varying degrees of correlation, depicted in the correlation Matrix reported in Figure 8. Its primary function will be to furnish users with estimated consumption values, allowing Facility Managers to evaluate the impact on the consumptions of temperature changes in the buildings.

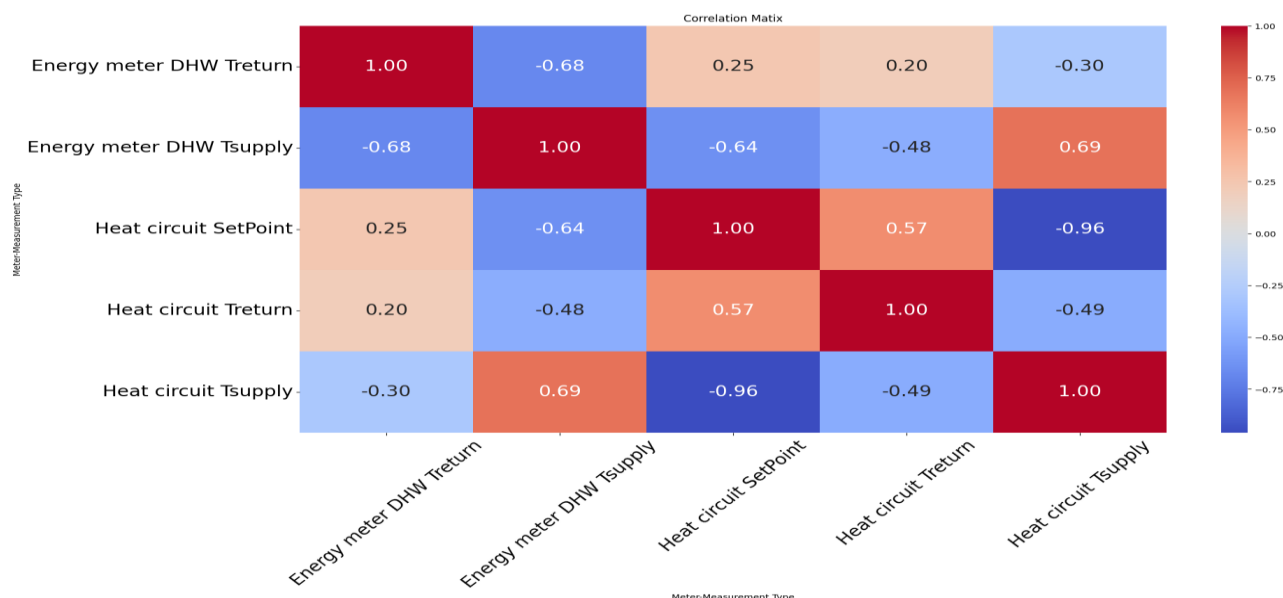


Figure 8: Correlation Matrix of the DHN Temperatures

The Jupiter Notebook with all the aforementioned analysis is available in the project's [GitHub repository](#).

2.1.6.2. Pilot 3 - Greece

As anticipated, the development of Pilot 3 Digital Twin mainly consists in a deep analysis of pilot data set, as preliminary activity needed to proceed with the development of the Simulation Engine and the proper configuration of the Context Broker. The pilot has focused around studying the Simulation Engine and the Context Broker and the familiarisation of development teams with HERON dataset. Specifically, DEDALUS partners have been given access to HERON's Energy Control, a smart metering infrastructure that supports smart meters that monitor the consumption of the whole household, smart plugs of various specifications that monitor the consumption of appliances with significant load and flexibility potential (incl. HVAC, Dishwashers, Washing Machines), relay for remote electrical water boiler control and a motion/movement sensor. HERON Energy Control is an integrated platform (databases and APIs) that integrates smart metering and smart home monitoring activities and provides visualisations through dashboards and a mobile app as a means of guiding customers towards more efficient and environmentally aware consumption. HERON Energy Control consists of two building blocks:



1. Real-time Energy Metering and Actuation Platform (REMAP): the technical infrastructure upon which the metering equipment operates.
2. Dashboards and visualisations in support of services developed to engage consumers and reduce their carbon footprint.

Connectivity of the metering assets is assured via the user's pre-existing Wi-Fi network, using the Internet to connect the smart devices with the cloud server where the IoT back-end system is running. There are several cloud services to support the collection of measurements via MQTT and their storage as time series in an InfluxDB database. Data access control to third-parties (such as the DEDALUS partners) is provided through API, which maps IoT device to more than one project and user. Storage of the relevant data is done in the relational PostgreSQL database, and JSON Web Tokens (JWT) are used to provide secure access to user applications (smartphone, dashboards). Parts of the information may be encrypted when storing it in the relational DB. The overall system provides simultaneous access from: a) the provider for consumption monitoring, b) end consumers for consumption analysis and c) third party services communicating over specific interfaces (APIs). The list of the measured parameters for the smart meters and smart plugs, measurement units and sampling interval are given in Table 5

Table 5: List of measured parameters for smart meters and smart plugs

Parameters for smart meters	Measurement Unit	Sampling Period
Active Power	Watts	30 sec
Voltage	Volts	30 sec
Energy	Wh	30 min
Reactive Power	Watts	30 sec
Power Factor	-	30 sec
Parameters for smart plugs	Measurement Unit	Sampling Period
Active Power	Watts	30 sec
Status (on/off)	Volts	30 sec
Energy	Wh	30 min

To gain insights on the flexibility potential of the Greek pilot participants once the DR services are deployed, HERON's data from smart meters and smart plugs attached to specific flexible assets was queried. Specifically, a dataset with the daily aggregated consumption was compiled, sampled from the smart meters of 92 pilot participants every 30 second for April and May in 2024. Out of the 92 pilot participants, 48 were equipped with flexible assets. Table 6 lists the equipped flexible assets and Table 7 provides a pilot inventory. Figure 9 and Figure 10 show the daily total consumption and the consumption associated with the selected flexible assets for April and May 2024 respectively. Consumption of the flexible assets ranges from 3% to 20% depending on the day. In more detail, Figure 11 and Figure 12 show the breakdown of the daily consumption of the flexible assets. On average, Washing Machines account for 3%, A/C Units for



8%, Cloth Dryers for 5%, Washing Machine / Dryer Combined units for 3%, Dishwashers for 16% and Electric Boilers with Relays for 3%.

Table 6: List of flexible assets for the Greek pilot

device_appliance_type	Description
3	Washing Machine
4	A/C Unit
6	Cloth Dryer
7	Washing Machine and Dryer Combined units
8	Dishwasher
12	Electric Boiler with Relay

Table 7: Household participation and smart assets inventory for the Greek pilot

Description	#
Participating households	92
Households with Flexible assets	48
Total Flexible assets	93
Washing Machines	32
A/C Units	26
Cloth Dryers	7
Washing Machine and Dryer Combined units	5
Dishwashers	20
Electric Boilers with Relay	3
Avg Flexible assets	1,94



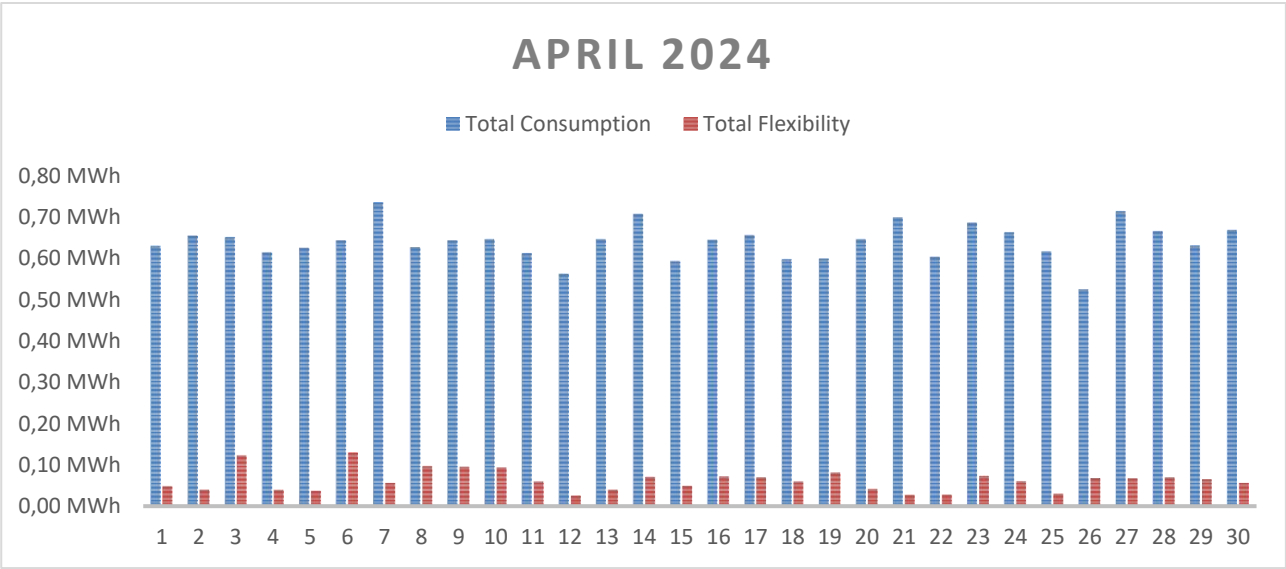


Figure 9: Total consumption and available flexibility for April 2024.

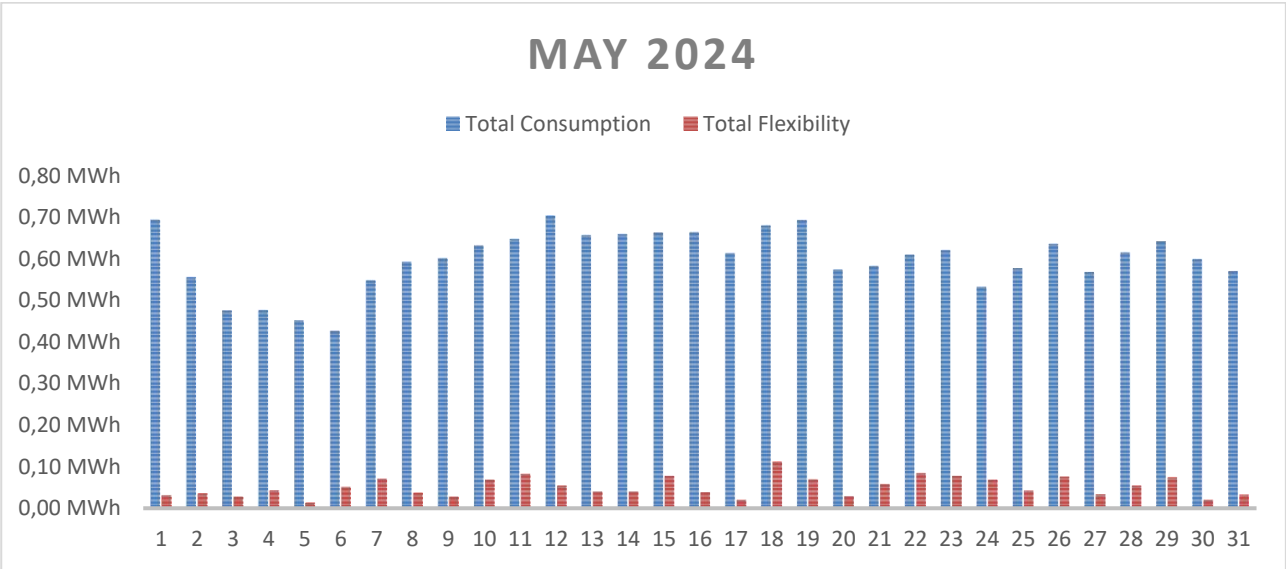


Figure 10: Total consumption and available flexibility for May 2024.



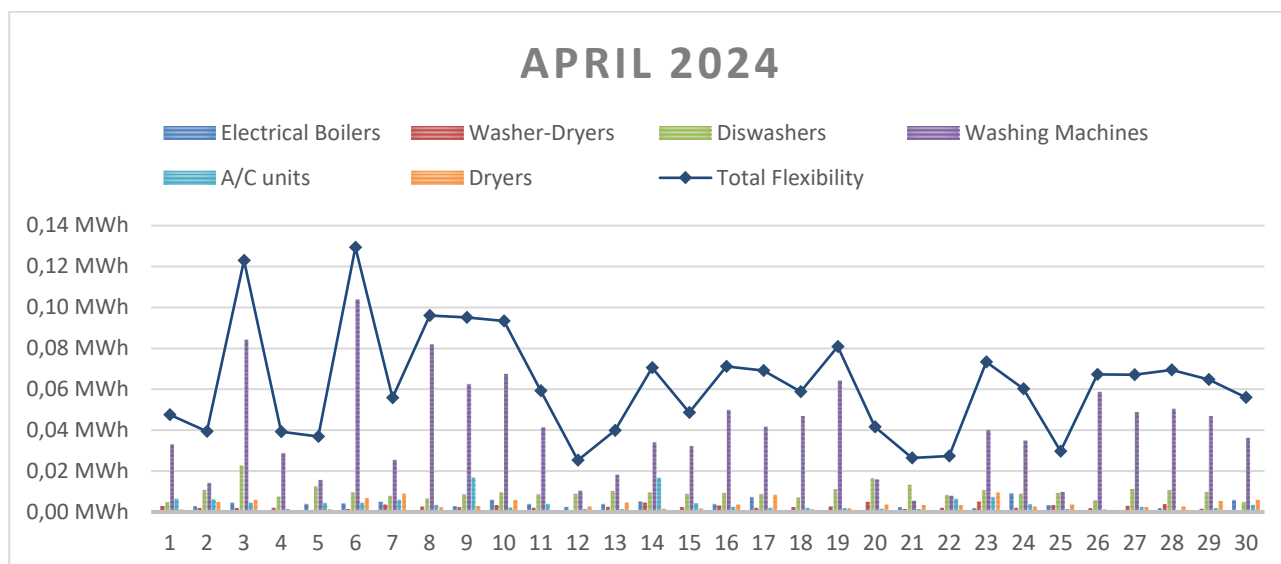


Figure 11: Consumption breakdown for flexible assets for April 2024.

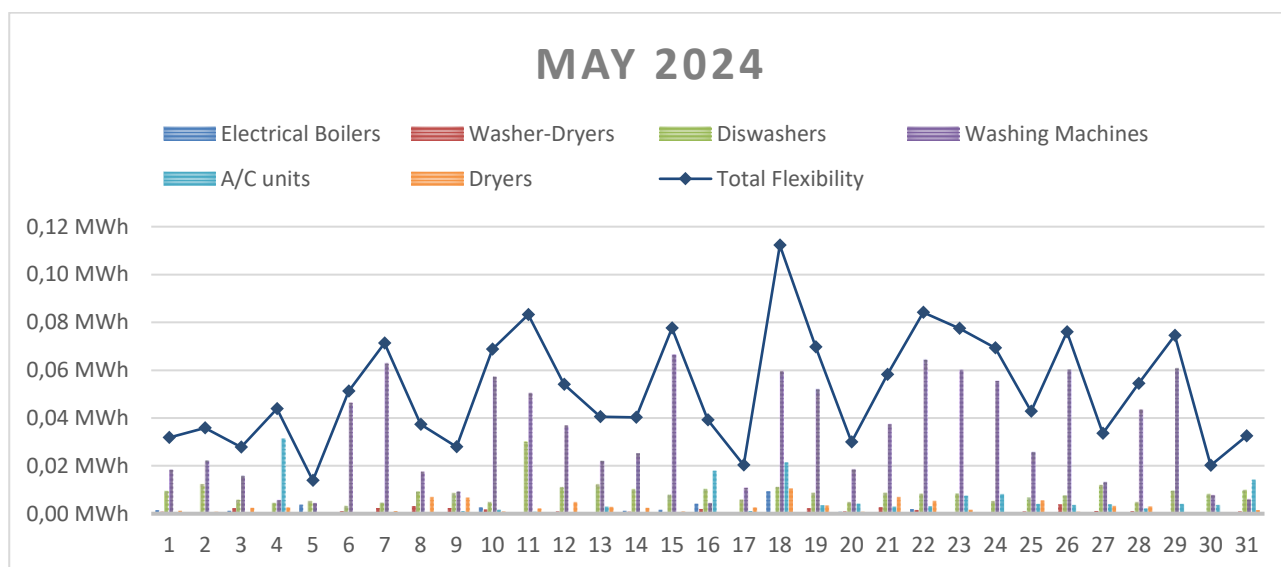


Figure 12: Consumption breakdown for flexible assets for May 2024.

2.1.6.3. Pilot 4 – Italy

The development activity in the Italian pilot mainly consists in the first version of Middleware API allowing the interaction with the service responsible for providing recommendations and alerts to the building manager concerning energy consumption while considering occupants' comfort. Indeed, Pilot 4 aims to seamlessly integrate flexibility models to furnish the Energy Manager with a comprehensive list of available recommendations. This list of recommendations will be provided through the DT. The middleware API is developed using FastAPI and it is available in the project's [GitHub repository](#).



Moreover, the development has started for the User Interface. A mock-up of the interface (Figure 13) has been prepared and agreed with the Pilot Leader.

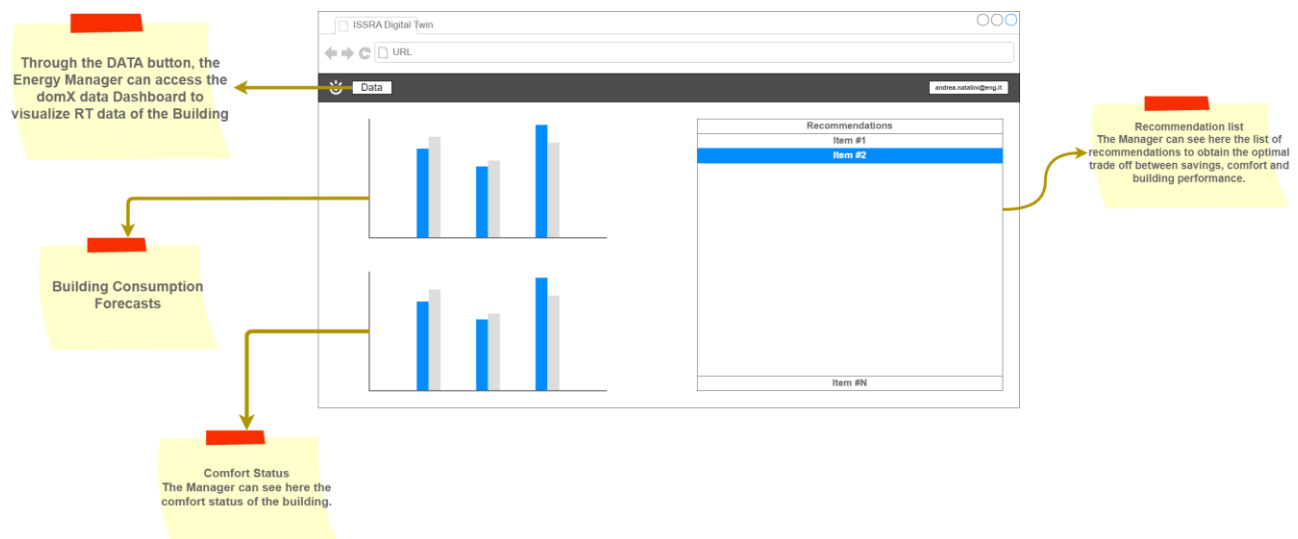


Figure 13: DT mockup – Pilot 4

The dashboard will be composed by one page allowing the Building/Facility Manager to exploit the list of recommendations giving the possibility to have all the flexibility options in one place. The comfort index of the inhabitant, as provided by TE-9, will be shown as well.

2.1.7. Outlook for the future expected release

The work planned for the next expected release, D4.2 and D4.3, respectively the second and the final way of DTs is specified per pilot in the following subsections.

2.1.7.1. Pilot 2 -Denmark

Following the data analysis process carried out in this development cycle, the following steps will be developed:

- Analysis of the .ifc file in order to produce a 3D model of the district.
- Heat Circuit Consumption Analysis.
- Development of a Deep Learning Regressor to estimate the district's consumptions.
- Development of Simulation Model, Middleware API and UI
- Integration of the Simulation Model with DEDALUS Interoperability Layer and with the UI
- Integration of flexibility and clustering models

2.1.7.2. Pilot 3 – Greece

For this pilot the following steps needs to be implemented:

- Analysis of available data from the Pilot Site.
- Storage System specification implementation: starting from the specification provided by the Pilot Leader, a preliminary version of the Storage System simulator will be constructed.
- Proper Mocks of the DT Interface should be provided (UC11).
- Integration with flexibility recommendation provided in the context of Task 4.3.

2.1.7.3. Pilot 4 – Italy

The work planned for the next expected releases, is specified as follows:

- Implementation of the first version of the UI according to the mock reported in section 2.1.6.3.
- Refinement of the Middleware API Layer to integrate Comfort Services

2.2. Apartment building Digital Twin for blockchain integration

2.2.1. Overview of the service

In the context of Romanian replication pilot (Pilot 7), digital twin model is being developed for the students' apartments building that incorporates data from heterogeneous sources with the purpose of supporting decision making in demand response programs and simulating the execution of apartment level DR actions (see Figure 14). The DT model is interacting with the DLT/Blockchain/Smart contracts solution to facilitate the simulation of DR actions regarding appliance activation in the context of DR settlement smart contracts. This integration allows for communication and data exchange between the DT components and the TE-5 smart contracts. Based on Apartment Demand Response (DR) actions (e.g., limiting the energy consumption to given values, peak demand reduction, etc.) will be received, the DT model is leveraged to determine which appliances should be activated or adjusted to meet the specified constraints. The development process will be guided by HLUC10 - Building-level decentralized flexibility aggregation and tokenization for DR, with specific use case UC21 defined in D2.3

Table 8: DT Functional Requirements for TE-5

Functionalities (brief description)	textual	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated site	pilot	Development Status



Digital twin of the building to simulate the potential DR actions that can be taken within the apartments and calculate the expected impact of these actions on energy usage levels	TE5-FR5	UC21	TE-5	Pilot 7 - RO	Started
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2.2.2. Architectural diagram

Figure 14 shows the architectural diagram of the DT and its interaction with the DLT/Blockchain/Smart contracts solution (TE-5). TE-5 technological enabler for blockchain-based interoperability for flexibility tokenization & DR settlement consists of several interconnected components that facilitate building-level DR by decentralized aggregation of apartment flexibility. A blockchain-based distributed ledger is set up at the building level to store and digitally fingerprint energy data. Smart contracts are defined to manage flexibility using decentralized market trading mechanisms: a building manager smart contract enables the publishing of flexibility bids, while apartments-associated smart contracts allow the publishing of flexibility offers. Several apartments' flexibility offers are matched with the building manager's bid, and flexibility transactions are generated. During actual delivery, smart contracts are used for settlement and token distribution. More details can be found in deliverable D3.1 - DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (1st technology release).



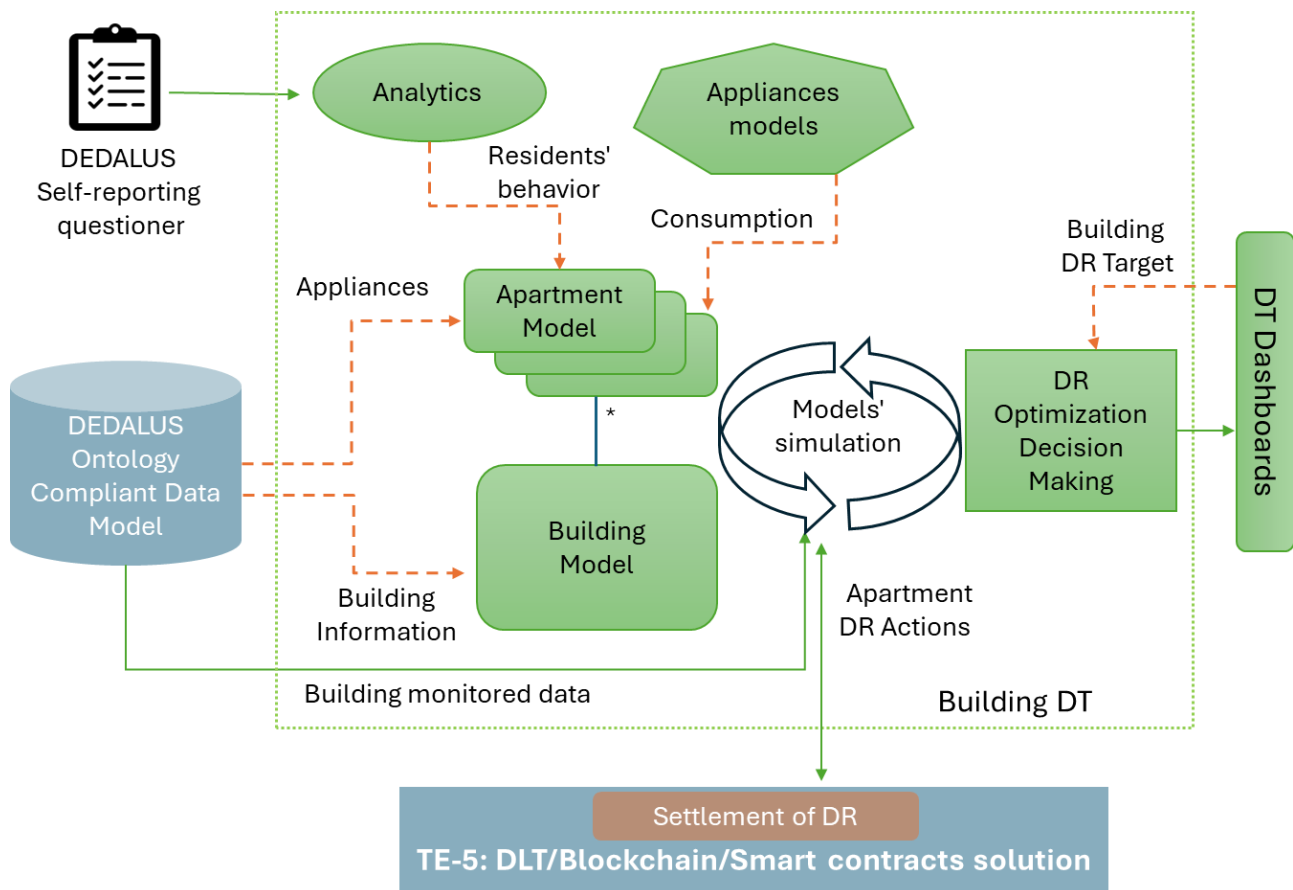


Figure 14: Apartments building DT, TE-5 integration

The building DT model integrates a diverse range of data sources, such as the DEDALUS self-reporting questionnaire and DEDALUS Ontology Compliant Data Model. The DEDALUS self-reporting questionnaire was filled in by the students with the purpose of capturing energy-related behaviours such as the time intervals of apartments occupancy and appliances' usage (i.e. when and how often appliances are used). Analytics are applied on top of this information to identify patterns of energy related behaviours and to construct models of the energy demand profiles of the apartments. We will simulate the daily energy consumption patterns of each apartment starting from the residents' behaviour by applying clustering algorithms to create initial consumption patterns that can be randomly varied, or by employing other machine learning algorithms such as neural networks. The DEDALUS Ontology Compliant Data Model provides the technical information about the building (e.g. the number of apartments, the number of floors and the organization of apartments on each floor), information about the apartments (e.g., number of residents living in the apartment, the number and type of appliances used by the residents), and monitored data about the total energy consumed at the level of the building acquired at a given rate. Also, the data model provides information about the appliances used in each apartment and their models that contain specific information about the energy consumption characteristics of each individual appliance. By integrating these data sources and models, the apartment's model simulates and determines energy consumption profiles that accurately represent how appliances contribute to the overall energy consumption within the apartment. The building model uses the information provided by the DEDALUS Ontology

Compliant Data Model to represent the building as resulted from the technical information as well as the daily hourly total energy consumption obtained after processing the raw monitored consumption data. The apartment and building models, along with the building's target energy specifications provided by the building manager through the digital twin's front-end, are processed by the DR Optimisation Decision Making component. This component aims to identify apartments that should adjust their energy consumption profile to meet the building's target energy profile while optimising appliance usage patterns within those apartments. The target energy can be expressed either as an hourly consumption profile or as a percentage of the building's total energy usage. The problems of identifying these apartments and optimising appliance schedules are combinatorial optimisation problems complex to solve due to the complex interactions and large search space involved. To address these challenges, the DR Optimisation Decision Making component uses bio-inspired meta-heuristics, such as the Harris Hawks Optimisation algorithm. This approach leverages principles from nature to efficiently explore and optimise the solution space, ensuring that the building's energy consumption is adjusted effectively while meeting specified targets. The use of bio-inspired meta-heuristics is particularly advantageous in handling the complexities of real-world energy optimisation tasks within a dynamic building environment.

2.2.3. Data for service execution

The data to be employed for the development of the DT are derived from Pilot 7 as shown in Table 9.

Table 9: Data Requirements for Pilot 7 - RO

Dataset Name	Protocol	Format	Used By	Available	Notes
Pilot7_energy_consumption	FTP/ MQTT	CSV	Building model and DR optimisation decision making	Yes	
Pilot7_questionnaire_data	-	CSV	Analytics	YES	

2.2.4. Service output

The output of the service consists of insights on which apartments should adjust their energy consumption profiles to help the building meet demand response (DR) goals. Additionally, it includes information on potential appliance activations in the context of DR settlement smart contracts. The output profiles will be displayed on the DT dashboard, while the appliance activation details will be integrated with the smart contracts of TE-5.



2.2.5. Main benefit and innovation

Essentially, the innovation lies in combining digital twin technology with blockchain and smart contracts to simulate DR actions regarding appliance activation within the context of DR settlement smart contracts. Additionally, the use of nature-inspired heuristics helps in finding the most effective solutions to meet the specified constraints, such as matching a demand curve, while considering various factors like occupancy patterns, appliance usage, and external environmental conditions. Moreover, combinatorial simulations with nature-inspired heuristics allow for the exploration of diverse solution possibilities within the DT, often leading to solutions that are close to optimal in a reasonable amount of time.

2.2.6. Development activity

The development activities within the reported timeframe of the project have primarily focused on conceptualization and the integration of its various components. Therefore, the data flows between components, such as the movement of energy consumption data from the building energy management system to the model's simulation and decision-making, were implemented. Additionally, the initial versions of the following components were implemented:

- **Analytics component** use algorithms for clustering the intervals when the apartments' residents are at home and when they use the appliances consuming electricity, producing patterns of residents' behaviours.
- **Apartment models** simulate energy consumption patterns and profiles based on appliances, their energy models, and residents' behaviours.
- **DR Optimisation Decision Making:** applies the Harris Hawks Optimisation Algorithm to determine how apartments should adjust their energy consumption profile by simulation of DR actions regarding appliance activation in the context of DR settlement smart contracts.

This stage was realized through the implementation of the component's functionalities in Python function, available on the TUC private GitLab repository.

2.2.7. Outlook for future activities

For the next steps we intend to perform the following:

- Implement the integration of the DT with the smart contracts for DR settlement to simulate the actions enforcement.
- Implement the digital twin's dashboards component and finalize the Building Model.
- Refine and integrate the **DR Optimisation Decision Making** component with optimisations to determine the minimum number of apartments that should adjust their energy consumption profile by simulation of DR actions regarding appliance activation in the context of DR settlement smart contracts.



3. Optimal trade-off among comfort, energy and building performance

3.1. Overview of the service

The service aims to ensure comfort in indoor environments by harmonizing Indoor Environmental Quality (IEQ) factors such as thermal comfort with energy consumption and costs. Additionally, it facilitates Demand Response (DR) services while emphasizing the preservation of health and well-being during their implementation.

This goal is achieved through the utilization of data collected from sensors installed in various buildings. To accomplish this, the service develops and applies AI-based algorithms capable of quantifying and predicting comfort-related factors, such as the Thermal Sensation Vote (TSV), using building data. This prediction relies on historical data collected from the involved pilots, as well as real-time data acquired from the sensor network provided by the pilot buildings. By employing this AI-based approach, the service assists stakeholders in managing buildings more efficiently and sustainably while ensuring occupants' comfort levels. Additionally, alongside maintaining acceptable comfort levels, reducing energy consumption and costs is a primary objective of the service. Furthermore, the service provides a coaching solution aimed at facilitating the proper management of indoor building conditions. The comfort prediction generated by the developed AI model serves as feedback for end-users (building manager, occupant) to adjust indoor environmental conditions and restore optimal comfort status. Moreover, the service collects feedback directly from end-users regarding indoor environmental and comfort conditions using survey-based approaches. This feedback is utilized to refine the AI model implementation and ensure that comfort predictions align with occupants' preferences.

The development process involves several steps:

1. **Definition of user requirements and technical specifications** (step 0): this initial phase serves to map user requirements and technical specifications. Technical actions required for service application in the building are determined. Specifications related to installing missing data sources (e.g., sensors) are provided.
2. **Assessment model for comfort-based flexibility** (step 1): this phase involves analyzing the status of the building to benchmark pilot building features and verify the availability of necessary data sources and technologies. This assessment informs the strategy and methodologies for service implementation, considering factors such as energy efficiency, digitalization level, structural characteristics, and existing technologies/sensors.
3. **Interoperability and quality** (step 2): upon gathering necessary information about the building and available data sources, the collected data is integrated into the service platform to ensure seamless compatibility and high data quality.
4. **Baseline monitoring** (step 3): this phase involves tracking energy consumption and comfort parameters over historical data to establish a baseline for evaluating building performance. Data analysis prepares and harmonizes input datasets for AI model training.



5. **Profiling and identification of anomalies** (step 4): analyzed data is utilized to create profiles of energy consumption and indoor environmental conditions, enabling the identification of anomalies. This step enhances understanding of building dynamics and supports targeted interventions for improved performance.
6. **Integration of service during DR events** (step 5): during DR events, the service seamlessly integrates with building operations to implement comfort-driven actions. This ensures that energy demand is dynamically reshaped while preserving occupant comfort and well-being.
7. **Delivery of coaching solution for comfort-based flexibility** (step 6): the final step involves delivering a coaching solution aimed at enhancing comfort and well-being. Leveraging insights from previous steps, an AI-based model is trained and tested with real-time data to provide personalized recommendations. Additionally, user feedback gathered through surveys informs model refinement, ensuring ongoing optimisation and user satisfaction.

A dedicated interface within the Digital Twin will be provided to allow the visualization of the AI-model results concerning comfort quantification and to monitor the consumption condition of the building. Additionally, a dedicated tool will be used to collect end-users' preferences and their opinions on indoor comfort conditions. The input datasets will include building's occupants feedback, environmental parameters, and energy consumption data. The service will be implemented through the Python programming language, utilizing libraries such as NumPy [1], Pandas [2], SciPy [3], or Scikit-learn [4]. Furthermore, the mentioned dashboard will also present KPIs related to the service. Hence, all the necessary data for their calculation will be collected.

Table 10 shows the functional specification of the service mapped with FR and Use cases defined in D2.1 and D2.3.

Table 10: Service's overview

Functionalities (brief textual description)	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated pilot site	Development Status
The service provides customised energy-saving recommendations for the building, prioritising occupants' comfort preferences.	TE9-FR1	UC4	TE-9	Pilots 2, 3, 4, 5, 6	Ongoing

UNIVPM outlined the necessary software components and visualization tools essential for implementing the service within pilot scenarios. Two comfort models are considered: the simplified Predicted Mean Vote (sPMV) [5] and the adaptive comfort model. While the developments for the sPMV model are detailed in this document, work on the adaptive model is scheduled for the upcoming period.

Both the sPMV and adaptive models utilize environmental parameters such as indoor temperatures, relative humidity, and occupants feedback as key inputs. UNIVPM chose to



employ a combination of machine learning (ML) and Artificial Intelligence (AI) algorithms. Baseline ML models, including Random Forest, Decision Trees, Naïve Bayes, K-Nearest Neighbor, Adaboost, and Bagging algorithms, are integrated for comfort classification. Additionally, the Long Short-Term Memory (LSTM) Network is selected to forecast comfort levels by leveraging historical data from pilots, considering various forecasting horizons. Furthermore, ML and AI approaches will be utilized to predict user feedback in terms of TSV.

In addition to these software components, the service implementation includes the development of a tool to collect occupants feedback on thermal sensation, and a dashboard serving as a graphical user interface (GUI). This dashboard is seamlessly integrated into the Digital Twin (Task 4.1), providing an overview of indoor environmental conditions and energy consumption, incorporating outputs from the implemented AI/ML-based models for enhanced monitoring and analysis.

3.2. Architectural Diagram and Algorithm Specification

The comfort-based flexibility model aims to achieve an optimal trade-off among comfort, energy efficiency, and building performance. Figure 15 provides the conceptualised service architecture that will be integrated in the DEDALUS architecture.

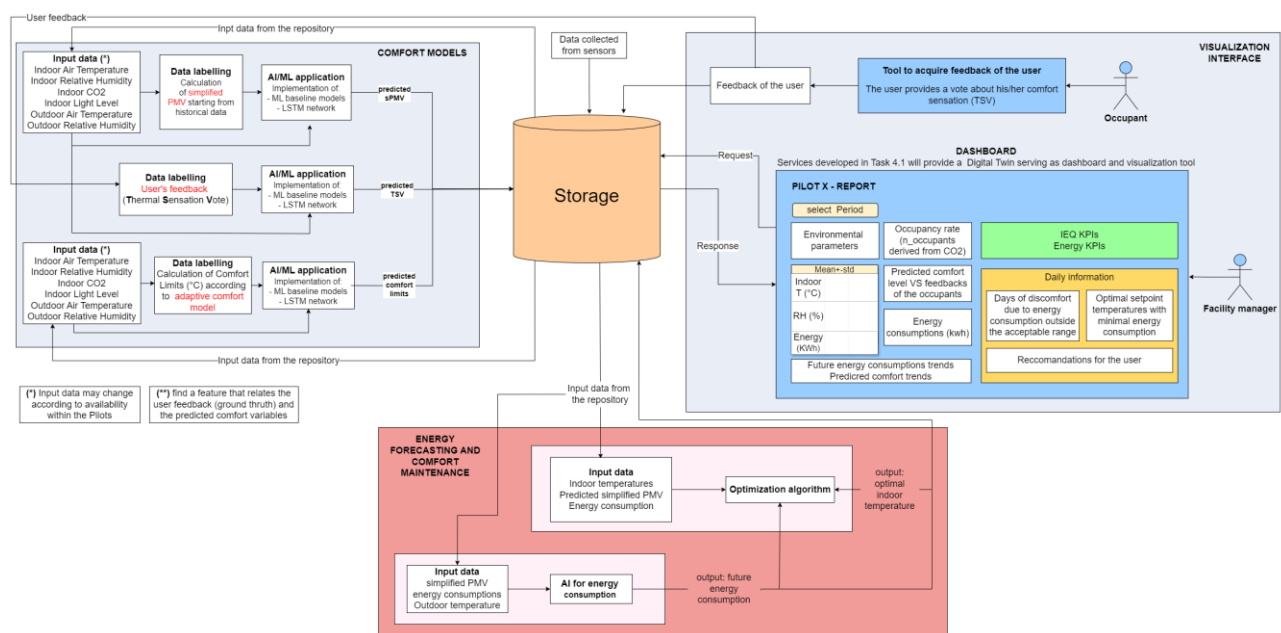


Figure 15: Architectural diagram of the service

The “Comfort models” box (Figure 16) represents the integration of two comfort models within the service framework. These models draw input from the DEDALUS local business repositories, identified as “Storage” in the figure, which contains environmental parameters collected from sensors installed at the pilots’ sites. Additionally, users’ feedback is sourced from an online repository. The simplified Predicted Mean Vote (sPMV) model, chosen for its simplicity and applicability in real-life contexts, relies on only two environmental parameters, i.e. indoor

temperature and relative humidity. Furthermore, the adaptive model, based on upper and lower comfort limits expressed as temperatures, guarantees optimal thermal comfort for occupants within a specified temperature range. The adaptive model also incorporates user feedback to observe its influence on comfort measurement.

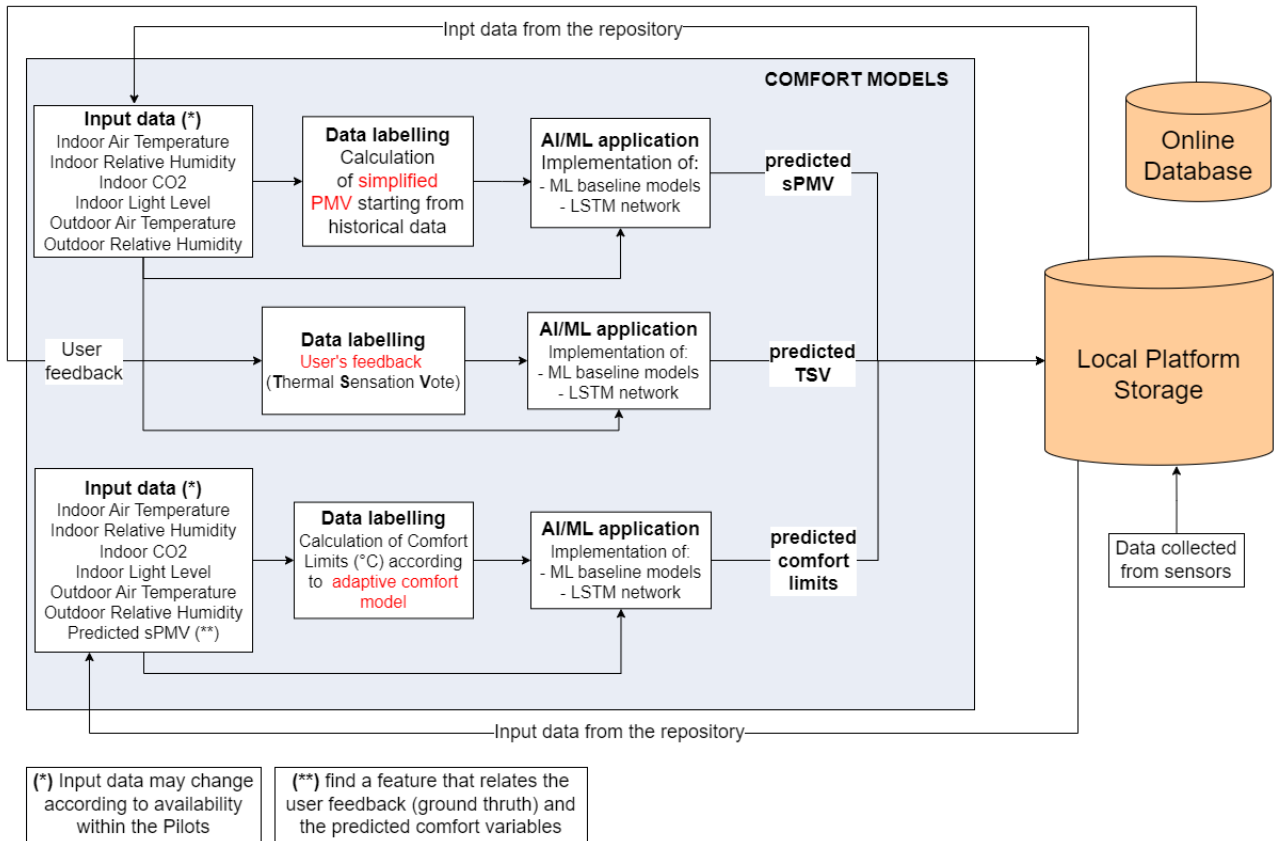


Figure 16: Section of the architecture dedicated to comfort models

Figure 17 provides an overview of the visualization interface integrated into the technical architecture of the service. The first interface is designed to collect users' feedback on their Thermal Sensation Vote (TSV). Additionally, as shown in Figure 16, the feedback is incorporated as a label in the comfort models software block, aiding in model refinement. To simplify feedback collection, a database will be created to store user-entered data. A dedicated webpage will be developed, prompting users to input details such as pilot, flat, and their respective TSV. Accessible via a designated link, this webpage guarantees interaction that prioritizes user privacy and ensures anonymization of data. Following submission, all data will be securely stored in the specified database. Leveraging PHP, JavaScript, and HTML, the development process will create a webpage optimised for efficient data handling. For transparency and collaborative development, the source code will be available on GitHub.

The second interface is a dashboard accessible to users, offering graphical representations of predicted comfort levels and building environmental conditions through graphs, charts, and

tables. Furthermore, the dashboard interface will include visualization of recommendations. This dashboard will be integrated within the Digital Twin, provided by Task 4.1.

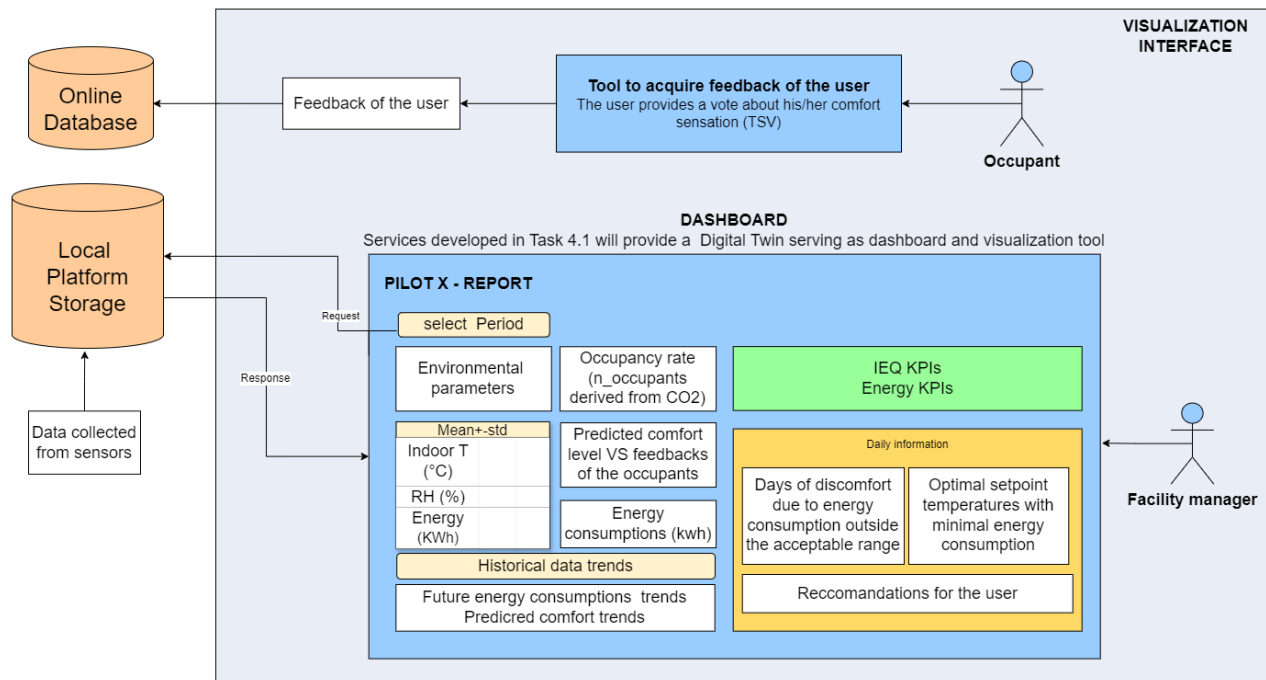


Figure 17: Section of the architecture dedicated to the visualization interface

The final component highlighted in Figure 18 focuses on developing an optimisation algorithm that considers forecasted energy consumption and comfort levels. Specifically, the goal is to determine an optimal temperature that minimizes energy usage while maintaining a satisfactory comfort level. This component includes sections dedicated to energy forecasting using specialized AI algorithms like the LSTM network, as well as the optimisation process itself. Recognizing the critical role of the energy forecasting component, collaboration with DOM-X (Task 4.3 leader for "Optimal DR management strategies (implicit and explicit) adaptable to different segments of residential energy consumers") was considered, as both services aims to provide recommendations to shift demand and offer flexibility to the grid. Further technical discussion on this collaboration will be held in the upcoming months.

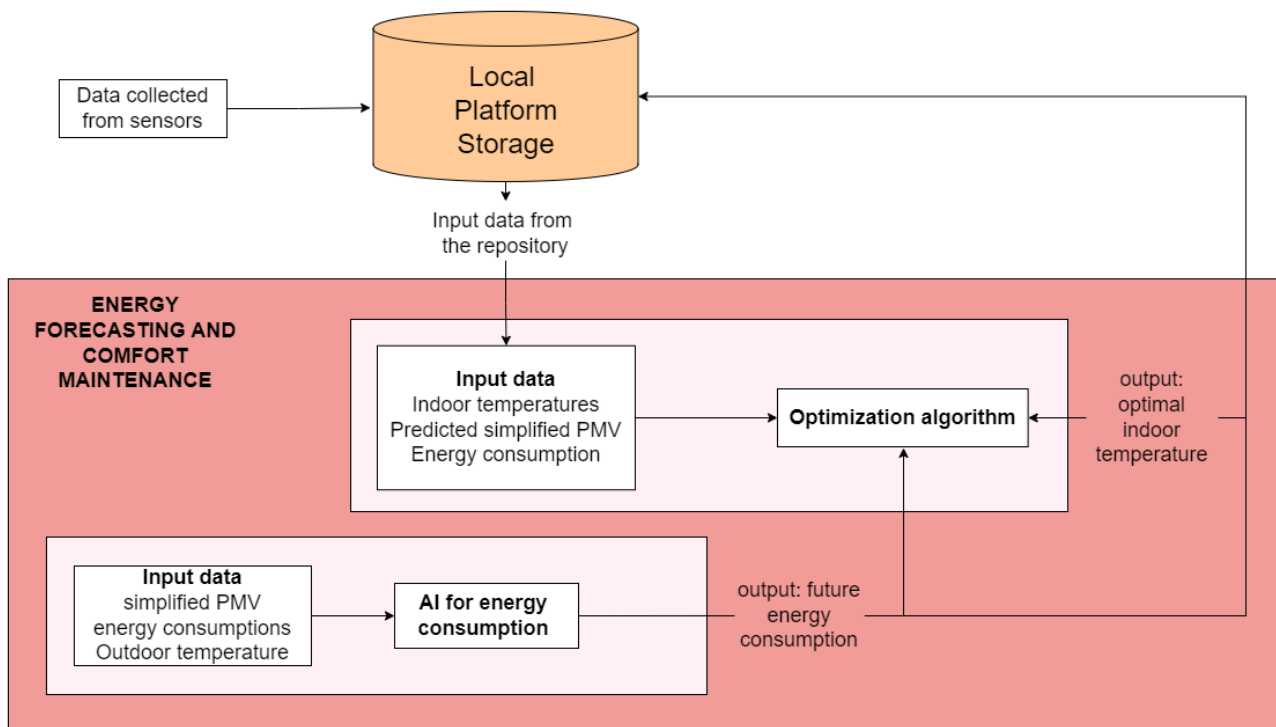


Figure 18: Section of the architecture dedicated to the energy models

3.3. Data for service execution

The successful execution of the comfort-based flexibility model relies on the input data collected from the buildings involved in different pilots. These inputs play a crucial role in shaping the model's understanding of indoor environmental quality (IEQ) and energy consumption dynamics. The primary data sources include:

1. **IEQ-related parameters:** sensors and systems are integrated to measure various indoor comfort parameters, including thermal, visual, and air quality factors. Air temperature, relative humidity, CO2 concentration, and lux levels are collected from the buildings. Additionally, data from a weather station is collected to provide contextual information.
2. **Energy consumption metrics:** data collection involves monitoring energy consumption factors such as metering, sub-metering, and smart plugs. This encompasses information on heating systems, lighting systems, and electrical systems within the buildings. Real-time data on on-site renewables production is also gathered. Understanding energy consumption patterns is essential for optimising comfort while minimizing energy usage.

Data collection may involve retrieving information from the pilots' local platforms, ensuring comprehensive coverage of relevant factors influencing comfort and energy consumption.

Table 11 provides an overview of the pilots' datasets defined in D2.3 relevant to the service.



Table 11: Pilots dataset overview.

Dataset Name	Description	Pilot country
Pilot2_electricity_consumption	Electricity consumption data of the ventilation system is collected for individual apartments.	Denmark
Pilot2_heating_consumption	The data is collected for individual apartments, with readings recorded every 5 minutes.	Denmark
Pilot2_indoor_climate	The data (Temperature, Humidity, CO2) is collected from individual apartments and rooms with readings recorded every 5 minutes	Denmark
Pilot3_heating_consumption	Heating data collected by individual households both from gas boiler systems and from heat-pump systems. The resolution of the collected data is less than 1 minute.	Greece
Pilot3_electricity_consumption	Electricity consumption data collected by individual households. The resolution of the collected data is less than 1 minute.	Greece
Pilot3_indoor_air_quality	Indoor air quality data collected by individual households. The resolution of the collected data is less than 1 minute.	Greece
Pilot4_electricity_consumption	Electricity consumption data collected for individual apartments in two buildings. Data can be monitored real-time	Italy
Pilot4_indoor_air_quality_and_light_intensity	The indoor conditions data (temperature, humidity, CO2, particulate matter, barometric pressure, light intensity) are collected for individual apartments in two buildings. Sampling rate 10 seconds.	Italy



Pilot5_electricity_consumption	Energy consumption collected from the users collected daily with DSO meters, with hourly sampling rate, and real-time data collected with smart meters.	Spain
Pilot5_DER	Data set information regarding the battery status and operation and photovoltaic panels generation.	Spain
Pilot5_IoT_devices	Data set from IoT devices including temperature, humidity and consumption of the smart plugs.	Spain

Before the collected data can be utilized for model training and analysis, it undergoes an exploratory data analysis (EDA) phase, followed by a preparation phase to refine the raw data. This includes formatting the data into suitable datasets, ensuring consistency and compatibility for use in AI algorithms.

Utilizing a tool, user profiles and preferences are gathered to enrich the data collection process with insights into individual comfort requirements. This feedback is incorporated into the input datasets, providing valuable insights into user expectations.

3.4. Service Output

The output of the service encompasses multifaceted contributions to both building management and grid operation. Through meticulous data analysis and leveraging insights from a range of sensors including environmental parameters and energy metrics, these services formulate actionable Demand Response (DR) strategies. These strategies are tailored to dynamically adjust building energy demand while upholding optimal comfort levels for occupants. The systematic orchestration of DR decisions within the service framework ensures the efficient dispatch of flexibility actions to meet the requirements of grid operators or aggregators. This seamless integration enables buildings to actively participate in DR programs, thereby bolstering grid stability and sustainability while safeguarding occupant comfort and well-being. Furthermore, the outcomes of these service seamlessly integrate into the Digital Twin, providing a comprehensive platform for monitoring indoor environmental conditions, comfort levels, energy consumption, and relevant Key Performance Indicators (KPIs).

3.5. Main benefit and innovation

The main benefits of the service can be summarised as follows:

1. **Enhanced comfort:** the service aims to provide a personalized comfort-based model, leveraging advanced learning techniques and considering user behavior and profiles. This



ensures that comfort levels are optimised while balancing energy consumption and overall building performance.

2. **Energy efficiency:** by dynamically reshaping energy demand based on comfort-driven actions, the service contributes to improved energy efficiency in buildings.
3. **Health and well-being:** the emphasis on preserving health and well-being during the application of DR services ensures that occupants remain comfortable and healthy throughout the process.
4. **Robust data collection:** the service ensures robust data collection, including indoor conditions monitoring, energy consumption metrics, and real-time data on on-site renewables production.
5. **Seamless communication:** the Digital Twin developed in Task 4.1 facilitates seamless communication of flexibility actions from the DSO/Aggregator to end-users, enhancing the user experience and ensuring effective implementation of DR measures.

The innovative aspects enclosed in the service touch different dimensions of digital service development. From a technical perspective, leveraging ML and AI techniques to classify and predict comfort, specifically in terms of sPMV, distinguishes the service from the state of the art. Additionally, the pivotal role of the user in its development and application represents another important novelty. In many cases of comfort assessment, user feedback is sought for a direct but subjective evaluation of thermal sensation, yet usually without considering a coaching solution. In this context, users' feedback is crucial for implementing ML and AI algorithms effectively. Furthermore, the significant role of the end-user is based on the recommendations that should be delivered to actively adjust the environmental surroundings and maintain or guarantee a proper comfort level.

1. **Personalized comfort-based model:** the service innovatively crafts a personalized comfort-based model, taking into account user behaviour and profiles, to optimise comfort levels while minimizing energy consumption.
2. **Integration of sensors and systems:** the seamless integration of sensors and systems designed to measure various indoor comfort parameters ensures comprehensive data collection and analysis.
3. **Utilization of advanced learning techniques:** advanced learning techniques are employed to create the personalized comfort-based model, contributing to its effectiveness and adaptability.
4. **Comprehensive approach:** the service takes a comprehensive approach, considering various factors such as user preferences, indoor conditions, energy consumption, and real-time data, to orchestrate the optimal dispatch of flexibility actions for DR.

Overall, the service resulting from Task 4.2 represents an innovative approach to integrating comfort considerations into DR services, with a focus on enhancing energy efficiency and overall well-being.



3.6. Development activity

This paragraph outlines the progress achieved through activities conducted from M5 to M13, providing preliminary results from early baseline monitoring based on historical data from Pilot 2 (Denmark) Demonstrator. These outcomes culminated in MS4 (M13) 1st technology development cycle completion. Here are presented the development and application of ML models with preliminary results.

The analysis of requirements revealed a lack of data across most pilots. Consequently, UNIVPM opted to concentrate on Pilot 2 Demonstrator for the initial service development, leveraging historical environmental data for ML algorithm training and testing. This process was carried out using Python, and the associated codes are available in the DEDALUS GitHub repository. The deployment of the service will take place at the pilot site. In terms of Technology Readiness Level (TRL), the model stands at TRL5 with a proprietary license.

3.6.1. Pilot 2 - Denmark

This paragraph reports the preliminary results of the application of the ML models in two Pilot 2 demo buildings. Figure 19 shows an example of the floorplan of a flat ("Flat 1") and the location of the installed sensors within it.



Figure 19: Floorplan of the Flat 1 and installed sensors



The apartments are outfitted with the NEOGRID sensor platform, which includes sensors that collect indoor environmental, meteorological, heating and electrical data (Table 11). These data are accessible through an online platform (<https://app.neogrid.dk>), where users can select the desired time frame and download historical data. Figure 20 and Figure 21 serve as examples, displaying temperature and relative humidity data captured by sensors installed in Flat 1 from January 2020 to the present day and in Flat 2 from January 2020 to July 2023, respectively.

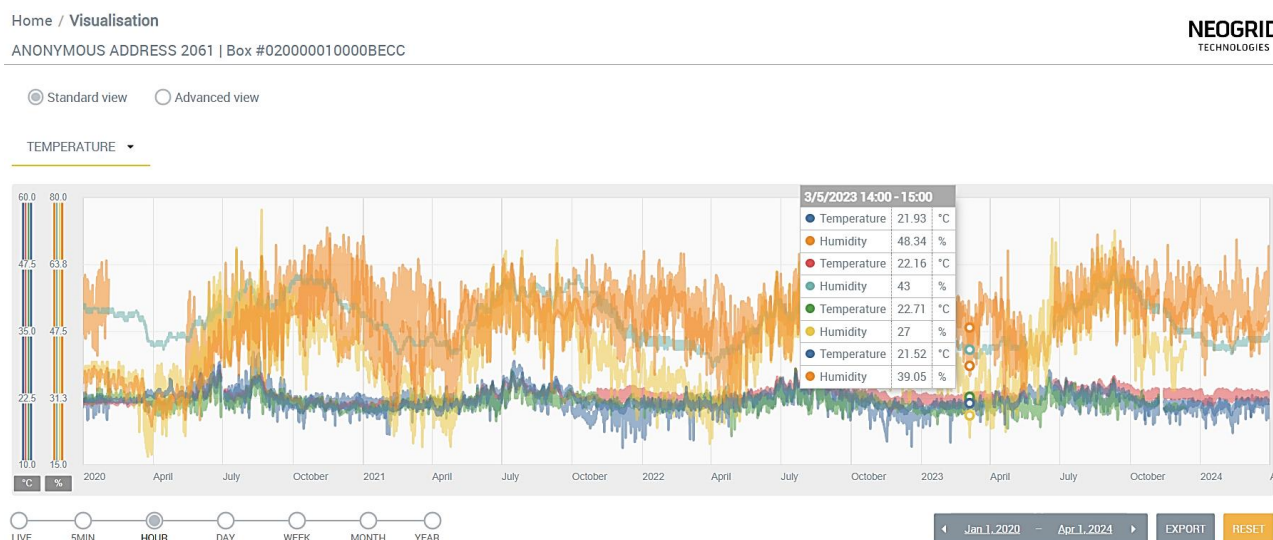


Figure 20: Historical temperature and humidity data of Flat 1.

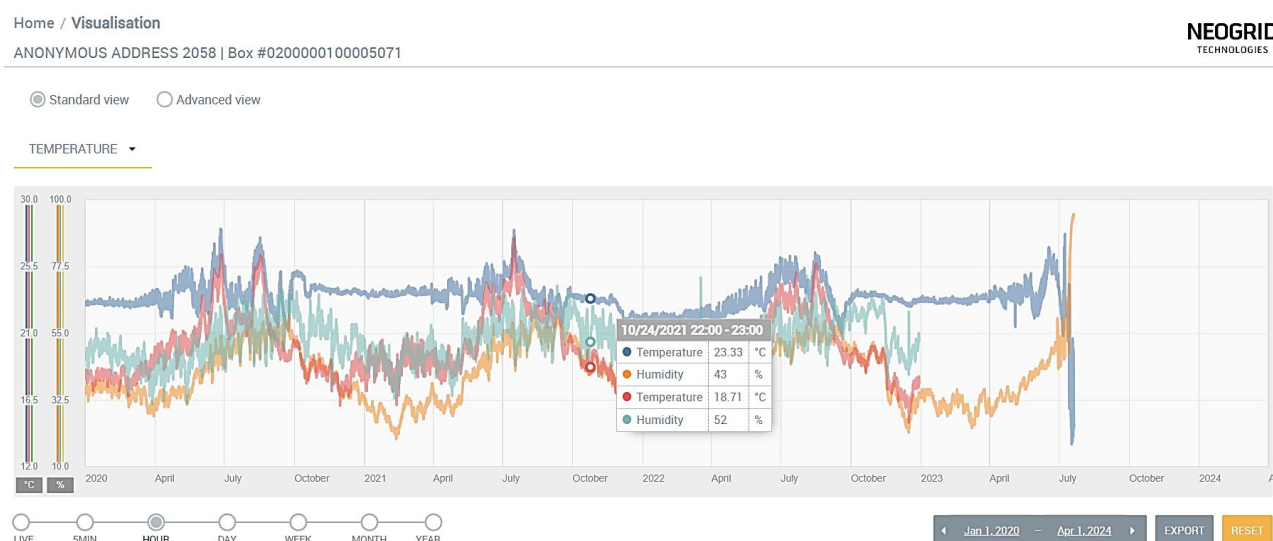


Figure 21: Historical temperature and humidity data of Flat 2

To initiate the services' first version, the primary objective was to implement ML-based algorithms for comfort level classification, specifically Random Forest (RF), Naïve Bayes (NB), K-Nearest Neighborhood (KNN), Decision Tree (DT), Adaboost (ADA), Bagging (BAG), after creating



a suitable input dataset. This dataset incorporated historical environmental data captured by the installed NEOGRID sensors and the labels generated in accordance with the simplified Predicted Mean Vote (sPMV) model (Equation 1). The aim was to employ ML techniques to classify comfort levels based on environmental parameters.

$$sPMV = aT + bP_v - c, \quad (1)$$

where T denotes the indoor temperature and P_v represents the water vapor pressure, which is contingent on relative humidity (RH) and the saturation of vapor pressure (Equation 2). Specifically, P_v is defined as:

$$P_v = P_{ws} \times \frac{RH}{100} \quad (2)$$

This stage was realized through the implementation of a dedicated Python function for sPMV calculation, available on the project's GitHub repository. The result of this process was the inclusion of labels in the input dataset for the Python-based ML algorithms, facilitating the classification of comfort levels using historical data from Pilot 2.

The sPMV values are mapped to comfort levels based on predefined thresholds. For example, an sPMV value close to 0 typically indicates a neutral comfort level, positive values indicate warmth, and negative values indicate coolness. The thresholds for these categories are defined as follows:

- $sPMV < -3$: Very Cold
- $-3 \leq sPMV < -2$: Cold
- $-2 \leq sPMV < -1$: Cool
- $-1 \leq sPMV \leq 1$: Neutral
- $1 < sPMV \leq 2$: Warm
- $2 < sPMV \leq 3$: Hot
- $sPMV > 3$: Very Hot

These labels are then used as the output classes for the ML algorithms.

The inputs for the ML algorithms include various environmental parameters such as indoor temperature (T) and relative humidity (RH). The output is the classified comfort level, which falls into one of the predefined categories (e.g., Very Cold, Cold, Cool, Neutral, Warm, Hot, Very Hot).

The added value of using ML algorithms is their ability to learn complex relationships between multiple environmental parameters and comfort levels, potentially improving the accuracy and robustness of comfort level predictions compared to using the sPMV model alone. ML algorithms can also handle a larger variety of input data and adapt to new patterns that may not be explicitly captured by the sPMV equation, thereby enhancing the overall prediction of comfort levels in different scenarios.

The datasets for Flat 1 and Flat 2 comprised 36,940 and 31,119 data points, respectively. The labelling process was conducted by creating a dedicated Python function, accessible in the



GitHub repository. Figure 22 and Figure 23 showcase examples of indoor temperature, relative humidity, and the corresponding calculated sPMV values for Flat 1 and Flat 2.

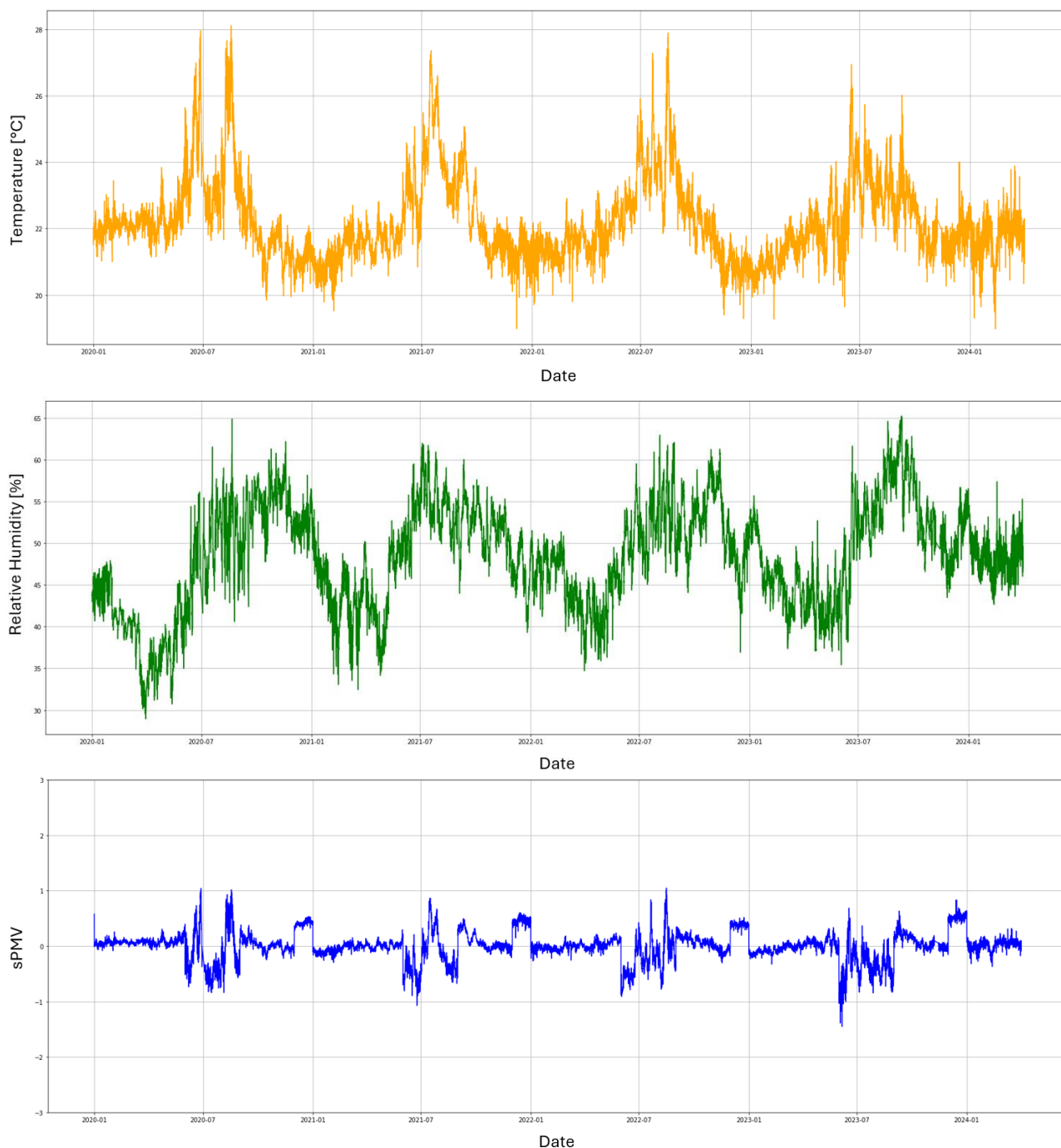


Figure 22: From top to bottom: temperature, relative humidity, and calculated sPMV for Flat 1.



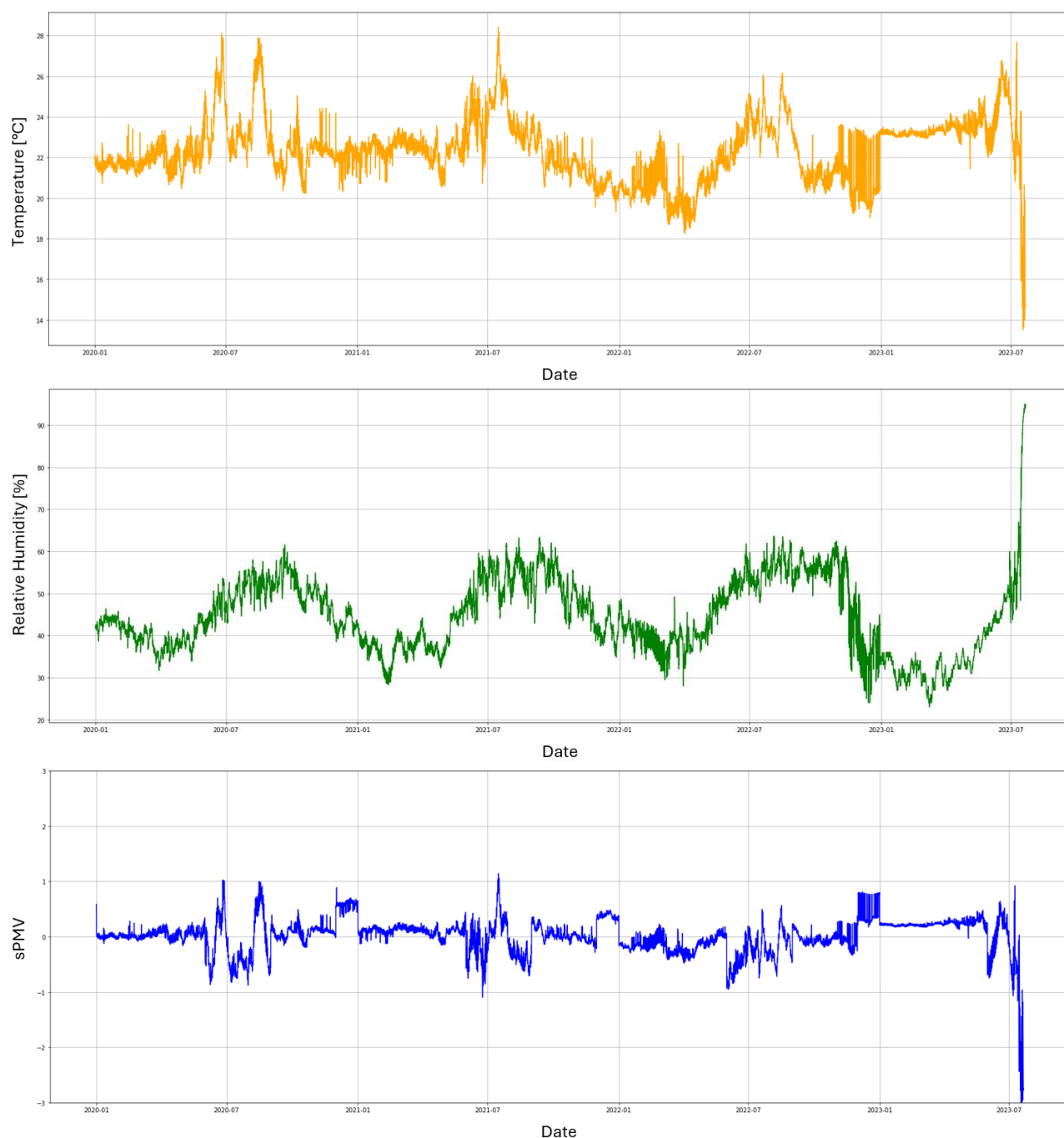


Figure 23: From top to bottom: temperature, relative humidity, and calculated sPMV for Flat 2.

Validation metrics (accuracy, F1 score, precision, recall) for ML-based algorithms are shown in Table 12 for Flat 1 and Table 13 for Flat 2. Accuracy measures the proportion of correctly predicted instances out of the total instances. It provides an overall indication of how often the model makes the correct prediction. The F1 score is the harmonic mean of precision and recall. It is particularly useful for evaluating models on imbalanced datasets, as it considers both false positives and false negatives in its calculation. Precision indicates the proportion of positive identifications that were actually correct. It reflects the accuracy of the positive predictions made



by the model. Finally, recall measures the proportion of actual positive instances that were correctly identified by the model. It provides insight into the model's ability to capture all relevant positive cases.

Table 12: Validation metrics of the baseline ML models for Flat 1.

ML model	Accuracy	F1_score	Precision	Recall
RF	0.93	0.64	0.73	0.59
DT	0.90	0.62	0.61	0.62
NB	0.91	0.46	0.45	0.46
KNN	0.92	0.61	0.72	0.55
ADA	0.92	0.56	0.75	0.49
BAG	0.92	0.64	0.71	0.60

Table 13: Validation metrics of the baseline ML models for Flat 2

ML model	Accuracy	F1_score	Precision	Recall
RF	0.91	0.79	0.81	0.78
DT	0.90	0.79	0.79	0.79
NB	0.91	0.63	0.73	0.62
KNN	0.91	0.79	0.82	0.77
ADA	0.77	0.61	0.61	0.66
BAG	0.90	0.80	0.81	0.79

The preliminary results of the ML algorithms have yielded promising outcomes, with mean accuracies reaching 0.92 and 0.88 for Flat 1 and Flat 2, respectively, in classifying comfort levels. Such high performance underscores the efficacy of these algorithms in analyzing data effectively. Moving forward, these algorithms will be deployed to process data collected from the various sensor platforms employed by pilots.

3.7. Outlook for the future expected release

The following are the steps to be taken to proceed with the development of the service:

1. Apply the previously described ML-based algorithms to all pilots.
2. Finalize the feedback tool (setting up the database and web page) and implement it in the pilot sites.
3. Initiate the real-time implementation of the service.
4. Utilize ML and AI models to predict the TSV based on the feedback collected from the buildings' occupants.



5. Consider exploring additional comfort dimensions (e.g., visual comfort, IAQ, etc.) as a valuable next step to achieve a comprehensive assessment of comfort and enhance its overall evaluation.
6. Start the integration of the service with Task 4.1 (Digital Twin interface), involving the implementation and development of the graphical user interface.
7. UNIVPM will collaborate with DOM-X in Task 4.3 “Optimal DR management strategies (implicit and explicit) adaptable to different segments of residential energy consumers” and NTUA in Task 3.4 “Multi-energy flexibility modelling for data-driven residential DR” to synergize expertise for the effective development and implementation of innovative DR and flexibility solutions tailored to diverse segments of residential energy consumers.



4. Optimal DR management strategies (implicit and explicit) adaptable to different segments of residential energy consumers

The aim of this chapter is to provide information about software expected to be developed as outcome of Task 4.3. This development corresponds to several services that aim to develop optimal DR strategies in the Greek and Austrian pilot sites and are linked to TE-10 *Nudging application* and TE-11 *Multi-value DR service*. These services are detailed in the following subsections.

A quick overview can be found in Table 14 below where the various services under development linked to DR strategies are listed.

Table 14: Service's overview

Functionalities (brief textual description)	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated pilot site	Development Status
Design of optimal DR strategies for the gas pilot	TE10-FR1	UC15	TE-10	Pilot 3 - GR	In progress
Building performance and end-user profiling for the gas pilot	TE10-FR1	UC15	TE-1, TE10	Pilot 3 - GR	In progress
Energy disaggregation for the electricity pilot	NA	UC1, UC2, UC11, UC13	TE-4, TE-10	Pilot 3 - GR	In progress
Integration of tariff system for implicit and explicit electricity DR services	TE11-FR1	UC13, UC14	TE-11	Pilot 3 - GR	In progress
OurPower Nudging Application	TE10-FR2	UC22	TE-10	Pilot 3 - GR	In progress
Integration of Implicit Demand Response service - Recommendation in the DT	TE8-FR5	UC11	TE-8	Pilot 3 - GR	In progress

4.1. Service A - Design of optimal DR strategies for the gas pilot

4.1.1. Overview of Service



Service A	Description
<i>Service / Software title</i>	Design of optimal DR strategies for the gas pilot
Responsible partner(s)	domx
Associated pilot site	Pilot 3, Greek – gas domain
Service brief overview	In this service we try to find out the gas pilot households with high energy savings potential in order to apply DR strategies to support flexibility. More specifically, each household that is equipped with the domx smart heating controller, supports a special parameter that defines how aggressively or how economically the heating system will operate when requested to heat up the place. Apparently, the same parameter results also in the energy consumption of the system. The end-users have the ability to parametrize this value according to their needs. The aim of this service is to calculate/simulate the energy consumption performance of each household under all possible configurations in order to identify the households that present potential of energy savings, without of course violating the comfort levels of the end users. To this aim domx is developing a new version of its simulator which will be trained using the datasets collected by the different households in order to calculate their energy under different configuration settings. By doing so it will be practical to identify the households with high energy savings potential and implement specialized nudges for these households.
Associated TE	TE-10
Associated Use Case	HLUC7 / UC15
Functional Requirements (input by other activities)	TE10-FR1 Input has been received by the D2.3, from the definition of the parameters of the pilot site
Data Input	Data provided by the domx smart heating controllers –this service is being developed with the aid of the data collected by the domx smart heating controllers that are installed in the gas boilers of the Greek pilot site. The smart heating controller collects all the parameters that are available from the gas boiler and is able to calculate the energy consumption for a given timeslot which is the main parameter used for the development of this service.
Service Output	Optimal DR strategies for the gas pilot



Service A	Description
Repository Location	domx cloud system
License information	Proprietary
TRL	TRL 5-6
Current version number	V.02
Expected date of finalization	May 2025, M25
Development Status	The first functional wave of the service has been released. Development for the upcoming waves is in progress.

4.1.2. Architectural diagram

Figure 24 shows the architectural diagram of the domx service. We use the data collected by the pilot households in order to feed the simulation algorithm that calculates the energy-consumption for every household under different settings. The output will be used to identify households with high energy-savings potential and finally to be applied in the Greek pilot.

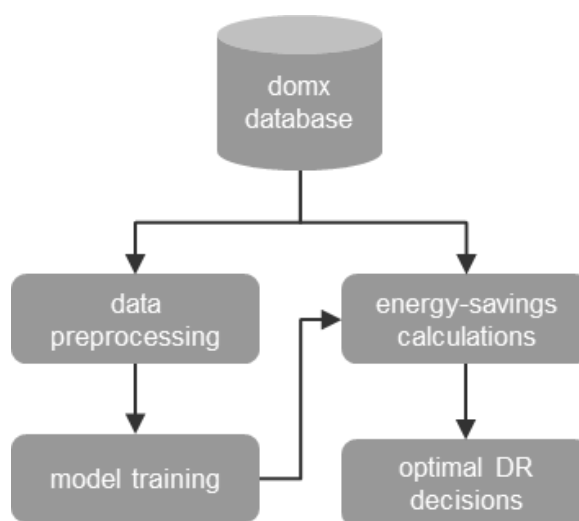


Figure 24: Service A - Architectural diagram



4.1.3. Main benefit and innovation

Benefit: The main benefit of this service is the development of a new simulator that will be able to precisely calculate the energy consumption of each household. This simulator can be used in various cases, such as the forecasting of the entire gas boilers portfolio under specific configurations, so that to derive to important decisions when required.

Innovation: The simulator that is under development for this service takes into consideration the thermal-dynamics of each household which provides an extra dimension in the prediction of the power consumption. This physics-based time series forecasting model is the latest version of AI in this field that increases the resulted accuracy.

4.1.4. Development activity

In this service it is being implemented a new version of the domx simulator to simulate the energy consumption performance of the various households in order to be able to calculate the energy consumption of an entire heating season under different configurations.

The existing domx simulator was presenting several limitations in the resulted performance, thus a new version that takes into consideration the thermodynamics of each household is being developed. In this version we consider an RC (Resistance-Capacitance) principle that actually models the thermodynamics of the buildings in order to derive to more accurate results. In Figure 25 we present the model that we are using where the various temperatures are annotated with T, while the pace of their alteration is modelled with the Resistors in the scheme. Finally, the two capacitors emulate temperature storage in this principle. By using this scheme, we have developed a first version of the simulator, which is trained with data provided by the Greek pilot households. Currently, we are in the process of evaluating the performance of the algorithm and adjusting some parameters to achieve better performance.



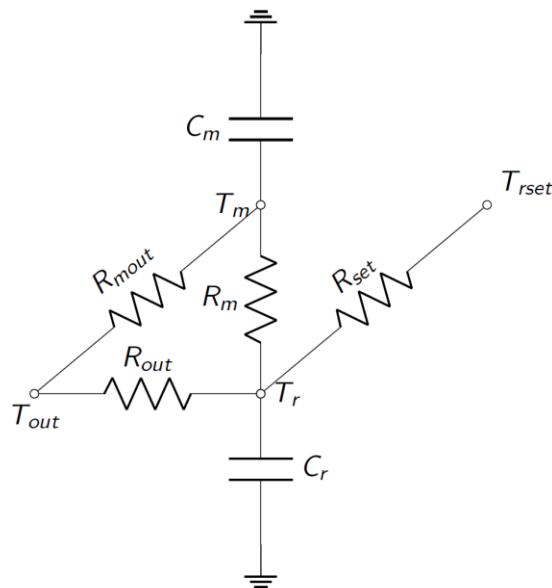


Figure 25: RC model used for the development of the new domx simulator

4.1.5. Outlook for the future expected release

It is foreseen to improve the performance of the developed simulator by testing more features in the current version of the simulator. As next step it is envisioned to train the simulator with the various households of the Greek pilot and next to use it in order to estimate the energy consumption of the trained households under different configuration settings in order to identify the ones with high energy savings potential. These households will be used for the nudging activity along with the identified configurations.

4.2. Service B - Building performance and end-user profiling for the gas pilot

4.2.1. Overview of the service

Service B	Description
Service / Software title	Building performance and end-user profiling for the gas pilot
Responsible partner(s)	NTUA (main contributor), domx (supporting)
Associated pilot site	Pilot 3, Greek – gas domain



Service B	Description
Service brief overview	The objective is to unveil concealed intricacies within heating load profiles, affording a more distinctive comprehension of energy consumption dynamics and heating system operations focusing on identifying houses with the highest estimated energy savings, paving the way for sustainable Demand Response (DR) programs
Associated TE	TE-1, TE-10
Associated Use Case	HLUC7 / UC15
Functional Requirements (input by other activities)	TE10-FR1
Data Input	Input Data are provided by Domx and the gas part of the Greek pilot.
Service Output	Clusters of houses according to their estimated energy savings
Repository Location	DEDALUS GitHub
License information	Apache 2.0 open-source
TRL	TRL 7-8
Current version number	v0.3
Expected date of finalization	May 2025, M25
Development Status	1st version ready and on-going development for the next version

4.2.2. Architectural diagram

The architectural diagram is as shown in Figure 26 and it spans from data collection to the creation of meaningful clusters according to each household energy-savings potential. The diagram is explained below in section 4.2.4.



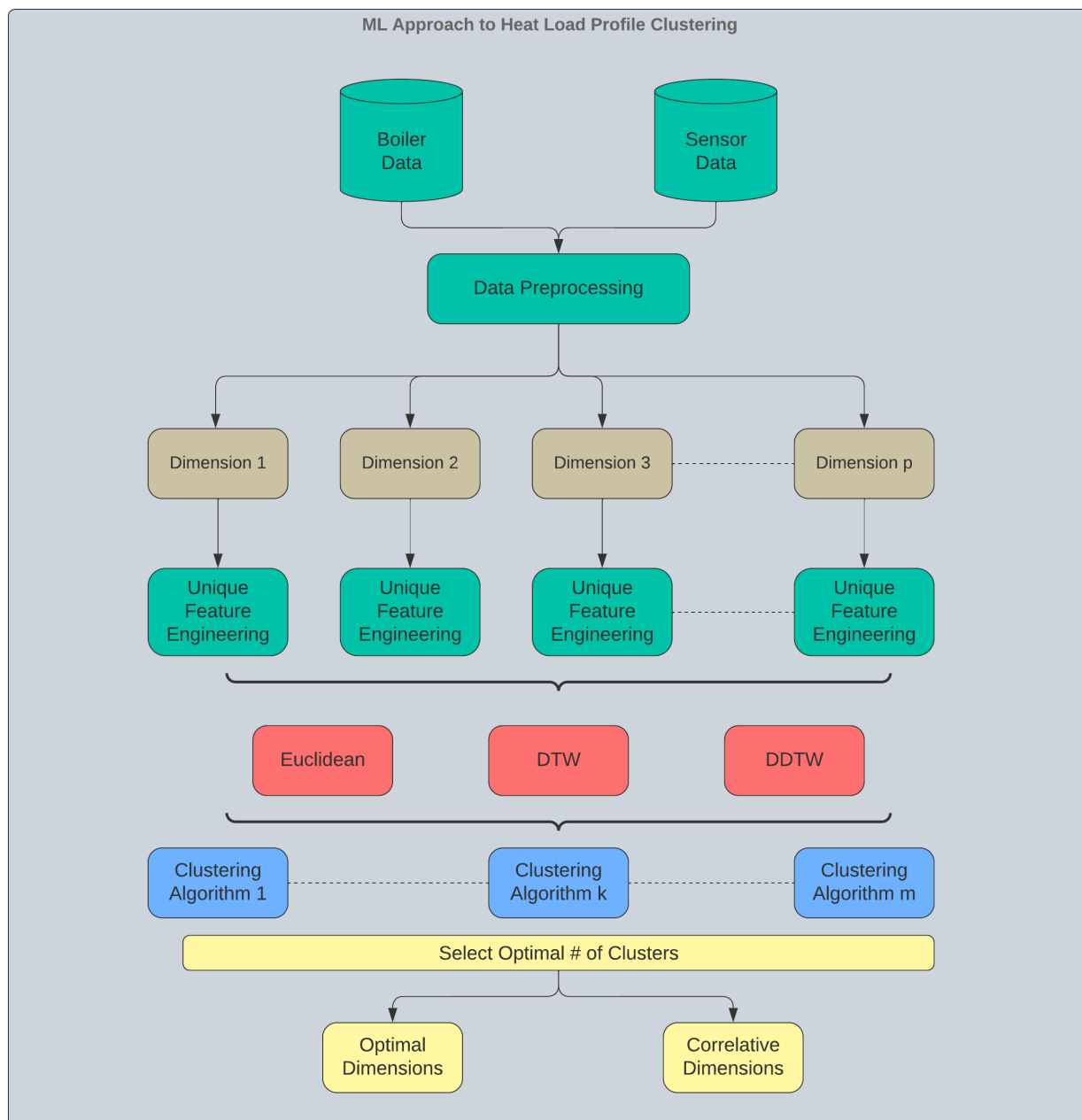


Figure 26: Proposed Methodology

4.2.3. Main benefit and innovation

Main Benefit: The ability to deliver accurate demand predictions rely heavily on considering all the key aforementioned parameters, such as weather, building, user and heating system profiles. However, the significance amplifies when it comes to the optimal management of large portfolios of residential heating systems, for achieving energy savings or implementing Demand



Response (DR) actions towards environmental sustainability. As the population of participating households and the amount of available historical data continue to grow, the application of dedicated forecasting models at household level becomes challenging and practically impossible. This fact emphasizes the need for implementing innovative approaches for segmenting participating households into more homogeneous and manageable groups, which can reduce the variability and uncertainty of demand and ultimately simplify the demand profiling and forecasting procedures. Indicative application scenarios may include the ability to pinpoint residences with the highest potential for energy savings and/or for DR participation. As a result, heating load profiles grounded in diverse facets of the problem, which enable a better understanding of consumption patterns and behaviors, are the solution to the identified problem and at the same time the motivation for this work.

Innovation: The current work extends existing literature by incorporating a comprehensive set of parameters, including heat demand, temperature, building characteristics, boiler specifications, and user preferences. It is noteworthy to acknowledge that, to the best of the author's knowledge, no prior work has delved into the specific domain addressed by the current study. This pioneering exploration sets a significant benchmark for future research in the field, offering novel insights and avenues for further investigation.

4.2.4. Development activity

The methodology depicted in Figure 26 follows this sequence: Initially, data collection involves gathering boiler and sensor data from indoor and outdoor spaces in the building, which undergoes processing to identify outliers and anomalies. Next, key aspects of the problem are pinpointed, and a customized feature engineering process is conducted to match the characteristics of each aspect, specifically tailored for use in the Euclidean distance (ED) approach. Simultaneously, raw time-series data are employed for the Dynamic Time Warping (DTW) and Derivative Dynamic Time Warping (DDTW) methods, respectively. Clustering algorithms are then deployed on the prepared data across varying cluster numbers, ranging from two to nine. Evaluation is based on established metrics of similarity and dissimilarity to determine the optimal number of clusters for each aspect, distance measure, and the overall problem. Ultimately, the results from each scenario are evaluated to identify the most effective approach for each aspect and their interrelations within the problem domain.

For the time being, the Heating Demand dimension of the problem has been addressed and the methodology will stand as it seems for the rest of the project. As mentioned earlier, specific feature engineering strategies have been developed for each identified dimension to improve the outcomes derived from the Euclidean distance (ED) approach, which will be comprehensively examined. Mean profiles for each dimension, for every minute of the day, have been extracted from the original data, forming the basis of the forthcoming analysis.

- **Time-of-Maximum Analysis (ToM):**
 - Identified peak periods of heat demand by determining the time of maximum load for each profile.



- Categorized data into distinct time ranges, namely early morning, morning, noon, evening, night, and late night, and created binary columns indicating whether the time of maximum load fell within each specified range.
- Analytically defined time intervals as follows:
 - Early Morning (5:00 - 8:00)
 - Morning (8:00 - 10:00)
 - Noon (10:00 - 17:00)
 - Evening (17:00 - 20:00)
 - Night (20:00 - 24:00)
 - Late Night (0:00 - 5:00)
- **Statistical Analysis:**
 - The statistical measures chosen for this analysis are as follows:
 - Mean
 - Standard Deviation
 - Maximum
 - Minimum
 - 25th percentile
 - 50th percentile
 - 75th percentile

Following feature engineering, the dataset underwent transformation to integrate the newly derived features. This involved expanding the original dataset to incorporate columns representing peak periods and statistical metrics computed for each heating load profile. The enhanced dataset formed the foundation for subsequent clustering analysis.

We experimented with two clustering algorithms, namely K-means Hierarchical Agglomerative clustering (HAC) and each algorithm demonstrated varying degrees of effectiveness in capturing the inherent structure of the data. Additionally, we evaluated the results based on three different distance metrics, Euclidean Distance (ED), Dynamic Time Warping (DTW) and Derivative DTW, to quantify the similarity between heating load profiles. Furthermore, utilizing evaluation metrics is crucial in determining the ideal number of clusters for the dataset. Commonly employed metrics in scientific research, including SIL (Silhouette score), DBI (Davies-Bouldin Index) and CHI (Calinski-Harabasz Index), are applied to assess the clustering results' quality. These metrics aid in identifying the optimal number of clusters that accurately depict the underlying data patterns. Leveraging such evaluation metrics empowers researchers and practitioners to make well-informed decisions regarding clustering outcomes, enabling them to refine their analyses and



improve the overall quality of load profiling processes. It is worth noting that the selection of the optimal number of clusters is based on choosing the clustering configuration that produces the highest SIL and CHI values, while aiming for a lower corresponding value of DBI. Following the necessary data preprocessing a visual example of a typical household can be seen in Figure 27.

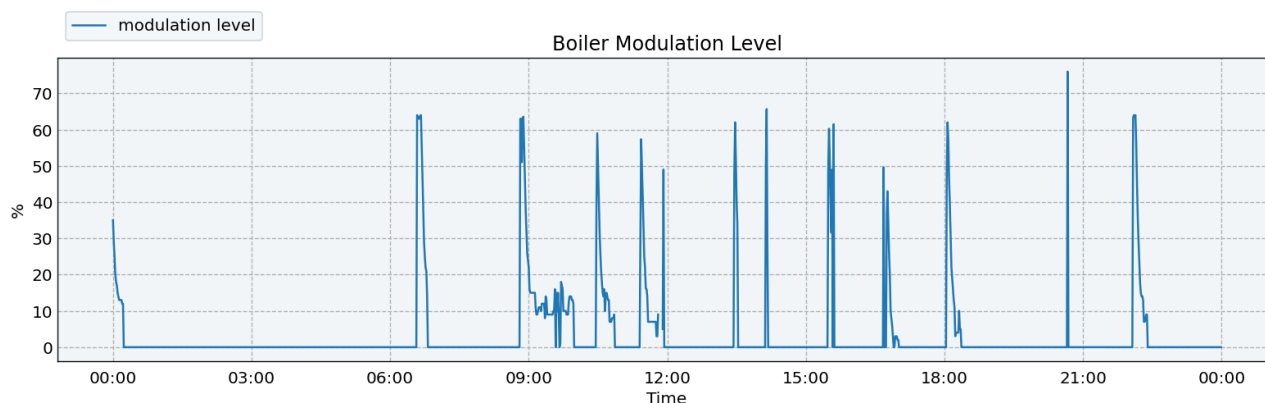


Figure 27: visual example of a typical household

Furthermore, the mean daily modulation levels for all 29 houses, for the heat demand dimension are showcased in Figure 28. Those can be characterized as the modulation level profiles that will be used in this study. For the DTW and DDTW distances these timeseries are going to be used for clustering, as for ED the extra features described earlier will conclude the input to the clustering algorithms.

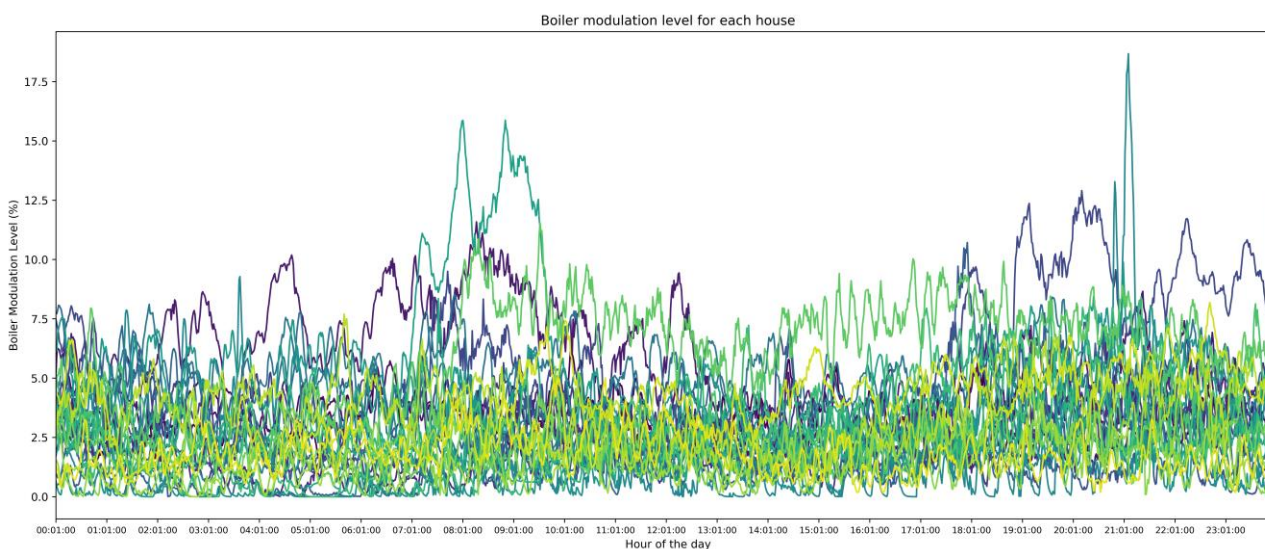


Figure 28: heat demand dimension - mean daily modulation levels for all 29 houses

The best number of clusters in this case is 3. This result can be evaluated in the Table below, that results can be seen for each evaluation metric for the cluster number 3.



As can be witnessed, the results for K-means with DTW are exceeding the rest and thus those results are going to be showcased later on.

Algorithm	Silhouette	Davies-Bouldin Index	Calinski-Harabasz Index
K-Means with ED	0.174537	1.536718	9.738654
HAC with ED	0.216338	1.497139	9.249262
K-Means with DTW	0.428794	0.759205	34.586499
K-Means with DDTW	0.121640	1.693275	4.394497

In the following figures, the clusters that are the output of this clustering attempt are illustrated and the differences in the mean and median load profile of each cluster can be seen with different colours.

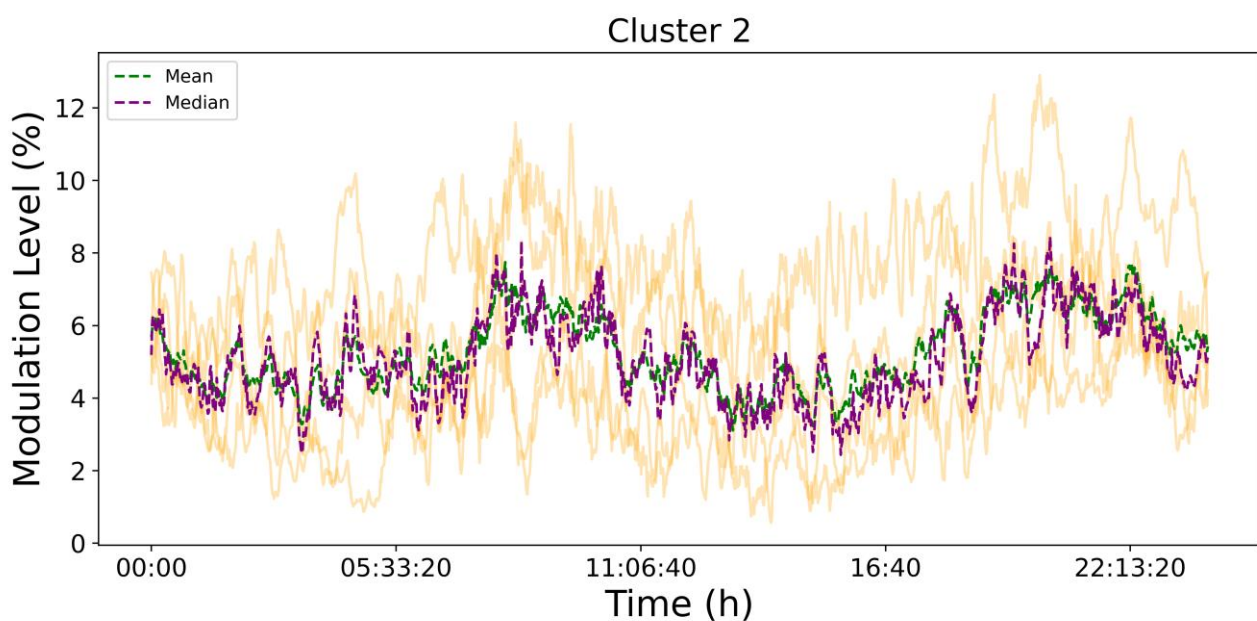


Figure 29: Cluster 2



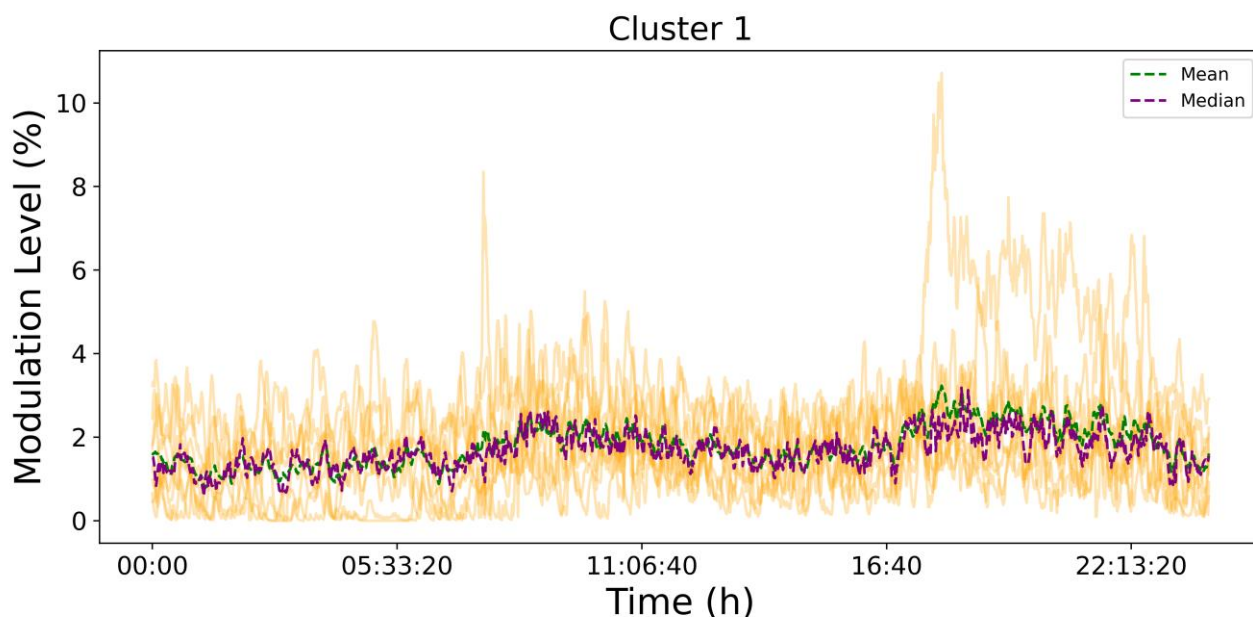


Figure 30: Cluster 1

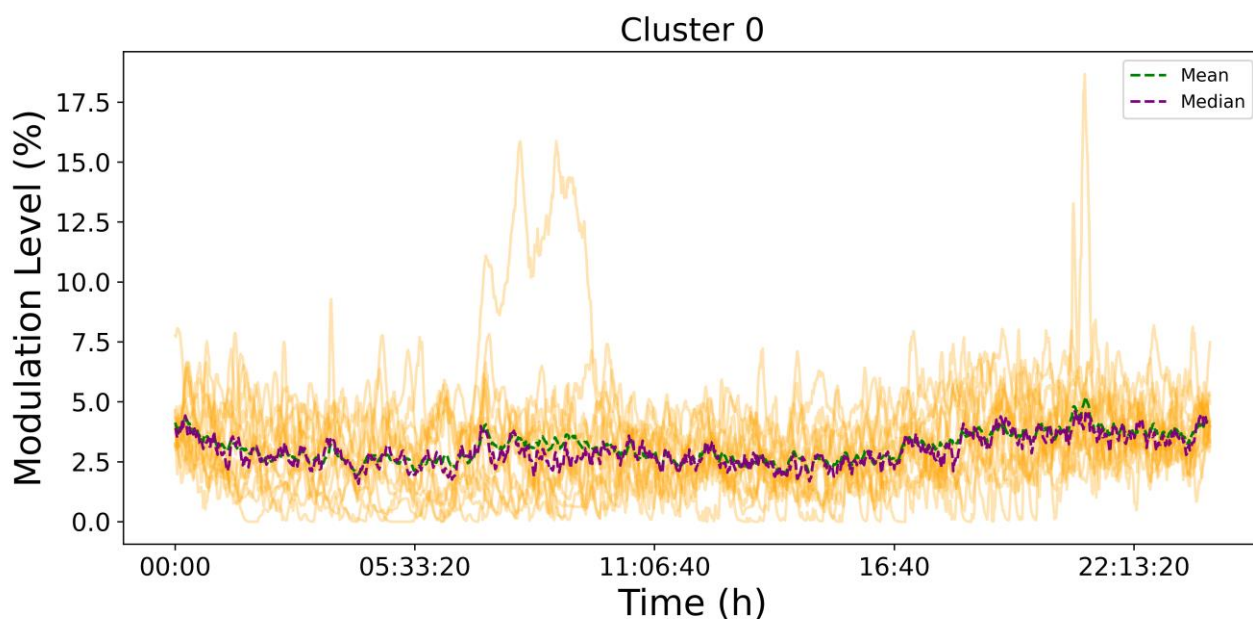


Figure 31: Cluster 0

4.2.5. Outlook for the future expected release

Summing up, the current research endeavour highlights the critical importance of meticulous analysis in understanding heating load profiles and optimising heating system operations, particularly in the context of energy savings and DR. Through the evaluation of clustering



algorithms and distance metrics across five dimensions, we have identified distinct patterns and correlations within the dataset, providing valuable insights for stakeholders.

In Figure 26 the methodology that we are going to follow into the next releases is presented. As can be clearly seen from the diagram, we have worked into one dimension of the problem, and evaluated the results for all 3 distance metrics chosen for this Task. Thus, the 4 dimensions that are going to be created, tested and evaluated are going to be the Temperature, Boiler, User and Building dimension. Furthermore, the optimal dimensions of the problem will be showcased when the analysis has already taken place for all the facets, with the corresponding similarities between them. That way, the true intricacies of heating load profiles will be present for all the dimensions of the problem.

4.3. Service C - Energy disaggregation for the electricity pilot

4.3.1. Overview of the service

Service C	Description
<i>Service / Software title</i>	Energy disaggregation for the electricity pilot (HERON)
<i>Responsible partner(s)</i>	NTUA (main contributor), HERON (supporting)
<i>Associated pilot site</i>	Pilot 3, Greek– electricity domain
<i>Service brief overview</i>	This service utilizes Non-Intrusive Load Monitoring (NILM) techniques to disaggregate energy consumption data at the appliance level
<i>Associated TE</i>	TE-4, TE-11
<i>Associated Use Case</i>	HLUC1 / UC1&UC2, HLUC4 / UC11, HLUC5 / UC13
<i>Functional Requirements (input by other activities)</i>	n/a
<i>Data Input</i>	- Smart meter data: power - Smart plug data: power (in support and for the validation of NILM models)
<i>Service Output</i>	Disaggregated load of appliances as input in DR services and the Flexibility Modelling Tool
<i>Repository Location</i>	DEDALUS GitHub
<i>License information</i>	Proprietary



TRL	TRL 7-8
Current version number	v0.2
Expected date of finalization	May 2025, M25
Development Status	1st version ready and on-going development for the next version

4.3.2. Architectural diagram

Energy disaggregation, also known as Non-Intrusive Load Monitoring (NILM), is a process that aims to break down the total energy consumption of a household or building into the individual energy usage of each appliance. This technique allows users to gain insights into their energy consumption patterns without the need for intrusive monitoring systems. The proposed methodology is depicted in Figure 32. This methodology outlines the systematic approach taken to collect, preprocess, train, and evaluate the data, ultimately leading to the identification of individual appliance-level energy consumption. The process is divided into several key phases, each contributing to the overall goal of accurate and efficient energy disaggregation.

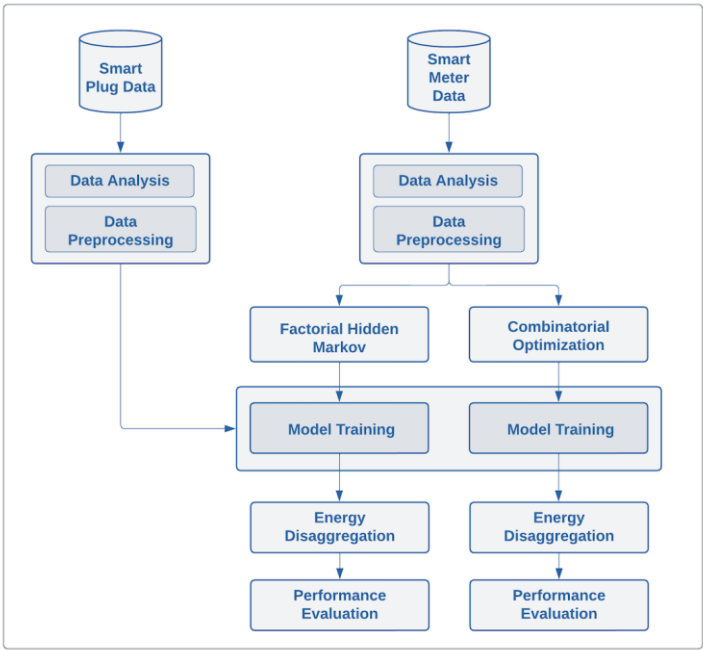


Figure 32; Energy disaggregation - Proposed Methodology

Explanation of Components:

1. **Data Analysis and Data Preprocessing:** This phase involves analyzing the collected data for patterns and anomalies. Data preprocessing includes cleaning, normalization, and feature extraction, ensuring the data is in optimal form for model training.
2. **Utilizing advanced machine learning models** such as Factorial Hidden Markov Model (FHMM) and Combinatorial Optimisation (CO), the system learns to recognize and predict the energy consumption patterns of different appliances.
3. **Energy Disaggregation:** The trained models disaggregate the total energy consumption into individual appliance-level data.
4. **Performance Evaluation:** The disaggregated data is evaluated using metrics like accuracy, precision, recall, and F1-score to ensure the models' effectiveness.

The entire process is designed to be automated, with real-time data processing and analysis being conducted without the need for constant expert oversight.

4.3.3. Main benefit and innovation

Main Benefit: The primary benefit of this energy disaggregation approach is the ability to provide detailed, appliance-level energy consumption insights in real-time. This granular data enables users to understand their energy usage patterns, identify high-consumption appliances, and implement energy-saving measures more effectively. Additionally, utility companies can leverage this information to offer personalized energy-saving tips and incentives, ultimately contributing to overall energy efficiency and cost reduction.

Innovation: The integration of advanced machine learning models such as the Factorial Hidden Markov Model (FHMM), and the Combinatorial Optimisation (CO) represents a significant innovation in the field of energy disaggregation. These models are capable of accurately disaggregating complex energy consumption data into individual appliance-level data with high precision. The use of open datasets like UK-DALE and REDD for training enhances the models' robustness and generalizability.

4.3.4. Development activity

The result of the development activity formally provided as achievement of MS4 1st technology development cycle completion is summarised as follows:

1. **Data Collection:**
 - **Smart Meter Data:** Collected overall power consumption data from smart meters installed at the pilot site, recording the total electricity usage of the entire household or building.
 - **Smart Plug Data:** Gathered data from smart plugs connected to individual appliances, measuring the power consumption of specific appliances to provide granular data for training and validation.
 - **Note:** As pilot data is currently not available, we tested the approach using publicly available datasets.
2. **Data Preprocessing:**



- **Data Cleaning:** Removed any anomalies, missing values, or outliers in the data to ensure accuracy and consistency.
- **Normalization:** Scaled the data to a standard range, facilitating better model performance and faster convergence.
- **Feature Extraction:** Identified and extracted relevant features from the raw data, such as power spikes, usage patterns, and appliance-specific signatures.

3. Training Phase:

- **Dataset Utilization:** Employed publicly available datasets such as UK-DALE and REDD for training purposes.
 - **UK-DALE (UK Domestic Appliance-Level Electricity):** This dataset consists of appliance-level power consumption data recorded from several houses in the UK. It includes detailed, time-stamped power usage information, enabling fine-grained analysis and model training.
 - **REDD (Reference Energy Disaggregation Data):** A comprehensive dataset that includes power consumption data from six houses in the US. It provides appliance-level and aggregate power measurements, serving as a benchmark for energy disaggregation research.
- **Model Selection:**
 - **Factorial Hidden Markov Model (FHMM):** A probabilistic model that represents multiple sequences of hidden states to capture the dynamics of power usage.
 - **Combinatorial Optimisation (CO):** An algorithmic approach that solves optimisation problems by combining various parameters to achieve the best energy disaggregation results.
- **Training Algorithm:** Each model was trained on the dataset to recognize and predict energy consumption patterns of different appliances.

4. Inference Phase:

- **Real-time Data Processing:** Fed pre-processed data from the pilot site into the trained models for inference.
- **Energy Disaggregation:** Each model disaggregated the total energy consumption into individual appliance-level consumption.

5. Evaluation and Comparison:

- **Performance Metrics:** Performance metrics such as accuracy, precision, recall, and F1-score were calculated for each model.
 - **Accuracy:** Measures the proportion of correctly identified appliances and their usage.
 - **Precision:** Calculates the number of true positive predictions divided by the total number of positive predictions.
 - **Recall:** Assesses the number of true positive predictions divided by the total actual positives.
 - **F1-Score:** A harmonic mean of precision and recall, providing a single metric to evaluate model performance.



- **Model Comparison:** Analysed the performance of each model (FHMM, CO) based on the calculated metrics to determine the most effective approach for energy disaggregation.
6. **Output Delivery:**
- **Appliance Identification:** Analysed disaggregated data to identify the operating appliances.

4.3.5. Outlook for the future expected release

In the upcoming release, data will be leveraged from HERON's Platform, providing access to a comprehensive repository of meter and smart plug data to enhance the accuracy and effectiveness of our energy disaggregation models. Once the data is retrieved via the REST API, the development, training, and optimisation of the energy disaggregation models will proceed. Subsequent iterations and refinements of the models will also utilize the initially retrieved datasets, enabling continuous improvement and validation of the models' performance.

Moreover, the disaggregated loads generated by our models will be served as valuable inputs for Demand Response (DR) services and the Flexibility Modeling Tool (TE-4).

4.4. Service D - Development of tariff system for implicit and explicit electricity DR services

4.4.1. Overview of the service

Service D	Description
Service / Software title	Development of tariff system for implicit and explicit electricity DR services
Responsible partner(s)	HERON
Associated pilot site	Pilot 3, Greek – electricity domain
Service brief overview	This service aims at generating a tariff system based on multiple conditions and their combination. For example, the tariff can be based on environmental (i.e. shares of renewable) or market (day-ahead market) signals or their combination.
Associated TE	TE-11
Associated Use Case	HLUC5 / UC13, UC14
Functional Requirements (input by other activities)	TE11-FR1
Data Input	Day-ahead and Transmission System Operator (TSO) data. For GR pilot, such data is prepared via APIs from HERON.



Service D	Description
Service Output	Files in JSON format with 24-hour tariff (electricity prices, RES shares or token points) as an incentive to engage in DR services
Repository Location	Business cloud system
License information	Proprietary for the hybrid version (combination of market and sustainability tariffs) and open-source versions for market data
TRL	TRL 7-8
Current version number	v0.1
Expected date of finalization	May 2025, M25
Development Status	First version of the service utilising data from the Integrated Scheduling Programming (ISP) is available

4.4.2. Architectural diagram

Figure 33 shows the high-level architectural diagram of the service.

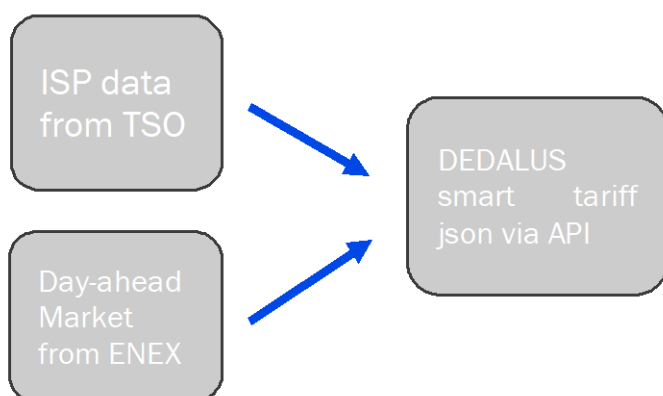


Figure 33: Architectural diagram of Service D

4.4.3. Main benefit and innovation

Benefit: The main benefit of this service is the development of a tariff that will be used for the DR service by utilizing publicly available data. It enables consumers to decrease their carbon footprint while they benefit from the wholesale market volatility and can be integrated seamlessly to existing DR services.

Innovation: There is no significant innovation on the current version of the DEDALUS DR tariff as it merely sorts the Renewable Energy Sources percentage in the system and identifies the hours



with highest and lowest shares of renewables. However, there is greater innovation potential in hybrid tariffs that combine market and system mix signals as indicated by the limited literature on dynamic market tariffs that take into consideration short-term forecasts of Renewable Energy sources.

4.4.4. Development activity

HERON has developed the first version of the service calculating the percentage of System Load satisfied by RES, drawn from Load and RES generation from the Integrated Scheduling Programming (ISP) provided by the TSO.

The ISP is solved as a co-optimisation problem taking into account the Balancing Energy and Balancing Capacity Offers of the BSEs as well as their respective constraints, Transmission System constraints and the TSO's needs, in order to minimize the cost of Balancing Energy and Balancing Capacity procurement. Clearly, RES and System Load forecasts during the ISP process are as accurate as they can be, given the importance of ISP in the energy stability of the system.

For the Greek system, the ISP is managed by the Greek TSO (IPTO) and is executed at three scheduled times for each Dispatch Day D for half hour blocks as follows:

- ISP1 is executed at 16:15 EET on calendar day D-1 and concerns all Dispatch Periods (48 Dispatch Periods) of Dispatch Day D – Forecasts are published
- ISP2 is executed after ISP1, considering the updated input data. It is executed at 00:00 EET on calendar day D and concerns all Dispatch Periods (48 Dispatch Periods) of Dispatch Day D.
- ISP3 is executed at 12:00 EET on calendar day D, considering the updated input data. The time interval taken into account is from 13:00 EET till the end of Dispatch Day D (24 Dispatch Periods).

It may also be carried out on demand, each time the TSO deems that there are significant changes that need to be balanced. For the purposes of the Implicit DR solution under development we are collecting RES (HV assets) and Load Forecasts as they are generated from the TSO. The TSO's ISP forecasts are not issued based on the aforementioned ISP schedule.

According to the [relevant code](#)¹, ISP forecasts are given as follows:

- ISP1 zonal Load and RES forecasts and system requirements are published on IPTO's website until 09:30 EET on calendar day D-1 (v1 forecasts & requirements)
- ISP1 zonal Load and RES forecasts and system requirements are updated and published on the website until 13:30 on calendar day D-1 (v2 forecasts & requirements)
 - *ISP1 is executed at 16:15 EET, D-1*
- ISP2 zonal Load and RES forecasts and system requirements are published on IPTO's website until 21:00, D-1

¹ IPTO Technical Decision: Integrated Scheduling Process (version 2.0, February 2021). Accessed from https://www.admie.gr/sites/default/files/users/dda/KAE/TECHNICAL%20DECISION%20ISP_eng_v2.pdf



- *ISP2 is executed at 00:00 EET, D*
- ISP3 zonal Load and RES forecasts and system requirements are published on IPTO's website until 09:00, D
- *ISP3 is executed at 12:00 EET, D*

The data can be found from the Greek TSO's [website](#) under [Market / Market Statistics / Data](#) by downloading the "ISP Requirements" which includes RES and Load forecasts and Mandatory Hydro declarations given that in Greece, Hydro, although a renewable energy source in principle, is allocated in the same category, in terms of system planning, with the dispatchable energy sources. Alternatively, the [File Download API](#) can be used to automate file downloads.

Although this information is publicly available, handling the data requires some specialist knowledge due to TSO APIs and csv files containing Greek market specific data points (such as the list of Pump storage Hydro plants without specifying whether they are used as generators or as flexible loads). To alleviate such complexities, HERON automated the process through the development of an API which provides information to DEDALUS partners from further developing the DR and Digital Twins services, without technical knowledge of the Greek market.

4.4.5. Outlook for the future expected release

- Development of a hybrid tariff combining market and environmental elements i.e. identify the intervals with lowest prices and relative high shares of renewables (+50%).
- Integration of the hybrid tariff as an API in order to be used in DR related services for frontrunner and multiplier.
- Integration of data sources for the multiplier pilots (IT, ES)
- Development of environmental, market and hybrid tariffs for multiplier pilots (IT, ES)

4.5. Service E - OurPower Nudging Application

4.5.1. Overview of the service

Service E	Description
<i>Service / Software title</i>	OurPower Nudging Application
<i>Responsible partner(s)</i>	OurPower, BluePrint
<i>Associated pilot site</i>	Pilot 1, Austrian
<i>Service brief overview</i>	The OurPower Nudging Application enhances user engagement and optimises energy consumption within communities. By providing personalized advice and enabling user responses to energy-saving opportunities, the service aims to foster proactive behavior changes



Service E	Description
	and enhance community-wide energy efficiency. Furthermore, the nudging application serves as a key information channel, offering data on energy production and consumption.
Associated TE	TE-10
Associated Use Case	UC22
Functional Requirements (input by other activities)	TE10-FR2 The system will deliver personalized energy utilization (saving) recommendations to users.
Data Input	• Energy Production Data: Information about the historic and forecasted energy generation from PV systems. • Energy Consumption Data: Data on historic and forecasted energy consumption by the community and individual users. • User Preferences and Historical Responses: Information on user behavior, preferences, and previous responses to nudging to tailor advice more effectively. • Community Event Information: Details about upcoming community dialogs and online events to be communicated to users.
Service Output	• Personalized Energy Consumption Advice: Specific recommendations for users to adjust their energy usage based on historic and forecasted data. • Improved Energy Management: Contribution to overall community energy optimisation by encouraging and facilitating individual actions in response to dynamic energy situations. • User Interaction Data: Records of how users respond to the advice and interact with the app, which can be used to further refine the nudging strategy and improve the app. • Enhanced Community Engagement: Increased user participation in community through direct communication and feedback opportunities within the app.
Repository Location	OurPower private Github repository
License information	Proprietary
TRL	6
Current version number	v0.1
Expected date of finalization	March 2025, M25



Service E	Description
Development Status	First version (skeleton application) has been developed, development for the next version is ongoing.

4.5.2. Architectural diagram

The architectural diagram specification of the OurPower Nudging application and its connected components/services is shown in Figure 34. At the core lies the Pocketbase-based OurPower Nudging application backend, serving as the foundational element for data management and processing. This backend integrates seamlessly with various data sources, including energy monitoring services and the EDA portal, to receive metering data from users. Additionally, the backend interfaces with forecast services to acquire energy and weather forecasts crucial for generating accurate nudging recommendations. On the frontend, the application utilizes Ionic, a versatile SDK, to create an intuitive interface for end-users across multiple platforms, including Android, iOS, and web. Through APIs, the nudging application connects to additional services for energy consumption data, weather forecasts, and nudging recommendations, ensuring comprehensive functionality. A landing page serves as the initial entry point for participants to onboard onto the OurPower Nudging application.

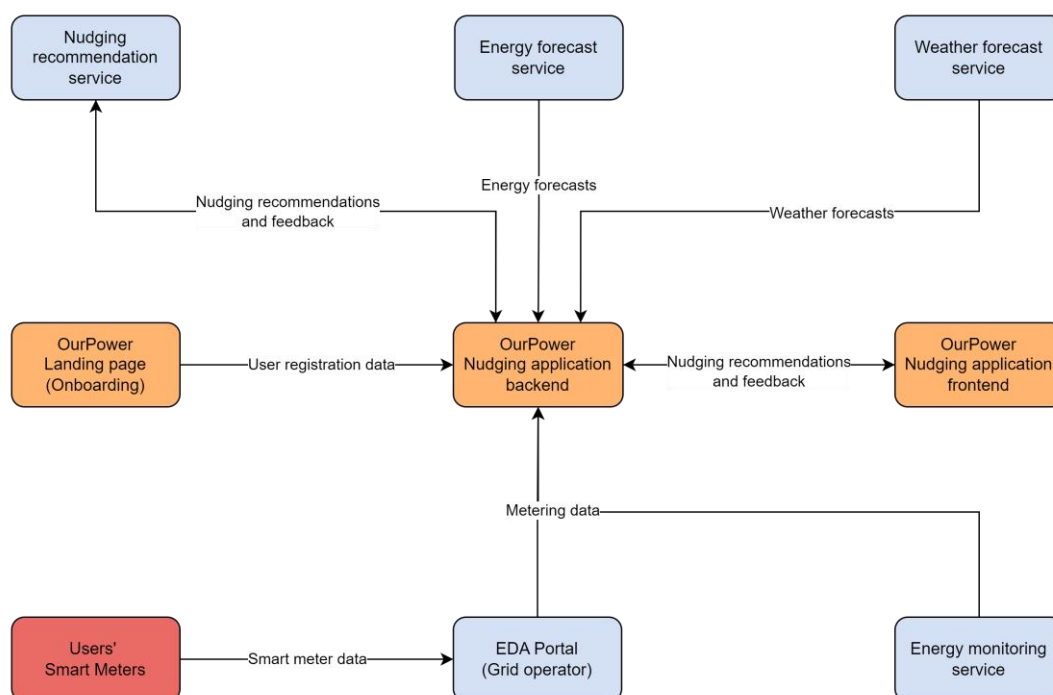


Figure 34: Architectural diagram of the OurPower Nudging application and connected components/services



4.5.3. Main benefit and innovation

The OurPower Nudging Application presents a transformative solution for optimising energy consumption and fostering proactive behaviour changes within communities. By leveraging personalized advice and user responses to energy-saving opportunities, the service enhances user engagement and community-wide energy efficiency. Its innovative approach serves as a key information channel, providing valuable insights into energy production and consumption. The main benefit lies in its ability to deliver personalized energy utilization recommendations, empowering users to make informed decisions based on historic and forecasted data. Moreover, the application contributes to improved energy management at the community level by encouraging individual actions in response to dynamic energy situations. Through user interaction data and enhanced community engagement, the OurPower Nudging Application not only drives efficiency but also cultivates a more responsive and sustainable energy ecosystem.

4.5.4. Development activity

The nudging application will utilize Pocketbase for backend services and Ionic for the frontend, ensuring a robust, scalable, and user-friendly experience. Pocketbase is an open-source backend solution that combines database and server functionalities into a single package. The backend will be hosted on servers managed by OurPower, leveraging OurPower's existing infrastructure. Ionic is a powerful HTML5 SDK that allows for the development of mobile applications using web technologies. It is particularly well-suited for cross-platform applications like the nudging application, which will be available to end-users as Android, iOS and web applications. Additional utilized services for the provision of, e.g., energy consumption data, weather forecasts and nudging recommendations, can be connected via APIs to the nudging application. Mock-ups of the user interface of the nudging application are shown above in Figure 35. These will be validated as part of the co-creation workshop in WP2 in May 2024. The user interface may change during the project depending on the received end-user feedback.



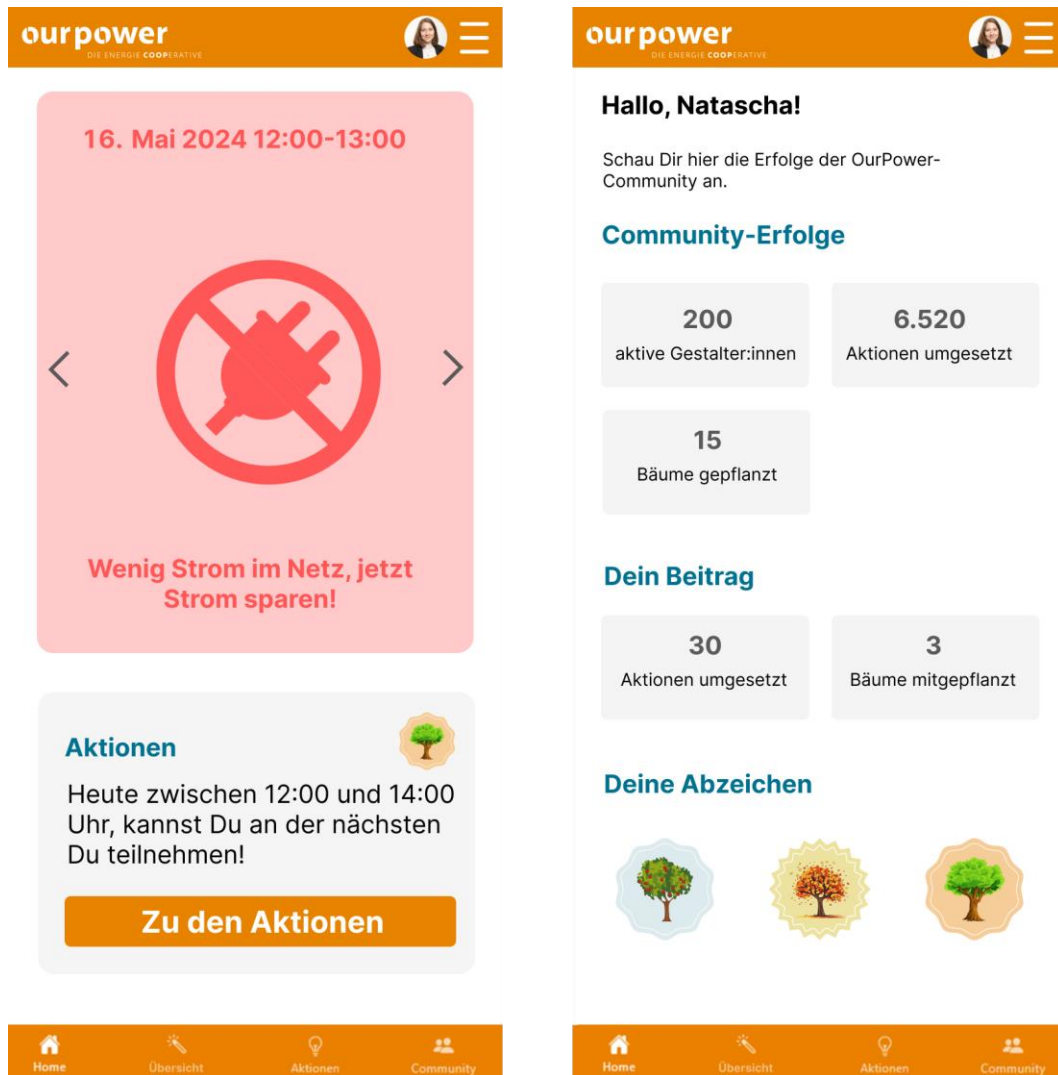


Figure 35: Mock-ups of the OurPower nudging application

4.5.5. Outlook for the future expected release

The planned developments for future expected releases of this service include:

- Implementation of the nudging application backend
- Integration of new nudging mechanisms in the application
- Improved user interface and user experience
- Enhanced community interaction features
- Integration with other components, e.g., Pre-aggregation framework



4.6. Service F - Multi-value Demand Response for virtual clusters of building

4.6.1. Overview of the service

Service F	Description
<i>Service / Software title</i>	Multi-value Demand Response for virtual clusters of building
<i>Responsible partner(s)</i>	HERON (main contributor), NTUA (supporting)
<i>Associated pilot site</i>	Pilot 3, Greek – Electricity domain
<i>Service brief overview</i>	This service provides DR actions as a stand-alone service that can be integrated in third-party applications and systems. DR events may refer to Implicit DR recommendations or explicit DR actions given remote control capabilities. The events are customizable in terms of duration and aim (upwards / downwards regulation) and are aimed towards clusters of consumers defined based qualitative and quantitative attributes.
<i>Associated TE</i>	TE-11
<i>Associated Use Case</i>	UC 13
<i>Functional Requirements (input by other activities)</i>	HLUC5
<i>Data Input</i>	• Energy Flexibility Indicator based on consumption data from IoT metering devices. • DEDALUS Tariff.
<i>Service Output</i>	json via an API. Json explained in section 4.6.4
<i>Repository Location</i>	DEDALUS GitHub
<i>License information</i>	MIT
<i>TRL</i>	TRL 5
<i>Current version number</i>	Not Applicable
<i>Expected date of finalization</i>	September 2025, (M29)



Service F	Description
Development Status	- DR Recommendations of data structures that can be used for the development of Digital Twins and services for runner-up pilots have been defined. - The optimisation modelling for the development of the DR service is in early stages.

4.6.2. Architectural diagram

Figure 36 shows the high-level overview of the service.

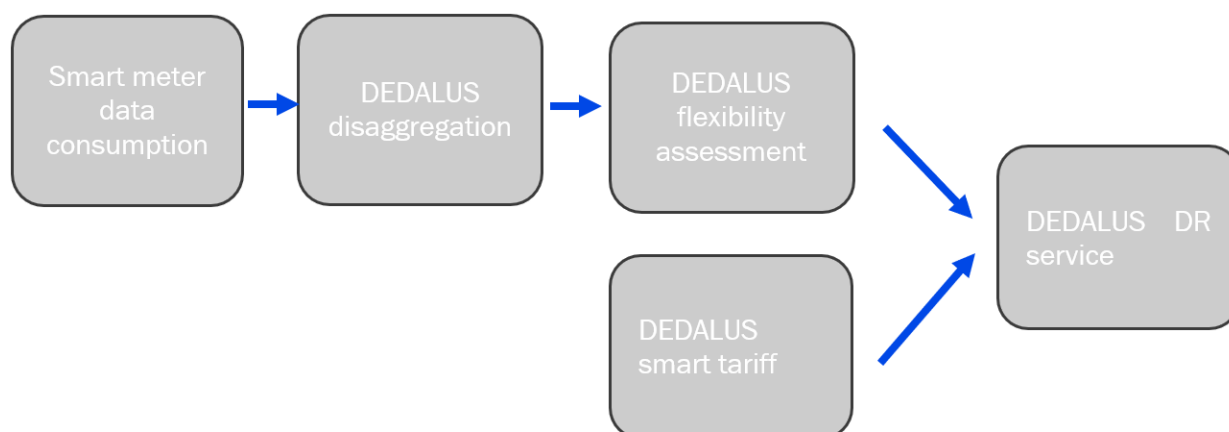


Figure 36 Architectural diagram – service F

4.6.3. Main benefit and innovation

Main Benefit: The primary benefit of this service is the ability to provide tailored DR recommendations and initiate specific actions aimed towards the consumers that are likely to follow them without requiring them to upload preferences or operation schedules. In doing so, interference is minimised, limiting the perception that DR services would require micromanaging for the consumers to maximise the benefits.

Innovation: The DR service under development extends the state of the art by integrating several optional components in the core model, allowing for multiple versions depending on the data and the technologies available. For example, the DR service could operate on “lighter” versions that do not require disaggregated data, and therefore give more generic recommendations (nevertheless based on smart meter data).



4.6.4. Development activity

For this development cycle, TE-11 is still in early stages. Setting the groundwork for further development there has been effort towards two directions:

1. Development of the data structure of the DR recommendation or action as a json that can be accessed via an API provided by HERON. Metadata on the data structure is provided in Table 15.

```
{
  "home_id" : "1",
  "event_type" : "1" or "0"
  "event_id" : "1_1",
  "asset_id" : "1_Dishwasher",
  "device_there" : "1" or "0"
  "device_id" : "domxplug-5432" or NaN
  "hours_start": "2024-04-16T01:00:00.000Z",
  "hour_end": "2024-04-16T03:00:00.000Z",
}
```

Table 15: Metadata definition of json output of the DR service.

API Parameter	Type	Explanation
home_id	Integer	ID of the household the device belongs in
event_type	Boolean	1: Increase consumption or 0: decrease consumption
event_id	String	ID of event combining home_id and an increasing integer
asset_id	String	ID of the flexible asset combining home_id asset type
device_there	Boolean	1: flexible asset has smart plug attached or 0: not
device_id	String	API device_id of smart plug
hours_start	Timestamp (UTC)	Start of DR event
hours_end	Timestamp (UTC)	End of DR event

2. Formulation of the optimisation problem for the DR service. As mentioned, the service interacts with various elements of DEDALUS services. However, in order to expedite the development of the service without being blocked from the parallel development of several DEDALUS components, the methodology from related academic literature is being reviewed and replicated. Most notably, the influential paper [8] on appliance specific residential DR is used as a reference for the first version of the DR model formulation:



4.6.5. Outlook for the future expected release

The planned developments for future expected releases of this service include:

- Finalization of the formulation of the DR model
- Integration of the DEDALUS components
- Development of MVP for integration testing with the Digital Twin, and the Front Runner pilot and Multipliers
- Integration with the Digital Twin and the Front Runner pilot (GR)
- Integration with Multiplier (IT, ES)
- Final Validation

4.7. Service G - Integration of Implicit Demand Response service - Recommendation in the DT

4.7.1. Overview of the service

Service F	Description
<i>Service / Software title</i>	Integration of Implicit Demand Response service - Recommendation in the DT
Responsible partner(s)	ENG
Associated pilot site	Pilot 3, Greek – Electricity domain
Service brief overview	This service aims at integrating the outcome of Service F (Implicit DR recommendations) within the context of the Digital Twin TE-8 in Task 4.1.
Associated TE	TE-8
Associated Use Case	UC 11
Functional Requirements (input by other activities)	TE8-FR5
Data Input	Service G will take as input the outcome of service F, so the main data to be provided are reported in Table 15.
Service Output	Visualization of DR strategies in the Digital Twin.
Repository Location	DEDALUS GitHub
License information	MIT
TRL	TRL 5



Current version number	Not Applicable
Expected date of finalization	September 2025, (M29)
Development Status	First version available

4.7.2. Architectural diagram

The Middleware API is one component of the Digital Twin developed in the context of Task4.1 and the related architectural diagram is shown in Figure 2, section 2. There isn't any specific architectural diagram associated with this service.

4.7.3. Main benefit and innovation

The main innovation brought by this tool resides in the core of the seamless integration of DR services with the Digital Twin. In a nutshell, the Facility Manager will be provided with an easy and intuitive way to understand potential DR strategies in the DT UI. The integration will also be provided with clear specifications, and easiness of communications, facilitating the onboarding of both sides of the chain (service side and DT's side).

4.7.4. Development activity

The integration of DR Programs in the Digital Twin will follow the implementation of the respective TE. Particularly, a Middleware service will be developed to send the incoming communications towards the DT interface. The current version of this component provides the possibility to perform some validity checks on the input received, that for now is structured as the planned JSON provided previously in table 15. Given this data, the API will validate them (e.g. the middleware will check if the home_id is valid and present in the accepted ids), providing errors in case of non-valid data or data that are not expected. The development is available in the GitHub repository.

Since this implementation is still early stages, additions to the functionality of the middleware may be reported in the next development cycle.

4.7.5. Outlook for the future expected release

For the following development cycle the development plan includes:

1. Refinement of the first version of the middleware.
2. Initial testing with DR recommendation services.
3. Initial testing with the Digital Twin.



5. Local flexibility management for energy communities and self-consumption/cost optimisation

The aim of this section is to provide information about the local flexibility management for energy communities and self-consumption/cost optimisation service as outcome of Task 4.4. In particular, the first release of the service developed in Task 4.4 will be described providing the following information:

1. High level description of the service: main goal, benefit and innovation
2. Functional specification of the service and mapping with use cases
3. Data input required for execution of the service
4. Service's output
5. Architectural diagram specification

5.1. Overview of the service

This service focuses on optimising and managing local energy flexibility within communities and districts, targeting economic, social, and environmental benefits. Leveraging OurPower's P2P digital platform (TE-12) and CARTIF's district level DR optimisation (TE-13), the service enhances demand response (DR) management through a sophisticated decentralized management framework and enables trading flexibilities on the flexibility marketplace. A critical component is the development of a pre-aggregation framework that interfaces with local assets, enabling the identification and utilization of flexibility opportunities. These flexibility opportunities are either showcased to a nudging app (TE-10 developed in Task 4.3 as explained in section 4.5) or executed by automated DR, e.g., by controlling EV charging stations and battery storage. As part of the service the Universal Secure Gateway (TE-6) and Secure and Privacy/Confidentiality Preserving Data Transactions (TE-7), both developed by COMS in Task 3.6, will be utilized for resource virtualization and real-time management, which automates the DR management processes, thus enhancing participation and supporting local sustainability goals. As the service will be tested in two demos, not all the components will be deployed in both demos. In this sense, The Austrian demo will focus on TE-12 while the Spanish demo will focus on TE-13, sharing the same approach detailed in this section.

Table 16 contains the functional specifications of the service and the mapping with the associated technology enablers and use cases.



Table 16: Service's overview

Functionalities (brief textual description)	Related Functional requirements (D2.3)	Related Use cases (D2.3)	Associated Technology Enabler	Associated pilot site	Development Status
<i>Textual description of the functionality</i>	<i>ID- Name</i>	<i>ID-Name</i>	<i>Name</i>	<i>Pilot</i>	<i>Delivered/Not delivered</i>
The system enables flexibility management / trading for pre-aggregated building flexibility.	TE-12-FR1	UC18, UC20, UC22	TE-12	Pilot 1 (AT)	Not delivered
The system provides optimal operation profiles for energy assets to maximize self-consumption or minimize operation costs at district level.	TE-13-FR1	UC18, UC19	TE-13	Pilot 1 (AT), Pilot 5 (SP)	Not delivered

5.2. Architectural diagram specification

Figure 37 and Figure 38 show the high-level service architecture designed to enhance local flexibility management within energy communities, with a focus on optimising self-consumption and cost-efficiency. This service integrates advanced monitoring, control mechanisms, and trading functionalities to ensure optimal energy usage and enhance community engagement.

Figure 37 focuses on the architectural diagram for the deployment of the service in the Austrian demo, while Figure 38 focuses on the architectural of the Spanish demo. The differences between both architectures are mainly related to the way the market is integrated in the service and to the monitoring platforms used in each demo.



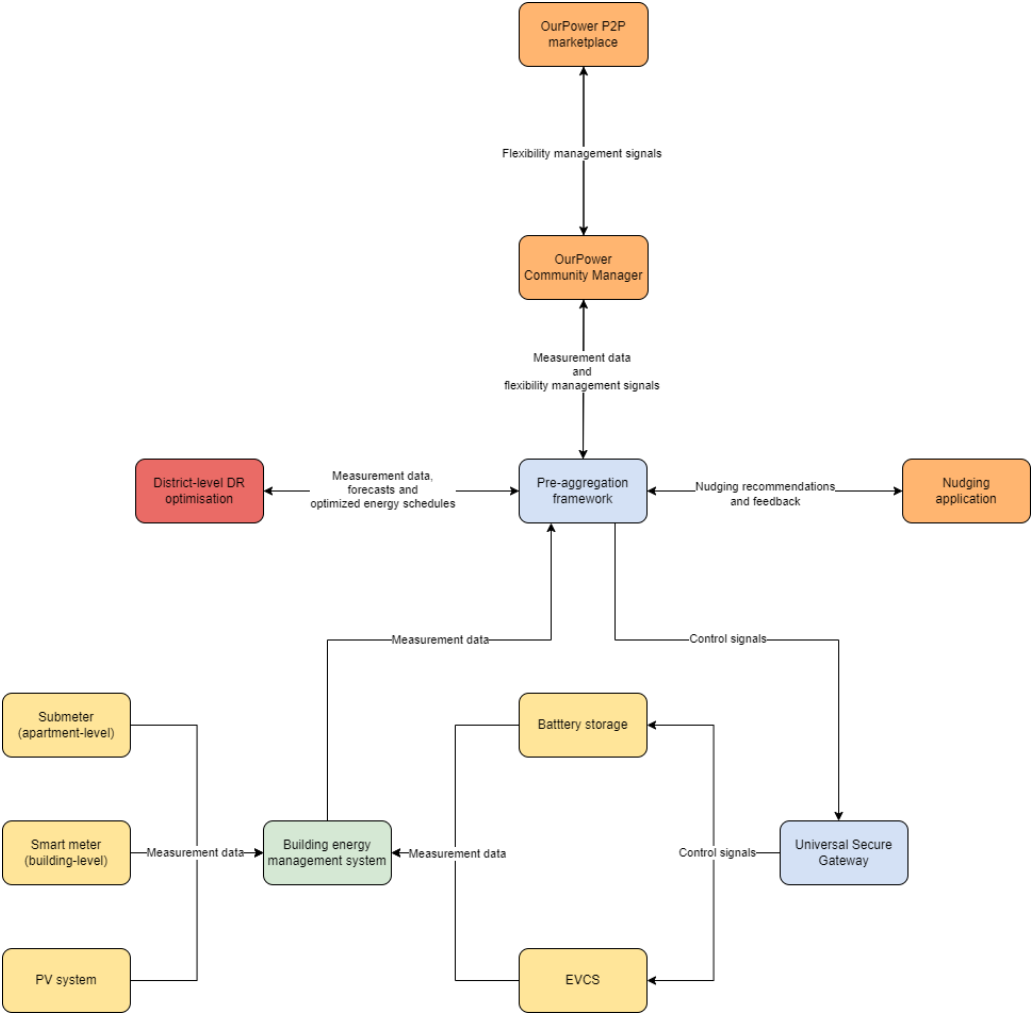


Figure 37: Architectural diagram for the deployment of the service in the Austrian demo

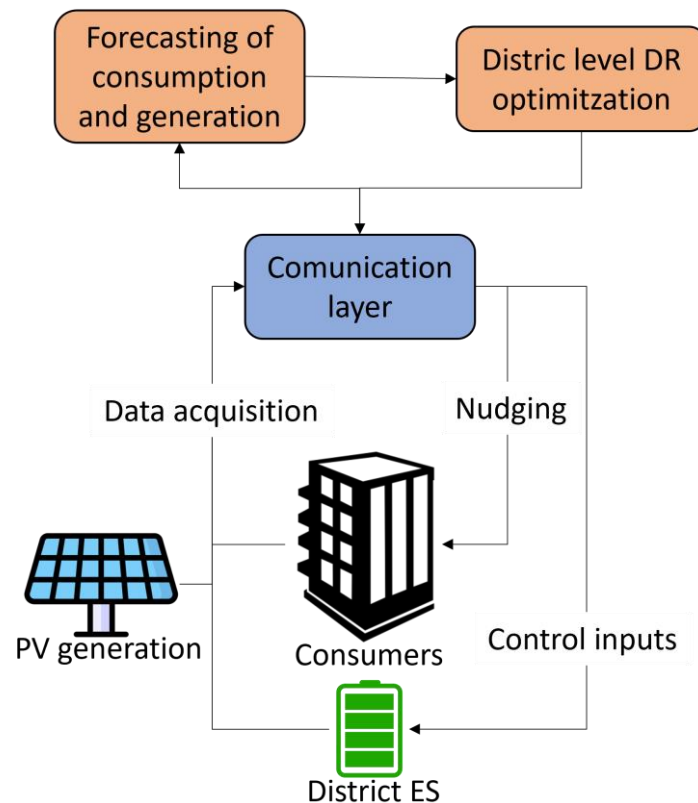


Figure 38: Architectural diagram for the deployment of the service in the Spanish demo

In the case of the Austrian pilot, the architecture of the service is centred around a few components which are seamlessly integrated to manage the energy dynamics of the community effectively. These include the Pre-Aggregation Framework, Universal Secure Gateway, District-level DR optimisation, OurPower Community Manager, and the OurPower P2P Flexibility Marketplace. Additionally, the Nudging App, developed under Task 4.3, interacts with this service, promoting manual demand response (DR) activities.

At the heart of the service is the Pre-Aggregation Framework, which is tasked with the continuous monitoring of energy metrics across the community. This framework gathers extensive data from various sources, such as energy consumption data from apartment and building meters, energy production data from photovoltaic systems, usage data from electric vehicle charging stations, and the operational state of battery storage systems. This rich dataset provides a detailed real-time view of energy usage and production, forming the basis for all subsequent optimisation processes.

The District-level DR optimisation complements the Pre-Aggregation Framework by specifically addressing the unique energy management needs at a broader district level. This system leverages the comprehensive data collected – from residential energy consumption patterns to the operational status of communal energy assets – to execute sophisticated DR strategies that are tailored to the district's scale and complexity. By optimising DR at the district level, the



system can manage peak load reductions more effectively, distribute energy resources efficiently across a larger network, and enhance the reliability and stability of the power supply.

Control within the service is twofold – automated and manual. Automated demand response is facilitated through the Universal Secure Gateway, which controls the operation of EV charging stations and battery storage based on predictive analytics and real-time data. This ensures that energy storage and consumption are optimised according to current network conditions and storage capabilities. Concurrently, manual demand response is supported by the nudging app, which receives personalized recommendations from the Pre-Aggregation Framework. These recommendations encourage users to adjust their energy consumption behaviours in real-time, enhancing the community's overall energy efficiency and participation in demand response programs.

The OurPower Community Manager plays a pivotal role in integrating community energy management with the flexibility marketplace in the Austrian demo. It processes and analyzes data received from both, the Pre-Aggregation Framework and the Distribution System Operator (DSO), crucial for effective community billing and accounting. It also interfaces with the OurPower P2P Marketplace, managing the offer and sale of aggregated community flexibility, responding to market demands, and forwarding flexibility signals back to the Pre-Aggregation Framework for execution.

Also, in the Austrian demo, the OurPower P2P Marketplace is responsible for flexibility management and handles the trading of flexibility through signals such as requests, offers, and responses. It efficiently matches these elements to optimise the utilization of energy resources within the community, fostering a responsive and economically beneficial energy ecosystem.

In the case of the Spanish pilot, the Pre-Aggregation Framework and the Universal Secure Gateway functionalities have been aggregated in the Communication Layer module. The platform used in both pilots to host these services will be different. It has also been included the Forecasting of consumption and generation service, as it will be developed during the project and has a key role in the operation of the DR optimisation service.

5.3. Data for service execution

For the execution of this service and further the effective operation of the local energy flexibility management service the acquisition and integration of following types of data inputs are required:

- Energy consumption data: This data is meticulously collected via advanced smart meters and sub-metering systems. These meters are crucial for monitoring real-time energy usage within the community, enabling detailed analysis and management of energy consumption at a very granular level. The precision of this data allows for the optimisation of energy distribution and load balancing, which are vital for effective demand response (DR) execution.



- Energy production data: Specifically focusing on photovoltaic (PV) systems, this data tracks the amount of energy generated by solar panels within the community. Accurate and timely production data is essential for optimising the integration of renewable energy sources into the grid and for facilitating effective flexibility trading within the P2P marketplace.
- Asset information: The service requires detailed and up-to-date information on the capabilities and availability of local energy assets, such as solar panels, batteries, and EV charging stations. This data is instrumental in optimising the utilization of these assets, maximizing their efficiency, and integrating them seamlessly into the overall energy management strategy.
- Weather data: The inclusion of weather data is pivotal for the comprehensive operation of the local energy flexibility management service. This data provides critical insights into environmental conditions that significantly affect energy production, particularly from photovoltaic (PV) systems. By integrating real-time and forecasted weather information such as solar irradiance, temperature, and cloud cover, the service can more accurately predict energy generation patterns. This predictive capability is essential for optimising the use of PV systems within the community. Additionally, weather data enhances the service's ability to perform effective load forecasting and demand response (DR) operations, as weather conditions directly influence energy consumption patterns and needs.
- Market data: This encompasses vital information about energy prices, demand, and supply dynamics. Access to real-time market data ensures that the service can make informed decisions, enabling users to engage in flexibility trading and DR activities that are economically advantageous.

5.4. Service output

The service outputs are designed to provide actionable information and tools for users, community managers, and energy providers, focusing on optimising energy use, enhancing sustainability, and supporting community engagement. These outputs are:

- Optimised Energy Schedules: the service provides detailed schedules that recommend optimal times for energy consumption and production, leveraging near real-time and forecasted energy consumption and production data. These schedules help users and building/community managers to maximize the utilization of renewable energy sources, aligning energy consumption with periods of high renewable energy production.
- Flexibility Utilization Schedules: Based on flexibility utilization schedules, users receive near real-time alerts and adjustments related to their energy usage and local availability of renewable energy. These alerts inform users about the best times to consume or conserve energy based on current renewable availability and demand dynamics, facilitating effective participation in demand response programs.
- Community Energy Reports: Comprehensive reports are generated detailing the community's overall improvements in energy efficiency, increases in self-consumption of produced energy, and reductions in carbon emissions. These reports provide insights into the effectiveness of energy management strategies and the community's progress toward sustainability goals.



- Flexibility Management Reports: Reports on flexibility management within the P2P marketplace are provided. These summaries help users track their flexibility activities and assess the impact of their decisions on promoting a greener energy grid. The service's outputs aim to empower users with comprehensive insights and practical tools needed to actively participate in and benefit from a sustainable energy ecosystem. This not only leads to enhanced energy efficiency and environmental benefits but also fosters a culture of active participation and awareness regarding energy sustainability within the community.

5.5. Main benefit and innovation

The service delivers significant benefits across economic, social, and environmental dimensions by optimising local energy flexibility management. By harnessing OurPower's P2P digital platform and CARTIF's district level DR optimisation (TE-13), it innovatively integrates demand response (DR) with a decentralized management approach, enhancing energy system efficiency and reducing costs. This service innovatively uses a pre-aggregation framework that interfaces with local assets to identify flexibility opportunities, making it a pioneer in enhancing residential customers' ability to participate in flexibility markets. Socially, the service fosters a more engaged and responsive community by utilizing non-financial incentives and a nudging app, which encourages participation in DR programs and promotes sustainable energy practices. Environmentally, the service contributes to reduced carbon footprints through optimised energy distribution and consumption, aligning with broader sustainability goals.

5.6. Development activity

The development activities in the reported timeframe of the project have focused primarily on conceptualizing the service and planning the integration of its various components, setting a solid conceptual foundation for future development and implementation phases. This approach ensures that all aspects of the service are thoroughly planned and designed to meet the specific needs of energy communities effectively. The next phase will build on this groundwork, transitioning from concept and planning to the actual development and implementation of the service components.

The initial phase on establishing a comprehensive concept for the service involved identifying the core objectives of the service, namely optimising local energy flexibility, enhancing self-consumption, and promoting cost optimisation within energy communities; defining the key components of the service architecture in each demo, including the Pre-Aggregation Framework (TRL5), Universal Secure Gateway (TRL5), District-level DR optimisation (TRL6), OurPower Community Manager (TRL6), and the OurPower P2P Flexibility Marketplace (TRL6); and outlining the roles and interactions of these components within the service to ensure a cohesive functionality that meets the needs of the communities it serves.

A crucial aspect is the integration of the various components. Therefore the data flows between components, such as the movement of energy consumption data from smart meters and production data from PV systems through the Pre-Aggregation Framework to other parts of the



system had to be mapped; the necessary interfaces and APIs that would facilitate seamless communication and data exchange between the different components of the service had to be identified; and the operational logistics for how the components would work together to enable effective monitoring, control, and marketplace interactions have been planned.

In addition to the foundational activities, specific arrangements have been made regarding the licensing and deployment of the developed components of the service. The project partners opted for commercial licenses for their software contributions and committed to maintaining their developed software in separate, privately managed repositories. This approach allows each partner to retain control over their intellectual property while facilitating collaboration and integration within the project framework.

Regarding the deployment of the service's software, the strategy varies for each component, reflecting the unique needs and constraints of the pilot sites. For instance, the OurPower Community Manager and the OurPower P2P Marketplace components are hosted on servers managed by OurPower, centralizing these functionalities and leveraging OurPower's existing infrastructure. As of the current stage of this deliverable, deploying the service's software in cloud environments has not been planned or initiated, with a focus maintained on local and centralized server solutions to meet the project's immediate operational and security requirements.

The service's software will be deployed on CIMNE's cloud as a microservice in the Spanish pilot. It will interact with the other cloud services already deployed, such as forecasting energy consumption and solar production.

5.7. Outlook for the future expected release

This section outlines a plan for the anticipated developments in future releases of the local flexibility management for energy communities and self-consumption/cost optimisation service. These enhancements and upgrades are aimed at refining the service's functionality, increasing its efficiency, and ensuring it continues to meet the evolving needs of energy communities. The planned developments for future expected releases of this service include:

- Enhancements to the Pre-Aggregation Framework
 - Integration of Secure and Privacy/Confidentiality Preserving Data Transactions component (TE-5) to enable identity and access management on an energy community-level
 - Extending user interface to enable community manager to manage the local flexibility
 - Integration of nudging recommendations
 - Extending functionality to provide data interfaces to/for other project service
- Upgrades to the Universal Secure Gateway
 - Extending functionality to support local asset virtualization and interactions
 - Improve service deployment and instantiation
- Improvement of District-level DR optimisation
 - Update optimisation problem formulation for both pilots



- Include interfaces for the service interoperability according to DEDALUS approach
- Enhancements to the OurPower Community Manager
 - Implement flexibility management/trading capabilities for pre-aggregated flexibility
 - Enhancement of community dashboard (user interface)
 - Upgrade of community billing and accounting
- Expansion of the OurPower P2P Marketplace:
 - Implementation of flexibility management framework (flexibility signals, matching, etc.)
 - Integration of matching algorithms to dynamically pair flexibility demands with available offers, thus optimising energy utilization across the community.
- Development of further necessary user interfaces and APIs
- Integration between components and with local assets



6. Optimal Aggregation of flexible energy resources for heat vs electrical flexibility valorization

The aim of this chapter is to provide information about the software expected to be developed as outcome of Task 4.5. This development can correspond to more than one services and has to be linked to the TE-14 *Aggregation of flexible energy resources for district heating management*.

The services developed under the headline *Aggregation of flexible energy resources for district heating management* will extended the Neogrid Platform with new services for optimal aggregation of thermal flexibility at individual buildings and at district level, where flexibility in how an energy-efficient building is heated can be used towards the district heating system, so either a building or a pool of buildings can move their energy consumption in time to minimise the collective load on the energy system. Furthermore, the solution will be extended to include ventilations systems connected to the electrical grid and include the flexibility from these systems toward the electrical grid, i.e. having one solution which can handle flexibility towards district heating and electricity systems and at the same time maintain the required indoor climate. The aim is to extend the existing system from building level to apartment level via smart thermostats in cooperating feedback from tenants with respect to comfort and flexibility requirements i.e. having humans in the loop. Having control possibility (thermostats and ventilation system) in each apartment, we expect to extract more flexibility from the building and at the same time honouring that tenants in the building can be more or less interested in providing flexibility.

The service will be closely tied to and tested in Pilot 2 (The Danish pilot).

High level services in focus

- Preheat control of DH flow temperature w/ valve feedback.
- In-apartment radiator set-point control for increased flexibility.
- Cost- & indoor climate-optimisation of ventilation systems (electricity).
- Flexibility forecast and aggregated control of district heating in apartment, building and pool of buildings (electricity and district heating)

The development process involves several steps, which in the following are presented as high-level steps, which will be carried out in relation to deliver the proposed services:

1) Initial phase:

- a) Complete physical installations
- b) Set-up building model / data collection /get meter keys from Lejerbo and placement of meters
- c) Start collecting baseline data
- d) Deploy Neogrid Preheat, collect “baseline” data (DEADLINE: SUMMER 24)



- e) Plan and execute co-creation workshops with tenants from the Danish Demonstrator (Pilot 2)

The rest of the steps are described below in paragraph 6.3.

The development activities in the reported timeframe of the project have focused primarily on conceptualizing the service and planning the integration of its various components, setting a solid conceptual foundation for future development and implementation phases.

The functional requirements for the service are listed in Table 17 below.

Table 17: Service's overview

Functionalities (brief textual description)	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated pilot site	Development Status
Control of heat in building via central control with apartment temperature feedback	TE-14-FR1	UC14	TE-3,8,9	Pilot 2 - DK	Ongoing
Control of heat in building via central control and using smart thermostat valves as feedback signal to controller	TE-14-FR2	UC14	TE-3,8,9	Pilot 2 - DK	Not Started
Monitor and Control of heat in apartment via central control and via Smart thermostats in apartments to minimize heat consumption and enable district heating flexibility	TE-14-FR3	UC14, UC17	TE-3,8,9	Pilot 2 - DK	Not Started
Monitor and Control of ventilation in apartment via central control to minimize heat loss via air change and enable to	TE-14-FR4	UC14, UC17	TE-3,8,9	Pilot 2 - DK	Not Started



Functionalities (brief textual description)	Related Functional requirements (D2.1, D2.3)	Related Use cases (D2.1, D2.3)	Associated Technology Enabler	Associated pilot site	Development Status
enable electrical flexibility					
Reduction and shifting of heat consumption for building or cluster of buildings	TE-14-FR5	UC16	TE-1, 3,4,9	Pilot 2 – DK	Not Started

Figure 39 shows the high-level architectural diagram of how the developed services are interacting in the DEDALUS setup having the Neogrid commercial platform as a main part of that.



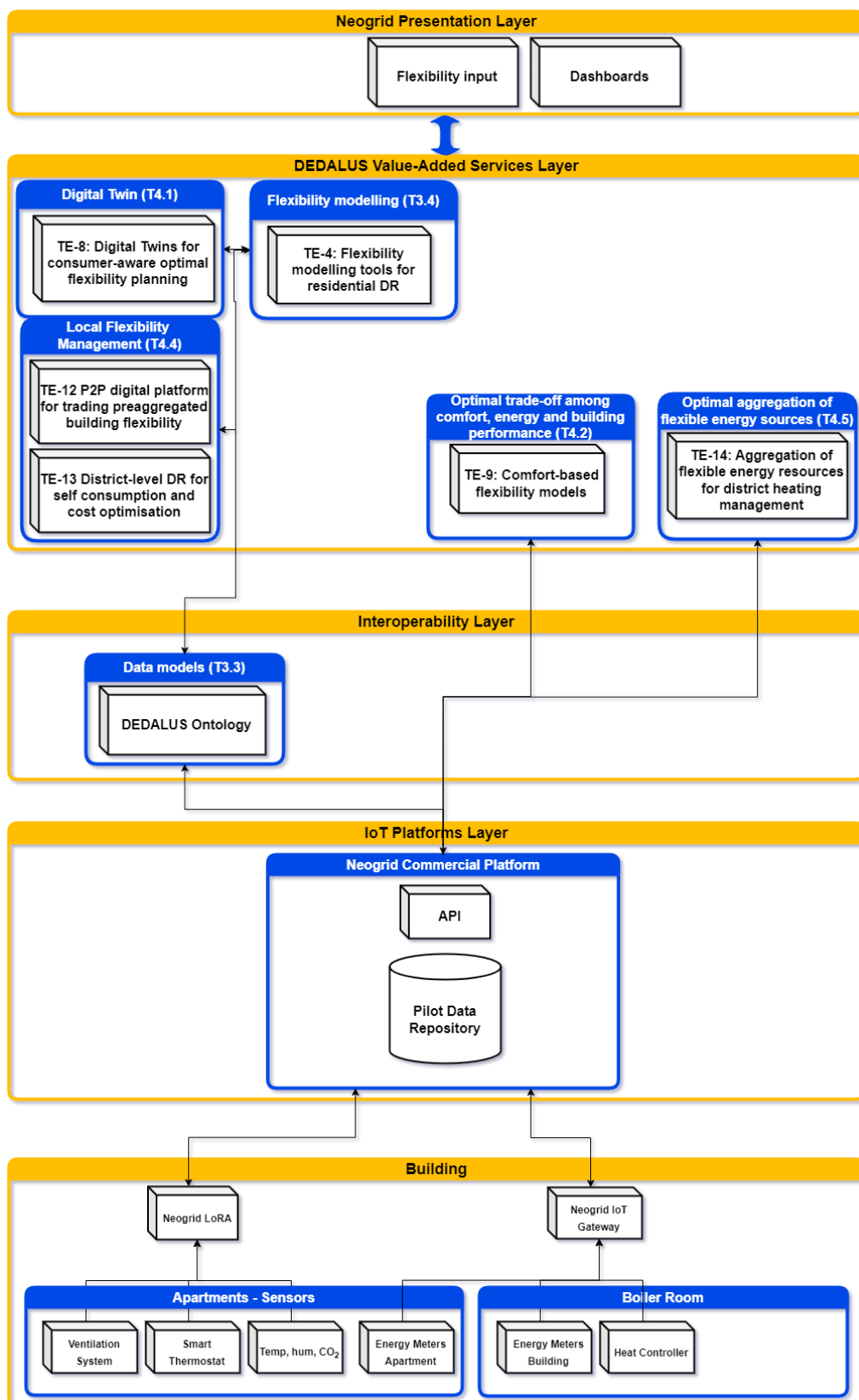


Figure 39: High level Architectural Diagram

In the following subsections, each functionality listed in Table 17 is linked to the related service that is further detailed providing information about both input and output data.

6.1.1. Service TE14-FR1

This service is about the control of heat in building via central control with apartment temperature feedback. This controller is based on one of Neogrid's commercial controllers, which will be refined during this project. The aim is to have a control running so the building manager from the beginning can start getting a feeling about the system so they can be involved and consulted during the project.

Required data for running this service is time series data from central energy meter, temperature data from controller, metrological weather data (historical and forecasts) and indoor climate data from the apartments.

Service output will be set point schedules which will be applied for the district heating water circulated around the building.

The measured indoor climate data and energy data will be used as base line for the pilot site.

6.1.2. Service TE14-FR2

This service is about the control of heat in building via central control and using smart thermostat valves as feedback signal to controller. The main goal is to extend the TE14-FR1 controller with feedback from radiator valves, to have a better view on how the heat is distributed around the building and to get information about which apartments are the limiting factor or not using all the radiators in the building.

Required data besides what is specified for In the following subsections, each functionality listed in Table 17 is linked to the related service that is further detailed providing information about both input and output data.

, valve opening signals from the valves are required.

Service output will be set point schedules which will be applied for the district heating water and domestic hot water circulated around the building.

6.1.3. Service TE14-FR3

Besides the goals for In the following subsections, each functionality listed in Table 17 is linked to the related service that is further detailed providing information about both input and output data.

) and Service TE14-FR2), this service aims to optimise and control the set points for the individual radiators in the apartments to estimate and activate flexibility for district heating across apartments and buildings and at the same time honouring the set points and tenant requirements for indoor comfort.



Required data besides what is specified for Service TE14-FR2, this service will require In the following subsections, each functionality listed in Table 17 is linked to the related service that is further detailed providing information about both input and output data.

) flexibility preferences (translated from requirement for indoor climate) and requested setpoints from apartments.

Service output will be a setpoint schedules for the individual radiators and for the central district heating water and domestic hot water production.

6.1.4. Service TE14-FR4

This service aims to optimise and control the air change rate in each apartment to estimate and activate flexibility for electricity (ventilation system) across apartments and buildings and at the same time honouring the tenant requirements for indoor comfort.

Requested data besides what is specified for previous services, this service will require data from ventilation systems in the apartments (time series data) and electricity price data.

Service output will be a setpoint schedules for the individual ventilation systems.

6.1.5. Service TE14-FR6

Based on the output from previous specified services, the aim for this service is to provide a tool to monitor, administer and execute in district heating domain for single building or cluster of buildings with focus on reduce and shifting the heat consumption (heating and domestic hot water production), to reduce the overall peak load of the building or a cluster of buildings. The tool should aim to provide services to district heating grid like:

- Peak reduction
- Reduce return temperature, maximizing cooling
- Optimise heat consumption with respect to constraints from district heating grid/company
- Optimise according to prices

Requested data from the previous services and configuration inputs from building manager. Furthermore, requirements from the district heating grid is required. Here this will be modelled and configured via the building manager inputs.

Service output will be set point schedules which will provide the requested flexibility.

6.2. Main benefit and innovation

The main benefit and innovational steps is to extend an existing heating control system from focusing on central control at building level to apartment level via smart thermostats in cooperating feedback from tenants with respect to comfort and flexibility requirements i.e. having humans in the loop.



Having control possibility (thermostats and ventilation system) in each apartment, we expect to extract more flexibility from the building and at the same time honouring that tenant in the building can be more or less interested in providing flexibility. Furthermore, tenants are actively involved in the control loop, since they can indicate via the provided App the level of comfort and level of flexibility. By moving to the apartment level with heating control, we expect to improve the energy savings compared to only controlling the building as a whole from the basement.

6.3. Outlook for the future expected release

This section outlines a plan for the anticipated developments in future releases of the aggregation and control services. The planned developments for future expected releases of this service include:

1) Preheat v2 - Valve feedback

- a) Forecasts models of building and apartments energy use, indoor temp., valve opening, etc.
- b) Develop & deploy Preheat with valve opening feedback, collect data.
- c) Visualisation / statistics on DH peak shaving for building managers (dashboard)

2) Ventilation Control

- a) Develop & deploy demand based ventilation control based on air change, co2, and humidity, collect data.
- b) Visualisation / statistics on ventilation, so building manager can confirm that QoS is maintained.

3) Preheat v2.5 - Individual apartment activation with aggregated activation

- a) Develop forecast models for flexibility in building thermodynamics
- b) Estimate thermal capacity of apartment and building to determine how long we can close heating without affecting the room temperature
- c) Develop UI with flexibility slider for tenants – based on input from co-creation workshops (Nudging work)
- d) UI for building manager to list flexibility settings and adjust them (people that calls in and complains about lack of heating, etc.) and expand dashboard with graph of building flexibility (how much of building thermo-capacity is used?)
- e) Develop & deploy Preheat with valve opening control, flexibility slide & maybe comfort profiles, collect data

4) Parallel steps to support and interact to DEDALUS

- a) Data API for digital twin, 1st step REST based interface as delivered today from the platform. 2nd step via the DEDALUS interoperability layer
- b) Implement DEDALUS ontology
- c) Develop interactions between flexibility components in DEDALUS which we are interfacing to



Conclusions

This deliverable provided information about the software artifacts belonging to the 1st technology development cycle completion, MS4 due at M13 (May 2024). Moreover, D4.1 was conceived as technical specification document of DEDALUS services as prerequisite for the proper achievement of the development activities. The software code was committed on the GitHub repository set up for the project or in the repository owned by partners responsible for the development. For each service, the link with the related TE is provided together with the information related to pilots in which services will be exploited. Indeed, some TEs, as they formally listed in the Grant Agreement, are extended and improved in the context of the project according to the needs of pilot and focusing on service's benefits and novelty.

This deliverable does not provide information about the development of services of CENTRICA due to the issue with the Irish pilot. Indeed, due to issues with the heat pump provider in the Irish demonstration, Centrica, as the pilot leader, lost this pilot and had to begin searching for an alternative pilot. According to the discussion with the project officer, the two options at this moment are either to move the new pilot to another country with a similar climate zone as Ireland, or to keep the pilot in Ireland while removing the context of social housing. Based on discussions with the project officer, after finalizing the pilot, an amendment will be submitted to the EC portal, and a newer version of this deliverable will be resubmitted.



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